

VEX Rank Engineering Notebook

Expanded Engineering History Edition



Website identity and public ranking interface

Website design development coding algorithm version history testing and test publishing

Project record from September 1 through September 23 2026

Prepared from repository source code, tests, deployment records, algorithm specifications, historical archives, and the supplied VCR v1.0 Lucidchart export.

Document Control and Revision Record

This edition follows the project from its local website beginnings through the September 23 GitHub release. Parts One through Nine preserve the detailed design and test record prepared through September 16. Parts Ten through Fourteen add the dated local and GitHub histories, current implementation, current checks, pull requests, and references. When a deployment or test status changed after September 16, the later dated record is the current one.

Field	Value
Document	VEX Rank Engineering Notebook, dated history edition
Coverage date	Local website history September 1-11; GitHub main through September 23 2026, commit 6a03cd0
Primary audience	Contributors, reviewers, robotics participants, and project maintainers
Architecture authority	VCR v1.0 six-page Lucidchart export dated August 2026
Executable rating model	VCR-3.0-candidate.2 in website/lib/vcr3.mjs
Historical dataset	2025-26 Push Back archive with complete fetched-calendar coverage
Revision purpose	Add complete dated GitHub commit and pull request record, local origin chronology, current architecture and verification evidence

Evidence convention

- Observed means the result was reproduced from source files, an automated test, a build command, a deployment record, or the published archive.
- Designed means the behavior is documented but may not be completely implemented or historically calibrated.
- Recommended means the notebook proposes the next engineering step and does not claim that work is finished.

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The table lists every part, chapter, engineering record, test case, code listing, appendix, and reference section. Page numbers are refreshed during the final Microsoft Word export.

How to Read This Notebook

Readers seeking the reason for the project should begin with Part One. Contributors changing code should use Parts Three and Four together with the annotated listings in Appendix A. Reviewers evaluating the rating model should read Part Five before using the historical statistics in Part Seven. Deployment maintainers should use Part Eight and the operational checklists in Appendix C.

Reader question	Recommended sections
Why was VEX Rank started	Part One and Project Origin Evidence
How was the design selected	Part Two Engineering Design Process
How does information move through the system	Part Three Architecture and Part Four Implementation
How did VCR change	Part Five Algorithm Version History
What was actually tested	Part Six Verification and Part Seven Historical Evidence
How is the test site published	Part Eight GitHub Firebase and Cloudflare
Where are the code and sources	Appendices A through E

Part One Project Origin and Purpose

This part records the problem that caused the project to begin and the reasons a combined website and rating system was selected.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

The Information Problem That Started the Project

VEX participants can obtain teams, events, schedules, matches, standings, brackets, skills, and awards from official systems, but those records answer separate transactional questions. A participant who wants to compare teams across many events must collect pages manually, interpret inconsistent event structures, and remember that a win-loss record does not describe opponent strength or alliance contribution.

The project began to provide one searchable public view that joins official records with a transparent estimate of competitive strength. The website and rating algorithm were treated as one engineering problem because a score without evidence is hard to trust and a data website without synthesis does not answer the comparison question.

Record field	Notebook entry
Evidence	Official data categories listed in the architecture export; website route map; VCR 1.0, 2.0, and 3.0 specifications.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	No formal user-interview dataset was available. The rationale comes from the information tasks the product was designed to support.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Can a reader trace every displayed conclusion to official or derived evidence
- Does the system state when data is partial or unavailable
- Can a contributor reproduce the same result without private knowledge

Why Official Standings Were Not Enough

Tournament standings measure performance within one event and one schedule. They are valuable, but they are not a season-long comparison because events differ in field strength, number of matches, alliance partners, elimination formats, and available statistical detail. A team may rank highly at a local event and face a substantially different field at a Signature Event or Worlds.

VEX Rank therefore preserves official standings as display data while computing a separate versioned rating. The rating is intended to aggregate evidence chronologically and adjust expectations using pre-event strength. It does not replace official qualification rankings, awards, or championship results.

Record field	Notebook entry
Evidence	Lucidchart data-system boundary; README design rules; event-strength branches in VCR v1.0 and VCR 2.0.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A rating remains a model. It can be biased by missing matches, incomplete attribution, classification errors, or uncalibrated parameters.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Are official results visibly distinguished from model outputs
- Are judged awards excluded from rating inputs
- Does the model version appear with archived data

Why the Project Needed to Exist as a Website

A spreadsheet or private script could calculate a number, but it would not make the assumptions accessible. The public interface gives users a way to search a team, see event history, inspect schedules and brackets, compare ranks, and read the method. These views expose the context needed to question a result instead of accepting a score without explanation.

The website became the evidence surface for the model. Navigation, loading states, error messages, event pages, team histories, and component explanations were included as engineering requirements rather than decorative features.

Record field	Notebook entry
Evidence	Website information architecture; React views; team, event, ranking, and skills routes; contributor guide.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The present interface is broad and the main page is large. Usability has fixture and inspection evidence but not a formal participant study.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Can a user move from rank to team history to event evidence
- Do errors explain whether official data is delayed
- Can the interface work from a narrow screen and keyboard

Why Transparency and Versioning Became Core Requirements

A ranking can influence expectations and team reputation. Silent formula changes or undocumented data substitutions would make results impossible to audit. The project therefore needed named model versions, component ledgers, frozen pre-event expectations, archive manifests, coverage records, and clear limits.

Algorithm changes are treated as version changes, not casual tuning. The historical builder records the model identifier, and the Worker includes the identifier in cache keys for rating responses so a new model cannot reuse old cached results.

Record field	Notebook entry
Evidence	VCR_VERSION constant; archive coverage envelope; Worker canonical URL; VCR 3 audit history.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The VCR 3 candidate still requires full historical calibration and therefore must not be described as a stable validated release.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Does each result carry a model identifier
- Can the input season and coverage be checked
- Would changing a formula require new tests and a new version

Primary Users and Decisions

The design serves several readers with different questions. Competitors want to locate teams and events. Coaches and analysts want season history and strength comparisons. Contributors need file ownership, data contracts, and tests. Reviewers need to distinguish observed results from planned work. Deployment maintainers need reproducible publishing and secret-handling procedures.

The notebook and website organize information by user task rather than by database table. The same source record may support a public event page, an archive replay, a regression fixture, and an algorithm audit.

Record field	Notebook entry
Evidence	Website navigation; CONTRIBUTING.md source map; deployment guides; inspection checklist.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The product has no account-specific workflow and does not claim to replace official competition management systems.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Does every page answer a recognizable user question
- Are maintenance instructions available without AI tools
- Are operational secrets kept outside the browser and repository

Project Goals and Measurable Success Criteria

The project goals were translated into measurable engineering criteria: complete paginated collections, deterministic chronological replay, searchable team and event views, visible error states, versioned archives, protected credentials, bounded retries, and repeatable tests. These criteria allow a trial to fail usefully because the failure can be tied to a requirement.

Success means that the system can publish a test website and a validated archive without silently presenting incomplete official data as complete. Predictive accuracy is a separate criterion that requires chronological calibration and held-out evaluation.

Record field	Notebook entry
Evidence	Regression suite; archive validation; Pages build; Worker health and integration counts; VCR backtesting plan.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Passing scenario tests proves implementation behavior, not predictive superiority.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Are completeness and prediction evaluated separately
- Does a failed later API page reject the collection
- Can the same season be rebuilt from recorded inputs

Project Non Goals and Ethical Boundaries

VEX Rank does not score judged awards, infer a team's character, promise future match outcomes, or declare that one region is inherently stronger. It does not place the Event VEX credential in browser code. It does not treat an unavailable historical season as if a recent live sample were a complete substitute.

The model is limited to documented competitive evidence. Geography is a filter and calibration context, not a permanent bonus. Missing contribution and autonomous evidence remain neutral and marked missing instead of being fabricated.

Record field	Notebook entry
Evidence	README design rules; VCR 3 revision; Worker archive fallback policy; contributor limitations.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Model outputs may still be misunderstood unless the interface continues to explain uncertainty, evidence age, and version status.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Could a displayed label be mistaken for an official award
- Are missing signals visibly neutral rather than estimated without evidence
- Does the site avoid deterministic claims about future results

Project Need Statement and Design Brief

VEX Rank exists to turn fragmented official competition records into a searchable, auditable, season-aware view while estimating competitive strength with a versioned model. The system must preserve the difference between source facts and derived judgments. It must fail visibly when a complete dataset cannot be obtained, keep credentials on the server, and provide enough documentation that a contributor can reproduce or challenge the behavior without relying on an AI assistant.

Design question	Requirement	Acceptance evidence
Can users find relevant records	Search teams and browse rankings and events	Team-directory fixtures and rendered views
Can data be trusted as complete	Load all pages or reject the collection	Pagination and later-page failure tests
Can ratings be audited	Version formulas and store event components	VCR ledger and historical archive metadata
Can the test system publish safely	Keep token server-side and validate before deploy	Worker secret binding and Pages health gate
Can contributors maintain it	Document source ownership and unusual assumptions	CONTRIBUTING.md, comments, tests, and this notebook

Part Two Engineering Design Process

This part applies the full engineering design process and records how failed trials changed the requirements and implementation.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Define the Problem and Scope

The team framed the problem as reliable interpretation of public competition data, not simply a new leaderboard. Scope included data ingestion, rating logic, interface views, archive generation, and test deployment because each layer could invalidate the result.

The first design boundary excluded judged awards and account features. It required server-side credentials, chronological processing, and explicit uncertainty.

Record field	Notebook entry
Evidence	Lucidchart pages 1 and 6; README design rules.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The original design contained unresolved formulas and therefore served as architecture, not an executable specification.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Research Data Sources and Constraints

The official Event VEX API provides authoritative public records but can paginate, rate limit, delay, omit fields, and represent organizer content inconsistently. Historical event formats and bracket naming also vary.

The implementation uses bounded concurrency, retry rules, validation, canonical records, and fail-closed collection semantics.

Record field	Notebook entry
Evidence	vex-api.ts, agenda.ts, bracket.ts, event-classification.ts, inspection records.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Availability and completeness still depend on the upstream service.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Establish Evaluation Criteria

Correctness, reproducibility, usability, security, reliability, cost, and maintainability were selected before deployment alternatives were compared.

Each criterion received an observable check: tests for data behavior, manifests for replay, browser inspection for usability, secret scanning for security, and build/deployment logs for reliability.

Record field	Notebook entry
Evidence	Criteria table in the earlier notebook and expanded test matrix in Part Six.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Predictive validity remains incompletely evaluated.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Generate Alternative Designs

Alternatives included a static-only site, a Firebase function and Firestore cache, a Cloudflare Worker and D1 cache, a local ranking script, and a complete historical archive served as an asset.

The selected test stack separates a GitHub Pages frontend from a Cloudflare Worker API and versioned static archives. Firebase remains documented as a trial.

Record field	Notebook entry
Evidence	Firebase and Cloudflare deployment records; Pages workflow.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A future production stack may change if cost, traffic, or operational requirements change.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Select the Architecture

The choice balanced free-plan feasibility, credential protection, cache limits, reproducible builds, and contributor access.

GitHub Pages hosts static assets, the Worker handles API routes and secrets, D1 caches modest JSON responses, and large archives are served as versioned assets.

Record field	Notebook entry
Evidence	Worker route allowlist, 1.8 MB D1 threshold, archive manifest, workflow health check.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Free-tier capacity is not a service guarantee.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Build Prototypes

The project built the interface, API routes, rating modules, historical builder, and two deployment adapters in stages. Early versions intentionally exposed gaps rather than hiding them.

Prototypes were kept in versioned files and pull requests so failures could become design evidence.

Record field	Notebook entry
Evidence	Repository history summarized in the notebook timeline; VCR candidate documents.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The local repository snapshot does not include a complete commit log, so repository documents and recorded pull-request outcomes are used.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Test Against Real Failures

Worlds pagination, bracket replays, organizer agendas, missing scored flags, rate limits, and deployment billing constraints produced concrete failures.

Each failure changed a helper, test, cache, or deployment decision. The archive builder now resumes cached progress and refuses incomplete output.

Record field	Notebook entry
Evidence	Nineteen fixture tests; archive coverage; deployment inspection counts.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Fixtures cover known failure classes but not every historical organizer format.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Communicate Iterate and Govern Changes

The project records formulas, file responsibilities, limitations, test commands, and deployment steps so later contributors can work without private context.

Changes that alter ratings require a model version, tests, and archive rebuild. Changes that alter response identity require cache-key review.

Record field	Notebook entry
Evidence	CONTRIBUTING.md; VCR version files; archive manifest; this notebook.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Governance is documented but not yet automated as a formal release process.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What observation triggered this decision
- What test can falsify the assumption
- What artifact must change if the decision changes

Engineering Decision Matrix

Alternative	Strength	Blocking issue	Outcome
Static frontend only	Simple and inexpensive	Cannot protect Event VEX credential or compute uncached live routes	Used only for frontend assets
Firebase Functions and Firestore	Integrated serverless platform and cache	Cloud Functions required Blaze for the secure secret path	Rules and indexes retained; function path superseded
Cloudflare Worker and D1	Secret binding, low-cost edge runtime, SQL cache	D1 row/string and Worker execution limits	Selected for the test backend
Direct browser API calls	No custom backend	Credential exposure and inconsistent CORS	Rejected
One live route for all seasons	No archive publishing process	Partial samples could be mistaken for complete history	Rejected in favor of versioned archives

Part Three System Architecture from the Lucidchart Design

This part explains every page of the supplied VCR v1.0 Lucidchart export and traces each design branch into the implemented system.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

VEX Rank Engineering Notebook Dated History Edition

Architecture Overview and Feedback Loop

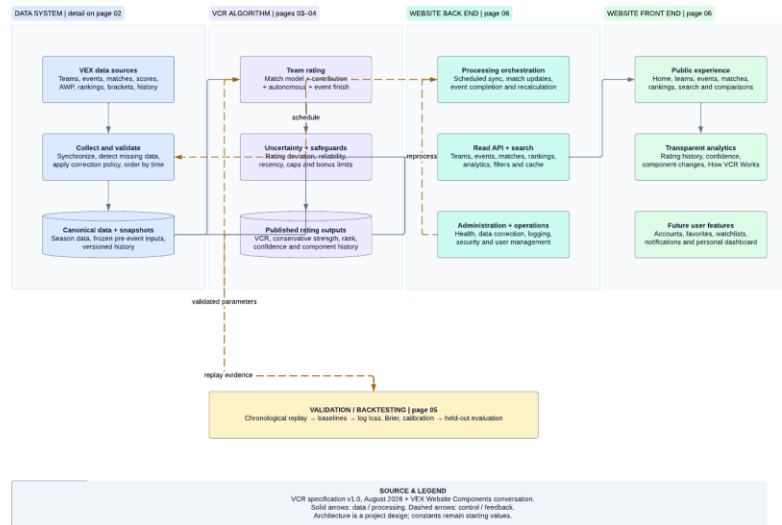


Figure 1 Original end to end system architecture

The first page separates official data, the rating engine, the backend, the public interface, and validation. Solid arrows carry records or calculations; dashed arrows return tests, policy decisions, or operational feedback. This separation became the central control against leaking credentials or allowing presentation code to silently redefine the model.

- Official data enters through validation before rating or display
- The rating engine publishes versioned outputs rather than mutating source records
- The backend mediates search, history, cache, and operations
- Validation can change parameters only through a versioned release

Data Ingestion Quality and Persisted Outputs

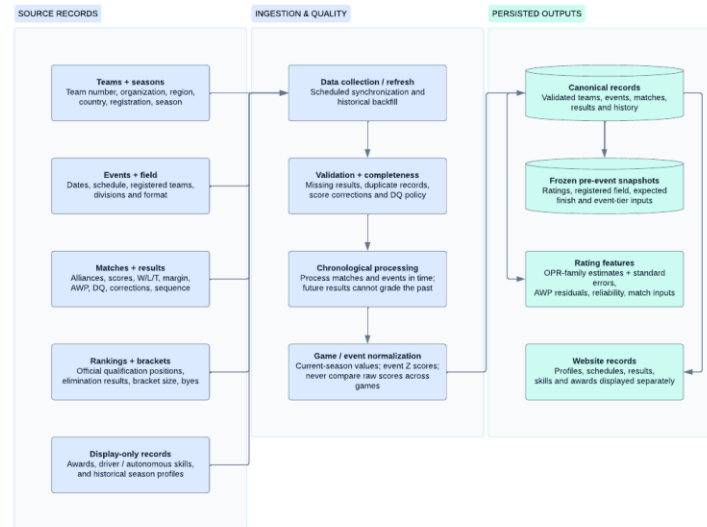


Figure 2 Data system detail

The second page defines source records, ingestion checks, canonical records, frozen snapshots, derived features, and website records. The most important design rule is chronology: a future result must never change an earlier expectation. Persisted snapshots and model identifiers make that rule testable.

- Teams and seasons establish identity
- Events and divisions establish context
- Matches and results provide performance evidence
- Rankings and brackets provide official position and progression
- Awards and unrelated skills remain display-only

VCR Version One Team Rating Flow

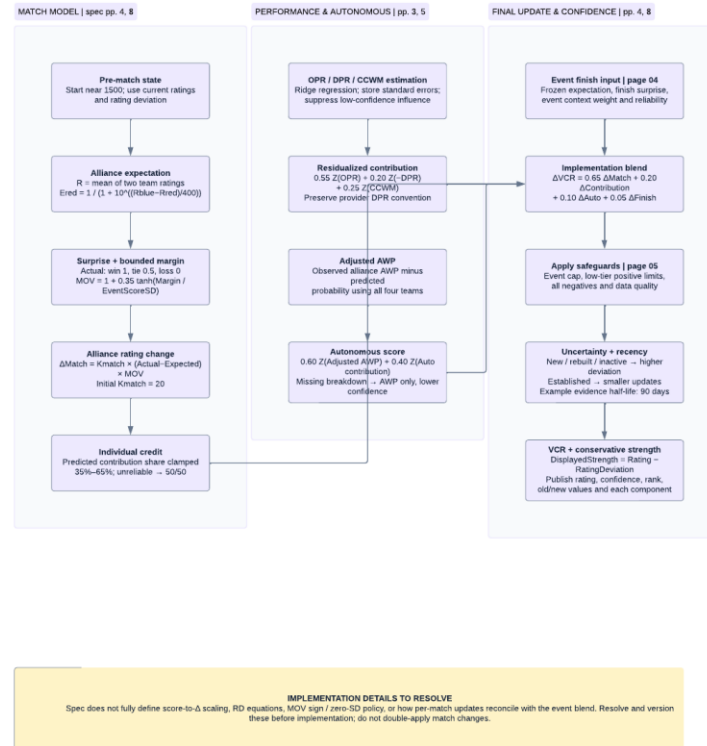


Figure 3 VCR v1.0 team rating flow

The third page shows pre-match state, alliance expectations, surprise and margin, partner credit, contribution estimates, autonomous evidence, uncertainty, and the final blend. It is valuable because it identifies every intended component, but several mathematical branches were left unresolved and later versions changed the weights.

- Freeze rating and uncertainty before the event
- Estimate alliance result from both partners and both opponents
- Bound margin influence
- Use reliability when evidence is incomplete
- Record the component path for later audit

Event Strength Finish and Asymmetric Adjustment

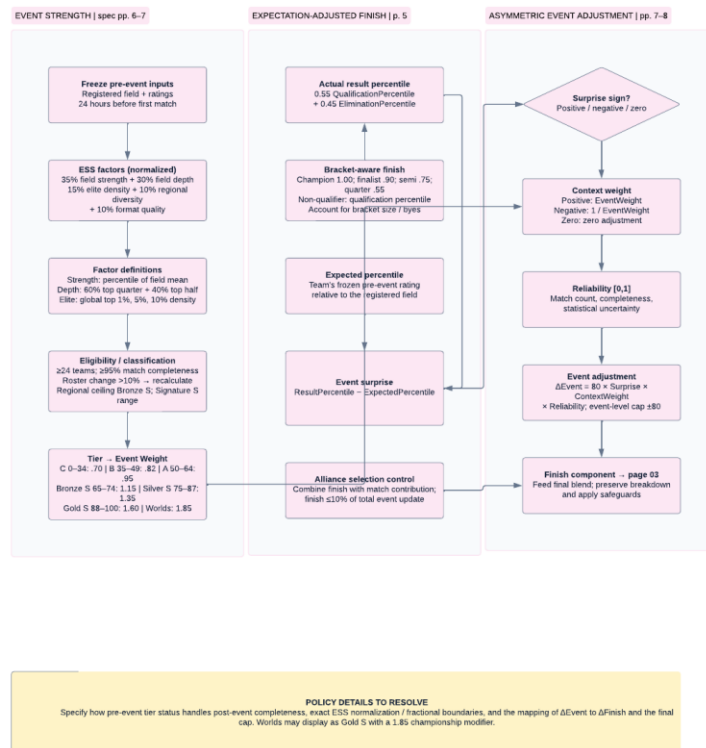


Figure 4 VCR v1.0 event outcome design

The fourth page treats event context separately from ordinary match updates. It proposes an Event Strength Score, expected finish, actual finish, asymmetric upside and downside, and a reliability factor. VCR 2.0 expanded this into numerical rules; VCR 3.0 simplified the executable candidate to tier-based settlement with a bounded reciprocal downside.

- Freeze event context before outcomes are known
- Compare actual finish with the same utility used for expectation
- Do not double-count opponent strength through match and event weight
- Keep champion policy explicit

Safeguards Backtesting and Parameter Governance

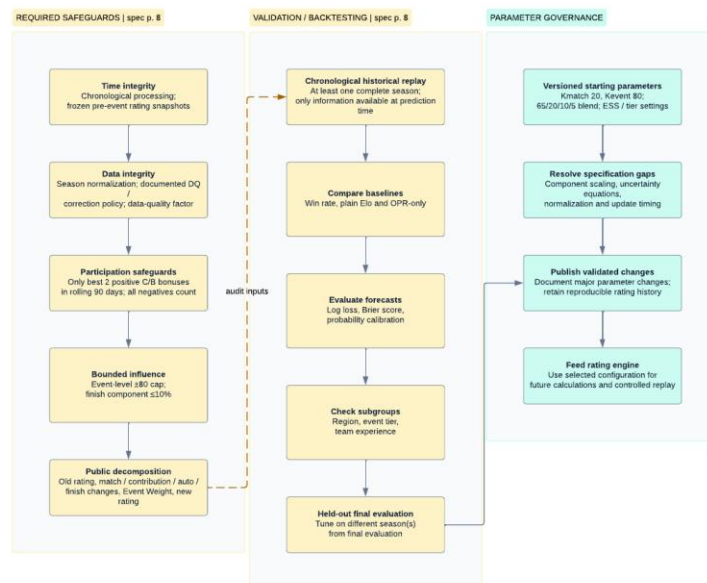


Figure 5 Validation and governance loop

The fifth page makes validation part of the architecture. It calls for time integrity, data quality, participation safeguards, public decomposition, chronological replay, baseline comparison, forecast metrics, subgroup review, held-out evaluation, and versioned parameter changes. These remain requirements even though the current candidate has only scenario and integration evidence.

- Train on the past and evaluate on later events
- Compare against simpler baselines
- Report calibration and ranking stability
- Separate data failure from model error
- Publish version and change notes

Frontend Backend Operations and Delivery

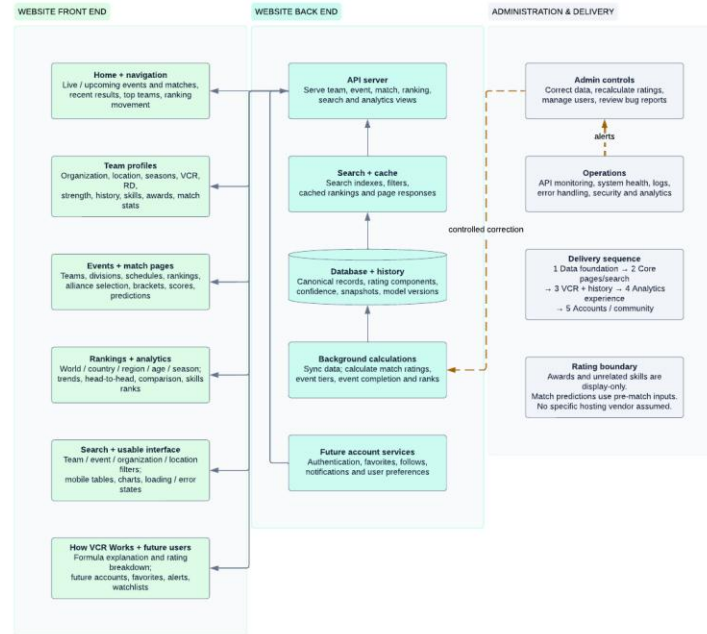


Figure 6 Website and delivery architecture

The sixth page links public pages to an API, cache, database, background evaluation, account services, and operations. The implemented test stack changed technologies but retained the boundary: the browser receives public JSON, the server owns credentials, archives are versioned, and deployment is gated by tests and health checks.

- Frontend contains no Event VEX token
- Backend allowlists routes and query fields
- Operations can inspect health and rebuild archives
- Delivery proceeds from data foundation to core pages to rating transparency

Architecture Traceability from Design to Code

The Lucidchart export expresses responsibilities rather than a fixed technology stack. Traceability requires mapping each design box to one or more current files and identifying where implementation is incomplete.

The notebook uses a traceability matrix so a contributor can find the present implementation without assuming that every diagram branch exists exactly as drawn.

Record field	Notebook entry
Evidence	Architecture figures; repository source map; deployment adapters; shared VCR module.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Accounts, full schedule simulation, complete attribution, and automated calibration remain absent or provisional.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Does every design responsibility map to a current file or an explicit gap
- Does a technology change preserve the responsibility boundary
- Can a test demonstrate the mapping

Architecture Traceability Matrix

Design responsibility	Current implementation	Status
Collect official data	website/lib/vex-api.ts and app/api routes	Implemented with retry and pagination
Canonical chronological records	historical rebuild scripts and archive validators	Implemented for published archive path
Rating engine	website/lib/vcr3.mjs	Candidate implementation
Historical snapshots	versioned archive JSON and manifest	One complete season published
Search and public pages	React page, team directory, and API routes	Implemented and fixture-tested
Cache and server mediation	Cloudflare Worker and D1	Test deployment
Validation laboratory	Node tests and rebuild coverage	Scenario and integration evidence
Predictive backtesting	Documented plan	Not complete
Accounts and saved views	Diagram only	Not implemented

Part Four Website Design Building and Coding

This part documents information architecture, interface decisions, data access, implementation boundaries, and contributor-facing code behavior.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Information Architecture and Navigation

The interface groups work into rankings, team discovery, events, event details, team profiles, and method explanations. A shared selected-season state keeps navigation coherent across views.

The main interface remains a single large React page with shared components. This simplified prototyping but created a refactoring need.

Record field	Notebook entry
Evidence	website/app/page.tsx; website/pages/main.tsx; browser inspection.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The main page is large and has an outstanding type and lint backlog.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Home View and Project Orientation

The home view must explain what VCR is, identify the selected season, expose primary navigation, and avoid presenting a model score as an official result.

The page uses the VEX Rank identity, summary cards, season controls, and direct paths to rankings and events.

Record field	Notebook entry
Evidence	website/app/page.tsx and website/public/og.png.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Formal usability testing has not been conducted.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Ranking View

The ranking view answers who is strongest under the selected archive and model version. It must show rank, team identity, rating, confidence, recent change, record, and filters without implying false precision.

Historical seasons load versioned archives; the live endpoint is explicitly a latest-event sample.

Record field	Notebook entry
Evidence	Ranking route, archive validator, and public archive assets.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The live route samples the latest 36 completed events and is not a complete season.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Team Directory and Search

Users often know a numeric family, suffix, partial team name, or organization. The directory must include inactive historical teams so older archive links remain discoverable.

Search normalizes case, spaces, country aliases, region, and grade filters. Numeric family searches return every suffix, including unregistered teams.

Record field	Notebook entry
Evidence	team-directory.mjs and four passing directory fixtures.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Directory freshness depends on rebuilding its published asset.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Team Profile and History

A team page connects identity, event history, match evidence, record, rating changes, and model components. It is the main audit path from a leaderboard row to underlying events.

Season identity is preserved in navigation and history entries include versioned event settlement where available.

Record field	Notebook entry
Evidence	team API route; VCR history ledger; archive rankings.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Champion inference and live-route formulas still have documented limitations.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Event Discovery and Classification

Event names and official level fields are inconsistent enough that display classification needs explicit rules. A World qualifier must not become the World Championship merely because its level field says World.

Classification combines exact championship-name evidence with official fields and separates level class from event format.

Record field	Notebook entry
Evidence	event-classification.ts and three regression tests.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Historical organizer names can still introduce unseen patterns.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Event Detail and Division Coverage

An event page must show every available division, match page, result, agenda entry, and bracket slot. Loading only the first page can omit most of Worlds.

The route requests complete collections and preserves division identity. Worlds inspection recorded 11 divisions and 865 teams for event 64025.

Record field	Notebook entry
Evidence	events/[id] route, vexCollection, inspection record.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Live upstream rate limits can delay large events.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Agenda Parsing

Organizer agendas arrive as structured tables or loose text. Flattening a table can detach times from activities and locations.

The parser retains start time, end time, activity, location, and day grouping while also supporting a single day without headings.

Record field	Notebook entry
Evidence	agenda.ts and two agenda fixtures.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Organizer HTML outside the recorded patterns needs new fixtures.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Bracket Slots Replays and Byes

A bracket is positional. Grouping only by list order moves replays and can place opponents in the wrong visual slot.

Slots are keyed by round and instance. A replay replaces the displayed match for that slot while preserving all games for detail, and absent byes remain null.

Record field	Notebook entry
Evidence	bracket.ts and replay regression fixture.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Unusual formats need explicit mappings rather than guessed positions.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Loading Empty and Error States

A failed API request must not look like a valid empty dataset. Users need to know whether no records exist, the upstream service is delayed, or a query is unsupported.

The interface separates loading, empty, and error states; the backend returns explicit 429, 502, and 503 responses where appropriate.

Record field	Notebook entry
Evidence	React state handling, Worker error mapping, inspection record.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Error copy and accessibility should be reviewed across more devices.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Responsive and Keyboard Design

Dense rankings and event data must remain usable on narrow screens and with keyboard navigation.

Responsive spacing, scrollable data regions, visible focus behavior, and compact cards were included in the inspection pass.

Record field	Notebook entry
Evidence	globals.css, UI components, manual inspection.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Automated accessibility tooling and participant testing remain future work.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Visual Language

The site uses a dark competition-field image, red accents, white text, and navy or neutral data surfaces. The visual system needs to support dense evidence without obscuring hierarchy.

Cards summarize; tables compare repeated records; charts show trends; warnings explain limitations.

Record field	Notebook entry
Evidence	og.png, globals.css, page components.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Color contrast and chart alternatives require a formal audit.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- What user question does this view answer
- What data state can fail
- What fixture or inspection proves the behavior

Repository Structure and Ownership

Source is divided among presentation, API routes, shared data helpers, the VCR core, historical jobs, deployment adapters, tests, and generated assets.

CONTRIBUTING.md names the first file to inspect for each area so a contributor can work without hidden context.

Record field	Notebook entry
Evidence	Repository source map and measured source inventory.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Generated archives are large and should not be confused with hand-edited source.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Browser Request Routing for Two Origins

GitHub Pages serves under /Vex-Rank while API requests use the Worker origin and archive requests may use a separate asset origin.

siteFetch selects the correct base and preserves the path. This keeps the shared interface usable in both same-origin development and split-origin testing.

Record field	Notebook entry
Evidence	website/lib/client-fetch.ts.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A new route category requires explicit routing and cache-key review.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Bounded Concurrency

Fetching every page or event serially is slow, while unbounded parallelism increases rate-limit failures.

mapLimit claims one stable index per worker, preserves output order, and caps concurrent work.

Record field	Notebook entry
Evidence	website/lib/vex-api.ts and archive rebuild scripts.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The best limit depends on upstream behavior and may require tuning.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Retry and Timeout Policy

Network errors, timeouts, HTTP 429, and server errors can be temporary. Client errors usually indicate an invalid request and should fail immediately.

Each request has a 15-second timeout and up to three attempts. Rate limits honor Retry-After within a bounded wait; other retries use exponential delay.

Record field	Notebook entry
Evidence	vexlson implementation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Retries do not guarantee success and can still consume free-tier execution time.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Complete Pagination

The first official page is metadata, not proof that the collection is complete.

vexCollection fixes per_page at 250, reads last_page, preserves all filters, loads remaining pages with concurrency two, validates each payload, and returns only after all pages succeed.

Record field	Notebook entry
Evidence	Pagination and later-page failure tests.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	If upstream metadata is wrong, completeness cannot be independently proven by this helper.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Process Local Cache

Repeated official requests waste latency and upstream quota.

Successful pages are cached for five minutes with a 600-entry bound. Failed responses are not cached as valid data.

Record field	Notebook entry
Evidence	vex-api.ts cache implementation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Process-local cache disappears when the runtime restarts.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

API Route Responsibilities

Routes translate public questions into official API collections and shared transformations. A route must not silently contain its own incompatible rating formula.

Events, event detail, team, ranking, and skills routes share data helpers and the VCR module where applicable.

Record field	Notebook entry
Evidence	website/app/api tree and contributor source map.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The team route and live ranking history still contain documented approximation limits.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Cloudflare Worker Dispatch

The Worker is the public server boundary for the test frontend.

A route allowlist rejects unsupported paths and query values before dispatch. CORS is limited to the GitHub Pages origin and only GET and OPTIONS are allowed.

Record field	Notebook entry
Evidence	website/cloudflare/worker.ts.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	CORS is a browser control, not authentication.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

D1 Shared Cache

The Worker needs a cache that can be shared across invocations.

Canonical request identity becomes the D1 key. Successful JSON smaller than 1.8 MB is cached for 15 minutes, leaving margin below D1's 2 MB value limit.

Record field	Notebook entry
Evidence	Worker cache code and Cloudflare D1 limits reference.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Large responses bypass D1 and can still consume upstream capacity.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Historical Archive Builder

A complete season cannot depend on a live endpoint that samples recent events.

The rebuild job downloads all fetched event payloads, caches pages, sorts by event end and ID, processes each event once, validates coverage, and replaces the final archive only after success.

Record field	Notebook entry
Evidence	rebuild-season-archives.mjs and archive tests.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Only Push Back had a complete published VCR 3 archive at the recorded milestone.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Archive Validation and Packaging

A wrong-season, wrong-model, or incomplete archive must not be published under a valid-looking filename.

Validation checks season identity, model prefix, ranking records, loaded versus expected event counts, and an empty failure list before packaging.

Record field	Notebook entry
Evidence	ranking-seasons.mjs, prepare-archive-assets.mjs, archive fixture.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Coverage confirms the fetched calendar, not every real-world competition beyond the upstream calendar.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Contributor Documentation and Comments

Comments are most valuable when they explain assumptions, units, side effects, security boundaries, or unusual behavior.

The repository adds comments around pagination, cache identity, model state mutation, bracket replay slots, and archive replacement, while CONTRIBUTING.md explains where changes belong.

Record field	Notebook entry
Evidence	Pull-request record and source comments.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Comments can become stale and must change with behavior.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Which input contract does this module assume
- What happens when the dependency fails
- Which test should change with this behavior

Measured Source Inventory

The measured implementation inventory contains 36 selected source and documentation files with 2,156 lines and 231,064 bytes across the application, shared libraries, Worker, scripts, and tests. Generated dependencies and large archive JSON files are excluded from this source count.

Category	Files	Lines	Bytes
app	8	1,182	149,681
cloudflare	7	224	14,954
lib	11	313	26,168
scripts	5	288	28,227
tests	5	149	12,034

Largest Maintained Source Files

Path	Lines	Bytes
website/app/page.tsx	521	109144
website/app/globals.css	309	7647
website/lib/vcr3.mjs	98	11440
website/scripts/rebuild-season-archives.mjs	97	8735
website/cloudflare/worker.ts	95	6163
website/app/api/teams/[number]/route.ts	83	9467
website/app/api/events/[id]/route.ts	80	6068
website/tests/data-regressions.test.mjs	76	4192
website/scripts/regrade-historical-ranking.mjs	67	5762
website/lib/vex-api.ts	65	3374
website/scripts/build-historical-ranking.mjs	64	8300
website/app/api/events/route.ts	62	5427
website/app/api/rankings/route.ts	57	8290
website/cloudflare/DEPLOYMENT.md	51	3482
website/app/api/skills/route.ts	43	2379
website/scripts/build-team-directory.mjs	40	3679
website/lib/agenda.ts	35	4008
website/cloudflare/archive-source.ts	32	2549

The large main React page is a maintenance signal. Splitting view-specific data adapters and presentation components is recommended, but it should be done behind regression tests so the refactor does not change ranking behavior.

Part Five VCR Algorithm Version History

This part documents the design intent, formulas, changes, unresolved questions, and validation status of VCR versions 1.0, 2.0, and the 3.0 candidate.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Why the Rating Algorithm Is Versioned

A rating formula changes the meaning of every published score. Versioning protects reproducibility by tying inputs, constants, settlement logic, and archives to a named model.

Every archive and audit entry should carry the exact model identifier. Cache identity must change when the model changes.

Record field	Notebook entry
Evidence	VCR_VERSION; archive modelVersion; Worker model query namespace.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A version name alone is insufficient unless the inputs and build process are also preserved.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Purpose

VCR 1.0 established the architecture for match surprise, bounded margin, partner credit, contribution, autonomous evidence, event strength, finish, uncertainty, and validation.

It is treated as a design proposal and traceability source, not a complete executable algorithm.

Record field	Notebook entry
Evidence	Supplied six-page Lucidchart export.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Several equations, scaling choices, and reconciliation rules were unresolved.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Match Model

The original match branch estimated alliance win probability from team ratings and used win, tie, or loss surprise with a bounded score-margin multiplier.

The key invariant retained by later versions is that expectations must use pre-match ratings and both alliances.

Record field	Notebook entry
Evidence	Lucidchart page 3.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The diagram did not fully settle score scaling across games or rating-deviation interaction.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Partner Credit

Alliance results must be divided between partners. The original design bounded shares between 35 and 65 percent and proposed local contribution evidence.

Weak evidence should move shares toward 50/50 rather than inventing precise attribution.

Record field	Notebook entry
Evidence	Lucidchart partner-credit branch.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The 35-65 bound was later widened in VCR 2.0 and VCR 3.0.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Component Blend

The original final update used 65 percent match, 20 percent contribution, 10 percent autonomous, and 5 percent finish.

This blend made tournament achievement a small part of the event change and left open how an event adjustment mapped into the five-percent branch.

Record field	Notebook entry
Evidence	Lucidchart page 3 and page 4.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The scale of each component was not defined well enough to make percentages alone meaningful.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Event Strength

The Event Strength Score proposed field strength, field depth, elite density, regional diversity, format quality, and frozen pre-event inputs.

The event tier should be set before matches so outcomes cannot make their own event appear stronger.

Record field	Notebook entry
Evidence	Lucidchart page 4.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The design needed numerical definitions and historical calibration.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version One Validation Plan

The design called for chronological replay, baseline comparisons, log loss, Brier score, calibration, subgroup checks, and a held-out season.

These tests remain the acceptance path for later versions because scenario tests cannot establish predictive quality.

Record field	Notebook entry
Evidence	Lucidchart page 5.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The full plan has not yet been completed for VCR 3.0.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Public Score

VCR 2.0 distinguishes internal strength μ , rating deviation RD, and a conservative public score. The public transformation subtracts uncertainty before scaling to 0-1000.

New or inactive teams remain conservative until reliable evidence reduces uncertainty.

Record field	Notebook entry
Evidence	README and VCR 2.0 full algorithm PDF.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The website's executable VCR 3 candidate currently publishes internal-style ratings rather than this complete VCR 2 public transformation.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Component Weights

VCR 2.0 specifies 55 percent match, 18 percent contribution, 10 percent autonomous, and 17 percent event result.

The larger event-result component responded to the weak five-percent finish influence in VCR 1.0.

Record field	Notebook entry
Evidence	README component table and full algorithm PDF.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Paper completeness did not guarantee that every API route used the same formula.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Match Evidence

The match engine uses alliance-average ratings, expected result, actual result, and bounded margin. Ordinary match updates are not multiplied by Event Weight because opponent ratings already encode difficulty.

Match difficulty and tournament achievement remain separate evidence branches.

Record field	Notebook entry
Evidence	VCR 2.0 specification.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Game-specific score normalization still requires calibration.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Dynamic Partner Shares

Partner credit can move between 20 and 80 percent using available contribution evidence, with global reputation pulling only when local evidence is weak.

The share rule tries to avoid equal-credit bias without allowing noisy estimates to dominate.

Record field	Notebook entry
Evidence	VCR 2.0 full algorithm PDF.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Reliable individual attribution is often unavailable from public match records.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Contribution and Autonomous

Regularized offense, defense, contribution, and autonomous models were proposed so raw OPR differences would not be treated as independent evidence.

CCWM equals OPR minus DPR and must not be counted as a third independent signal.

Record field	Notebook entry
Evidence	VCR 2.0 documentation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The VCR 3 adapter currently leaves contribution and autonomous neutral when reliable inputs are absent.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Event Result

Qualification and elimination performance are compared against frozen expectations. Elimination utility rewards stage reached rather than a single final-match label.

Byes should not count as wins, and expectation must use the same utility as the actual result.

Record field	Notebook entry
Evidence	VCR 2.0 event-result specification.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A complete schedule and bracket simulator was specified but not implemented in the current adapter.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Event Weight

Positive achievement is amplified at stronger tiers while downside uses a bounded reciprocal so weak-event underperformance still matters.

Ordinary match evidence remains unweighted at the event level.

Record field	Notebook entry
Evidence	VCR 2.0 event weight table.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Tier thresholds require historical evaluation.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Two Uncertainty and Recency

Rating deviation represents uncertainty, and older evidence decays in leaderboard calculations. Inactivity should increase uncertainty rather than directly declare a team weaker.

The season model partially carries prior knowledge while acknowledging a new game and robot.

Record field	Notebook entry
Evidence	README and VCR 2.0 PDF.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The current VCR 3 archive uses a 75-day half-life at season end outside settlement.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

Why VCR Version Three Was Proposed

The engineering review found competing formulas across the team-history route, leaderboard, and historical builder. Some paths assumed every opponent was 1500, omitted partners, or sampled incomplete event data.

VCR 3.0 centralizes completed-event settlement in one shared module and returns an auditable ledger.

Record field	Notebook entry
Evidence	docs/VCR-3.0-revision.md and website/lib/vcr3.mjs.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Candidate status remains until full historical calibration and deployment review.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Frozen Inputs

Before an event, the processor freezes entrant ratings, field distribution, tier, and expectation proxies. Match processing can update local partner evidence but must not rewrite the event's expectations.

Unknown teams receive a neutral 1500 prior. Future outcomes cannot leak backward into the frozen state.

Record field	Notebook entry
Evidence	processCompletedEvent implementation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The adapter uses a field-strength proxy rather than a full schedule simulation.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Match Formula

For each alliance, expected result comes from alliance-average ratings. Margin influence is 1 plus 0.25 times tanh of absolute margin over score scale. Residual multiplies surprise, margin, two, and partner share.

Six effective matches saturate match-sample strength so a longer schedule does not automatically accumulate more credit.

Record field	Notebook entry
Evidence	VCR 3 revision and scenario tests.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Score-scale selection is event-based and needs empirical review across games.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Tournament Evidence

Qualification contributes 21 points times actual minus expected. Elimination contributes 39 points times actual utility minus expected utility. Together they preserve the 35-65 split of a 60-point tournament scale.

The current adapter approximates expectation from frozen field position rather than simulating the exact schedule and bracket.

Record field	Notebook entry
Evidence	VCR 3 revision and shared module.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Published finish components should be described as provisional until the simulator exists.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Contribution and Autonomous

Contribution is bounded at plus or minus eight points and autonomous at plus or minus four using standardized residuals. Missing values contribute zero and remain absent.

Neutral missing evidence prevents fabricated precision and avoids double-counting OPR, DPR, and CCWM.

Record field	Notebook entry
Evidence	settleEvent component rules.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The adapter does not yet derive reliable individual contribution or autonomous residuals.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Event Weight and Cap

The base component sum receives positive tier weight or a bounded reciprocal downside, then event reliability. The final event change is capped at plus or minus 100.

The ledger records the raw base, weight, reliability, correction, cap adjustment, and rating after settlement.

Record field	Notebook entry
Evidence	settleEvent implementation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The weights are candidate constants and need chronological calibration.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Champion Policy

The product requirement is that a verified champion cannot receive a negative event change, and an unexpected champion receives a positive minimum when evidence reliability is positive.

This is an explicit achievement policy, not a disguised probability claim. A separate predictive rating should remain separate if required.

Record field	Notebook entry
Evidence	VCR 3 revision and champion fixtures.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Champion detection currently relies on named finals and win-loss counts and is incomplete for some multi-division championships.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Three Audit Ledger

Settlement returns every component, weighting step, correction, cap adjustment, version, and final delta so a team history can explain why a score changed.

Display rounding happens after raw values are recorded.

Record field	Notebook entry
Evidence	VCR history entries and tests.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The interface should expose more ledger detail without overwhelming casual users.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Comparison and Release Status

VCR 1.0 is an architecture proposal, VCR 2.0 is a detailed launch specification, and VCR 3.0 candidate.2 is the shared executable model used by the published Push Back archive.

A higher version number does not mean predictive validation is complete. Release status must state implementation and calibration separately.

Record field	Notebook entry
Evidence	Lucidchart, VCR 2.0 PDF, VCR 3 documents, archive metadata.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Historical comparison against VCR 2.0 and simple baselines is still required.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Is this rule architecture, specification, or executable behavior
- What data is frozen before the event
- What test or backtest is required before release

VCR Version Comparison Table

Design area	VCR 1.0	VCR 2.0	VCR 3.0 candidate
Status	Architecture proposal	Detailed specification	Shared executable candidate
Main blend	65 20 10 5	55 18 10 17	Additive 40 match 21 qualification 39 elimination plus bounded statistics
Partner share	35 to 65 percent	20 to 80 percent	Dynamic 20 to 80 percent from frozen rating and earlier local evidence
Finish model	Event surprise branch	Qualification and elimination utilities	Qualification and elimination components with provisional expectations
Champion policy	No explicit guarantee	Strong elimination importance	Verified champion nonnegative and unexpected champion positive floor
Audit	Planned decomposition	Detailed component design	Returned versioned event ledger
Validation	Backtesting plan	Chronological calibration required	Scenario and integration tests pass; calibration incomplete

VCR Three Settlement Formula Reference

Match and event components

```

E = 1 / (1 + 10 ^ ((opponentAllianceRating - ownAllianceRating) / 400))
MOV = 1 + 0.25 * tanh(abs(scoreMargin) / scoreScale)
residual = (actual - E) * MOV * 2 * partnerShare

n = sum(matchReliability)
M = 40 * weightedMean(residual) * min(1, n / 6)
Q = 21 * (qualificationActual - qualificationExpected)
L = 39 * (eliminationActual - eliminationExpected)
C = 8 * tanh(contributionResidual / 2)
A = 4 * tanh(autoResidual / 2)
B = M + Q + L + C + A

weighted = B * (B >= 0 ? positiveTierWeight : reciprocalDownsideWeight)
delta = clamp((weighted * reliability) + achievementCorrection, -100, 100)

```

These quantities are internal rating points centered near 1500. They are not multiplied by another percentage blend. Missing contribution and autonomous values remain zero-valued missing signals, and a completed verified champion may receive an explicit achievement correction.

Algorithm Calibration Plan

1. Freeze historical event inputs and replay strictly by end date and event ID.
2. Compare VCR 2.0, VCR 3.0, plain Elo, win-rate, and OPR-only baselines on the same later-event targets.
3. Measure log loss and Brier score for match probabilities, rank correlation for later event performance, and calibration by probability bin.
4. Report results by grade, region, event tier, data completeness, and activity level.
5. Use a held-out season for final evaluation after parameter selection.
6. Publish seeds, snapshots, model identifiers, and confidence intervals so results can be reproduced.

A scenario test can show that the champion floor is applied. It cannot show that the resulting rating predicts later matches better. The release decision must keep those claims separate.

Part Six Verification Test Data and Iteration Trials

This part records the current automated suite, measured build results, browser and live-integration evidence, lint backlog, and test limitations.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Verification Strategy

The test strategy uses layers because no single check can establish system quality. Pure fixture tests isolate deterministic data and rating behavior. Build checks verify that the static frontend compiles. Browser inspection checks rendered states. Live integration checks validate a real Worker, D1 binding, and public datasets. Historical replay checks coverage and scale. Predictive backtesting remains a separate unfinished layer.

Layer	Question	Recorded evidence	Limit
Unit and regression	Does a known input produce the protected output	19 passing Node tests	Only recorded fixtures
Build	Can the Pages frontend compile	2,362 modules and output artifacts	Does not prove runtime API availability
Lint and type	Is source maintainable and accessible	Known failure backlog	Baseline is not yet clean
Browser inspection	Are views usable and states visible	Worlds, loading, error, and narrow-screen checks	Not a formal user study
Live integration	Can Worker, D1, routes, and archives respond	Health and count observations	Point-in-time observations
Historical replay	Can a season be processed completely	1,681 loaded calendar events and zero recorded failures	One published candidate season
Predictive validation	Does the model outperform baselines	Planned	Not completed

Test Case 1 Historical Scores Remain Usable When The Upstream Scored Flag Is False

Field	Recorded value
Source	website/tests/archive.test.mjs
Focus	historical match eligibility
Fixture	played 35 to 20, ambiguous 0 to 0, unplayed negative scores, and an explicit scored 0 to 0 tie
Expected result	retain the played match and explicit tie; reject the placeholder, unplayed record, and duplicate
Observed result	Passed in 6.8675 milliseconds during the September 16 2026 run
Engineering significance	Prevents valid older results from disappearing because an upstream flag was not populated

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 2 Each Season Has Its Own Archive And Incomplete Or Wrong Season Results Are Rejected

Field	Recorded value
Source	website/tests/archive.test.mjs
Focus	archive identity and coverage
Fixture	four configured season paths plus valid and invalid archive envelopes
Expected result	require unique paths, matching season, complete coverage, expected model, and no failures
Observed result	Passed in 1.1143 milliseconds during the September 16 2026 run
Engineering significance	Prevents a partial or mislabeled archive from appearing as authoritative history

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 3 Semifinal Replays Retain Their Bracket Slot And Missing Byes Leave A Gap

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	bracket identity
Fixture	two semifinal instances, a replay for instance one, and a sparse round with a later instance
Expected result	show the newest replay in the same slot, retain its game history, and keep missing slots null
Observed result	Passed in 1.6892 milliseconds during the September 16 2026 run
Engineering significance	Protects bracket opponents and layout from list-order corruption

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 4 A Worlds Qualifier Is Not The World Championship

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	event classification
Fixture	a provincial qualifier, exact Worlds event, and a practice event
Expected result	classify only the actual championship as World Championship
Observed result	Passed in 2.7254 milliseconds during the September 16 2026 run
Engineering significance	Prevents inflated labels and downstream rating-tier mistakes

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 5 Mentioning A Signature Event Does Not Make A Scrimmage A Signature

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	event classification
Fixture	a scrimmage title containing the phrase and an official Signature field
Expected result	use official classification rather than substring coincidence
Observed result	Passed in 0.3512 milliseconds during the September 16 2026 run
Engineering significance	Protects event labels from incidental text

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 6 High School Names Do Not Imply School Based Tournament Format

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	event format classification
Fixture	High School Tournament and School-Based Tournament
Expected result	distinguish grade words from the official format phrase
Observed result	Passed in 0.3001 milliseconds during the September 16 2026 run
Engineering significance	Avoids changing filters based on a team grade phrase

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 7 All Pages Are Retained And Elimination Round Filters Survive Pagination

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	complete collection loading
Fixture	four mocked pages with repeated round array query parameters
Expected result	return all four pages while preserving every original filter
Observed result	Passed in 79.8972 milliseconds during the September 16 2026 run
Engineering significance	Protects large events and elimination evidence from silent truncation

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 8 Failed Later Pages Reject Rather Than Publish A Partial Collection

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	fail closed collection behavior
Fixture	page one succeeds and page two returns HTTP 403
Expected result	reject the whole collection
Observed result	Passed in 2.5680 milliseconds during the September 16 2026 run
Engineering significance	Prevents partial results from being labeled complete

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 9 Table Agenda Keeps Activity Start End And Location Together

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	agenda table parsing
Fixture	one organizer table row with day, two times, activity, and location
Expected result	produce one entry with the correct time range, activity, and location
Observed result	Passed in 2.0999 milliseconds during the September 16 2026 run
Engineering significance	Preserves the meaning of organizer schedules

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 10 Single Day Agendas Without Day Headings Still Retain Timed Entries

Field	Recorded value
Source	website/tests/data-regressions.test.mjs
Focus	agenda text parsing
Fixture	two timed lines and one untimed instruction
Expected result	create one day with all three entries
Observed result	Passed in 92.3847 milliseconds during the September 16 2026 run
Engineering significance	Supports common organizer text that lacks explicit day headings

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 11 Numeric Family Search Includes Every Suffix Including Unregistered Teams

Field	Recorded value
Source	website/tests/team-directory.test.mjs
Focus	team discovery
Fixture	2011A, 2011B, 2011C with mixed registration state
Expected result	return all suffixes for 2011 and exact normalized match for 2011B
Observed result	Passed in 4.0437 milliseconds during the September 16 2026 run
Engineering significance	Keeps historical and inactive teams discoverable

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 12 Names And Organizations Support Partial Case Insensitive Search

Field	Recorded value
Source	website/tests/team-directory.test.mjs
Focus	team discovery
Fixture	mixed-case organization query and partial team name
Expected result	match without case sensitivity
Observed result	Passed in 0.4119 milliseconds during the September 16 2026 run
Engineering significance	Makes the directory useful when users do not know an exact number

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 13 All Three Filters Combine And Countries Normalize

Field	Recorded value
Source	website/tests/team-directory.test.mjs
Focus	directory filters
Fixture	country alias, region, grade, and numeric query
Expected result	combine every filter and normalize USA to United States
Observed result	Passed in 1.1565 milliseconds during the September 16 2026 run
Engineering significance	Prevents one filter from silently replacing another

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 14 Locations Include Inactive Only Countries And Dependent Regions

Field	Recorded value
Source	website/tests/team-directory.test.mjs
Focus	directory filter values
Fixture	Canada and Iceland records that include inactive teams
Expected result	list their countries and regions
Observed result	Passed in 2.2824 milliseconds during the September 16 2026 run
Engineering significance	Supports historical browsing instead of only current registration

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 15 Single Active Robot Can Receive All Alliance Credit

Field	Recorded value
Source	website/tests/vcr3.test.mjs
Focus	partner share validation
Fixture	one robot with share one and a large upset
Expected result	produce finite positive evidence and settlement
Observed result	Passed in 2.8706 milliseconds during the September 16 2026 run
Engineering significance	Supports legal single-active-team evidence without invalid share assumptions

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 16 Complementary Shares Tolerate Ordinary Floating Point Rounding

Field	Recorded value
Source	website/tests/vcr3.test.mjs
Focus	numeric validation
Fixture	one minus 0.8 and an invalid share above one
Expected result	accept the rounded complement and reject 1.01
Observed result	Passed in 1.3181 milliseconds during the September 16 2026 run
Engineering significance	Avoids false failures while preserving parameter bounds

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 17 Unexpected Champions Gain At Every Tier

Field	Recorded value
Source	website/tests/vcr3.test.mjs
Focus	achievement policy
Fixture	a weak overall event record with verified championship across every tier
Expected result	produce a positive final delta for every tier
Observed result	Passed in 3.1027 milliseconds during the September 16 2026 run
Engineering significance	Enforces the disclosed champion minimum

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 18 Legitimate Non Champion Loss Remains Negative

Field	Recorded value
Source	website/tests/vcr3.test.mjs
Focus	achievement policy boundary
Fixture	the same weak event without champion status
Expected result	retain a negative event delta
Observed result	Passed in 0.4941 milliseconds during the September 16 2026 run
Engineering significance	Shows that the champion rule does not erase ordinary downside

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Test Case 19 Completed Event Processor Applies Champion Guarantee And Produces Versioned Audit

Field	Recorded value
Source	website/tests/vcr3.test.mjs
Focus	integrated event settlement
Fixture	one qualification loss followed by a final win for four teams
Expected result	increase the champions above 1500 and record version, champion state, and exact elimination component
Observed result	Passed in 28.0056 milliseconds during the September 16 2026 run
Engineering significance	Connects match parsing, champion inference, settlement, state mutation, and audit history

Failure interpretation

A future failure would mean that behavior protected by this fixture changed or a dependency changed. The contributor should first reproduce the isolated fixture, then inspect the named helper, and only update the expected result if the design requirement intentionally changed.

Additional coverage recommended

Add a fixture from a second organizer or event shape, include the boundary immediately outside the accepted range, and verify that the public response explains the error rather than returning a valid-looking partial value.

Automated Test Run Summary

The complete local fixture command ran all four test files together. Node reported 19 tests, 19 passes, zero failures, zero cancellations, zero skips, and zero todo tests in 2,103.2708 milliseconds. Individual test times show that agenda parsing and paginated collection tests were the slowest fixtures in this run, while most pure classification and rating checks completed in a few milliseconds.

Test file	Tests	Primary behavior
archive.test.mjs	2	Historical match eligibility and archive identity
data-regressions.test.mjs	8	Brackets, classification, pagination, and agendas
team-directory.test.mjs	4	Search, normalization, and location filters
vcr3.test.mjs	5	Partner shares, champion policy, and integrated settlement
Total	19	All passed

Static Frontend Build Evidence

The Pages-targeted Vite build completed successfully on September 16 2026. It transformed 2,362 modules and produced one HTML entry, one CSS asset, and one JavaScript asset. The build warning identified the JavaScript chunk as larger than 500 kB after minification, which is a performance concern rather than a build failure.

Artifact	Raw size	Gzip size
dist-pages/index.html	0.48 kB	0.33 kB
dist-pages/assets/index-CNoIUUYZ.css	217.35 kB	32.65 kB
dist-pages/assets/index-BLg3QZ9x.js	655.54 kB	195.89 kB

Recommended next build trial

Split rarely used views and large chart or UI modules with dynamic imports, compare the resulting chunks, and verify that navigation, loading, and GitHub Pages base paths still work.

Lint and Type Safety Baseline

The lint command failed on the recorded source snapshot. This is important test data and is not hidden by the passing fixture suite. Reported categories include explicit any types, unused expressions and variables, array construction and sorting rules, React hook dependencies, synchronous state updates in effects, accessibility roles and keyboard handling, floating promises, template-expression typing, possible undefined values, and archive-script type restrictions.

Category	Representative locations	Risk
Type precision	API routes, page.tsx, Worker adapters	Unknown payload shapes can escape compile-time checks
Accessibility	UI primitives, pagination, interactive elements	Keyboard and screen-reader behavior may be incomplete
React effects and hooks	team-directory.tsx and page.tsx	Cascading renders or stale dependencies
Maintainability	unused code, new Array, sort without comparator	Intent is harder to audit
Null and undefined handling	Firebase adapter and directory code	Runtime failure when data is absent

The backlog should be reduced in behavior-preserving batches with the fixture suite and Pages build run after each batch. A clean lint baseline should become a release gate before a stable VCR release.

Browser and Live Integration Evidence

Check	Observed result	Interpretation
Worker health	D1 connectivity returned ok	Worker and database binding were live
Current calendar	707 events	Paginated current-event route returned a substantial set
Current ranking	581 teams	Live sampled ranking route completed
Push Back skills	9,980 team entries	Large collection route completed
Team 3150D season 197	16 events and 12 qualification rows	Season-specific team path stayed connected
Worlds event 64026	585 teams and all eight divisions	Worlds pagination and division coverage were present
Repeated team request	about 80 milliseconds in one check	D1 cache reduced repeat latency
Event 64025 inspection	11 divisions and 865 teams	Website inspection caught incomplete-division risk

These are dated observations, not performance guarantees. A repeatable monitoring and load-test plan should record cold and warm latency distributions, response sizes, status codes, and cache headers over time.

Testing Limitations and Remaining Evidence

- The fixture suite protects known behaviors but does not enumerate every historical event format.
- The successful static build does not prove that the live API or archive host is available.
- Browser inspection does not replace a structured accessibility audit or participant usability study.
- The live integration counts are point-in-time observations and may change as official data changes.
- The published archive proves complete loading of the fetched calendar, not that the upstream calendar contains every real event.
- Scenario tests prove VCR rules execute as written; they do not establish calibration or predictive advantage.
- The lint baseline is not clean and therefore maintainability and accessibility risks remain open.

Part Seven Historical Replay and Data Analysis

This part documents the complete Push Back archive build and summarizes measured distributions without presenting them as predictive validation.

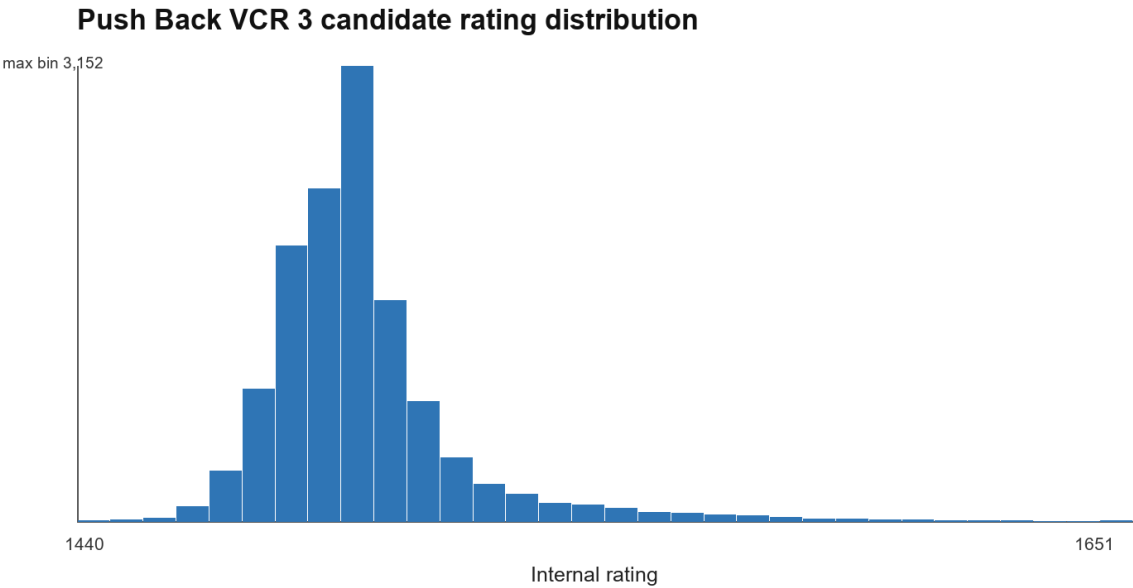
The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Published Push Back Archive Coverage

Metric	Recorded value
Season	2025–26 Push Back
Season ID	197
Model	VCR-3.0-candidate.2
Archive teams	12,573
Rated events	1,548
Eligible matches	110,150
Expected calendar payloads	1,681
Loaded calendar payloads	1,681
Recorded failures	0
Archive size	13.08 MiB
As of	2026-06-01

Rated events are fewer than loaded calendar payloads because some completed event records contain no eligible scored competition matches. The builder retains the distinction instead of inflating the rated-event count.

Rating Distribution

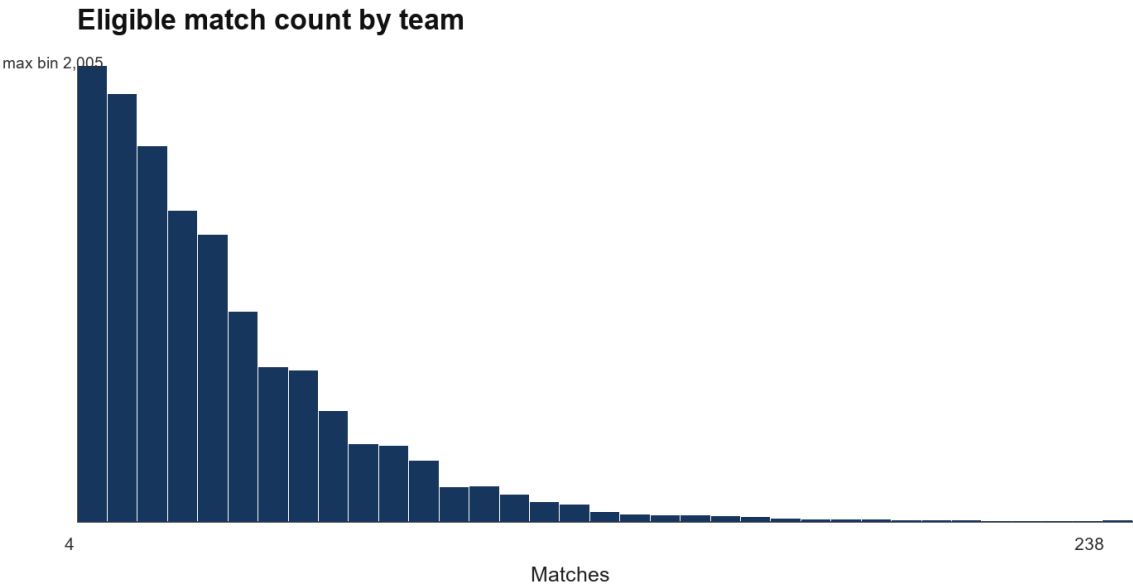


Measured rating distribution in the published Push Back archive

Statistic	Value
Minimum	1440.00
10th percentile	1479.00
25th percentile	1486.00
Median	1494.00
Mean	1496.51
75th percentile	1501.00
90th percentile	1515.00
Maximum	1651.00
Population standard deviation	18.36

The distribution is concentrated near the 1500 prior, with a long upper tail. This describes the candidate archive output and does not demonstrate that the scale is calibrated.

Eligible Matches per Team

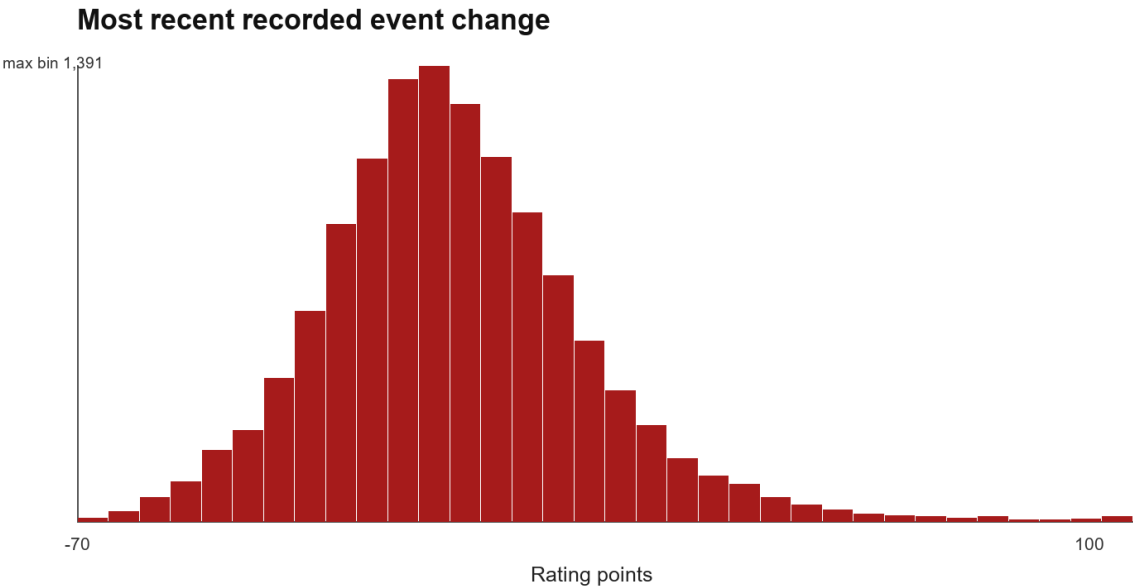


Measured eligible matches per team in the published Push Back archive

Statistic	Value
Minimum	4.00
10th percentile	8.00
25th percentile	15.00
Median	28.00
Mean	35.03
75th percentile	48.00
90th percentile	72.00
Maximum	238.00
Population standard deviation	27.14

Match counts vary substantially, so evidence volume and uncertainty should be interpreted together. More matches do not automatically mean a higher rating.

Most Recent Event Change



Measured most recent event change in the published Push Back archive

Statistic	Value
Minimum	-70.00
10th percentile	-34.00
25th percentile	-23.00
Median	-12.00
Mean	-9.87
75th percentile	2.00
90th percentile	16.00
Maximum	100.00
Population standard deviation	21.08

Negative recent changes outnumber positive changes in this archive. The pattern can reflect mean reversion, event weighting, survivor selection, or model behavior and requires backtesting before interpretation.

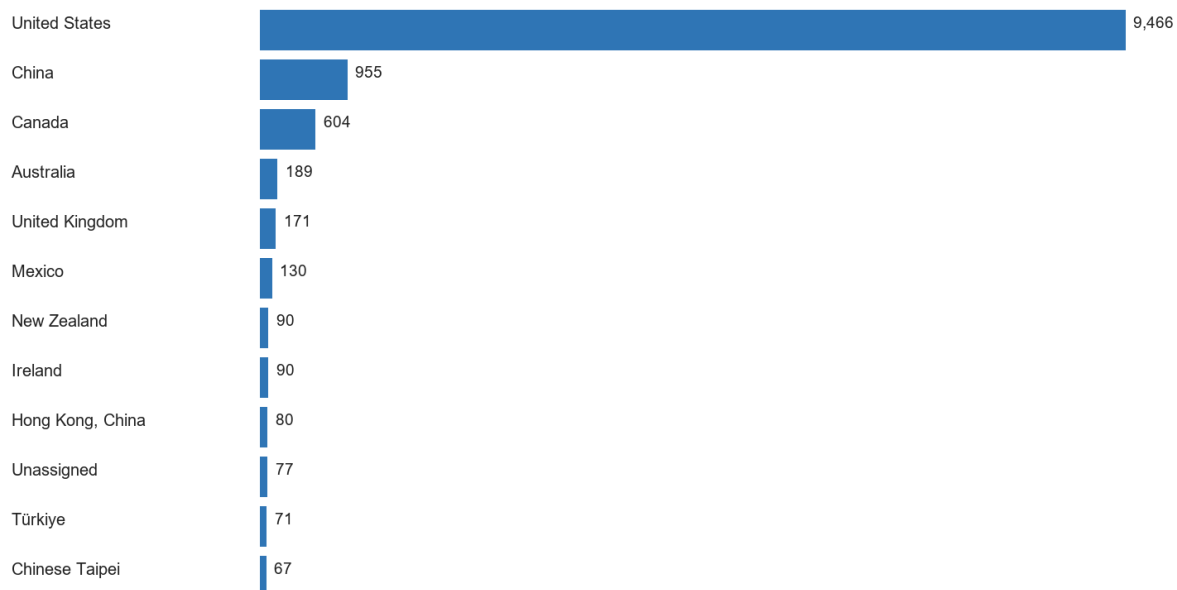
Event Count and Confidence Distributions

Statistic	Events per team	Confidence or RD field
Minimum	1.00	25.00
10th percentile	1.00	28.00
25th percentile	2.00	35.00
Median	3.00	45.00
Mean	3.97	50.84
75th percentile	5.00	62.00
90th percentile	8.00	85.00
Maximum	23.00	120.00
Population standard deviation	2.72	20.60

The archive field named confidence behaves as an uncertainty quantity in which lower values accompany mature high-evidence teams. The interface should name and explain the unit consistently so users do not interpret a lower number as lower confidence without context.

Country and Grade Coverage

Teams in the archive by country



Teams in the published Push Back archive by country

Grade	Teams
High School	8,520
Middle School	3,976
Unknown	77

Country and region counts describe archive participation and data availability. They must not become permanent rating bonuses or penalties.

Top Country Counts

Country	Teams
United States	9,466
China	955
Canada	604
Australia	189
United Kingdom	171
Mexico	130
New Zealand	90
Ireland	90
Hong Kong, China	80
Unassigned	77
Türkiye	71
Chinese Taipei	67
Vietnam	55
Korea, Republic of	54
Singapore	50
Thailand	43
Macau, China	28
Switzerland	27
Japan	26
Saudi Arabia	24

Top Region Counts

Region	Teams
California, United States	1,124
Texas, United States	1,060
Ohio, United States	495
Florida, United States	354
Indiana, United States	335
Pennsylvania, United States	317
Michigan, United States	306
Tennessee, United States	284
Colorado, United States	277
Virginia, United States	258
New York, United States	254
South Carolina, United States	238
Massachusetts, United States	236
Ontario, Canada	234
British Columbia, Canada	224
Minnesota, United States	224
Illinois, United States	218
Guangdong, China	197
Alabama, United States	195
Washington, United States	189

Displayed Event Tier Evidence

Tier in recent form entries	Entries
B	30,028
Bronze S	7,709
Silver S	2,006
Worlds	1,447

The recent-form records contain 2,081 champion-marked entries. These counts come from the displayed form histories embedded in the archive, not a complete independent event-award table.

Published Ranking Sample Rows 1 Through 10

Rank	Team	Rating	RD	Change	Record	Events	Region
1	1028A	1651	25	70	161–12–0	13	Nebraska, United States
2	471B	1630	25	38	162–19–1	14	California, United States
3	62880A	1626	25	32	116–22–0	12	New York, United States
4	10012Y	1623	25	67	126–23–2	14	British Columbia, Canada
5	2899A	1622	25	55	117–32–3	13	California, United States
6	10B	1620	25	34	128–20–1	11	Washington, United States
7	10C	1619	25	68	131–43–0	15	Washington, United States
8	334C	1617	25	82	143–47–2	16	California, United States
9	8838D	1615	25	75	96–24–0	11	California, United States
10	8682A	1615	25	53	126–17–0	13	Georgia, United States

These rows document the published artifact at the recorded date. They are examples for archive verification, not a claim that ordinal order is stable across future model versions.

Published Ranking Sample Rows 11 Through 20

Rank	Team	Rating	RD	Change	Record	Events	Region
11	79912C	1615	25	-33	141–16–0	16	Texas, United States
12	6210H	1612	25	37	137–29–0	16	Indiana, United States
13	2054V	1611	25	28	128–22–2	12	Utah, United States
14	15442A	1610	25	61	111–27–0	13	California, United States
15	58676R	1611	27	94	71–9–0	6	Quebec, Canada
16	9568W	1608	25	100	80–41–0	12	British Columbia, Canada
17	7760X	1607	25	67	81–9–0	7	Michigan, United States
18	7405R	1606	25	31	110–25–2	12	New Jersey, United States
19	54010A	1604	25	35	99–28–1	12	Pennsylvania, United States
20	2137A	1603	25	35	128–31–0	14	California, United States

These rows document the published artifact at the recorded date. They are examples for archive verification, not a claim that ordinal order is stable across future model versions.

Published Ranking Sample Rows 21 Through 30

Rank	Team	Rating	RD	Change	Record	Events	Region
21	4863A	1603	25	60	110–26–0	12	California, United States
22	9568Z	1602	25	48	140–25–0	14	British Columbia, Canada
23	2577X	1601	25	25	120–22–0	13	Alabama, United States
24	10085A	1600	25	100	94–18–0	9	Shanghai, China
25	1412E	1599	25	7	97–47–1	15	Texas, United States
26	2145Z	1598	25	15	182–50–0	19	Virginia, United States
27	9364E	1598	25	32	134–50–0	15	Tennessee, United States
28	5069Y	1598	25	15	145–22–0	14	Nebraska, United States
29	80001X	1596	25	92	69–42–0	12	California, United States
30	70407R	1595	25	79	73–24–1	9	New York, United States

These rows document the published artifact at the recorded date. They are examples for archive verification, not a claim that ordinal order is stable across future model versions.

Published Ranking Sample Rows 31 Through 40

Rank	Team	Rating	RD	Change	Record	Events	Region
31	2011B	1595	25	29	112–25–0	12	Ohio, United States
32	1412J	1595	25	13	134–30–0	15	Texas, United States
33	462V	1594	25	59	77–28–0	11	California, United States
34	91233A	1594	25	54	73–17–0	8	California, United States
35	74074Y	1594	25	-13	112–18–1	12	Oklahoma, United States
36	8110W	1593	25	39	90–21–1	9	Minnesota, United States
37	11101X	1593	25	48	103–25–0	12	California, United States
38	2145Y	1592	25	-24	154–45–0	19	Virginia, United States
39	1468A	1591	25	100	75–33–0	11	California, United States
40	6842K	1590	25	26	124–30–0	13	Indiana, United States

These rows document the published artifact at the recorded date. They are examples for archive verification, not a claim that ordinal order is stable across future model versions.

Published Ranking Sample Rows 41 Through 50

Rank	Team	Rating	RD	Change	Record	Events	Region
41	6374B	1589	25	48	72-25-0	9	California, United States
42	11101K	1589	25	79	114-35-1	14	California, United States
43	66666A	1593	29	88	61-9-0	5	Anhui, China
44	1727K	1588	26	45	68-19-0	8	Maryland, United States
45	1082M	1587	25	76	84-19-1	9	Texas, United States
46	72202Z	1587	25	-1	95-13-0	11	Arkansas, United States
47	8838C	1587	25	49	76-25-1	10	California, United States
48	57999D	1586	25	84	58-31-0	8	Ontario, Canada
49	12914X	1586	25	-1	178-54-1	23	Virginia, United States
50	6210M	1586	25	0	146-32-0	16	Indiana, United States

These rows document the published artifact at the recorded date. They are examples for archive verification, not a claim that ordinal order is stable across future model versions.

Archive Rebuild Trial and Rate Limit Failure

The first multi-season rebuild reached the upstream rate limit during High Stakes. Restarting the entire job would repeat completed downloads and increase load. The failure changed the builder rather than being treated as an operator inconvenience.

Observed failure	Design response	Verification
HTTP 429 during later season	Space requests by 1.8 seconds and stop on explicit rate limit	Worker and builder report rate-limit state
Repeated completed pages	Persist successful page responses by hashed request	Second run reuses complete cache entries
Two builders could overlap	Use a reservation lock	Only one reservation owns the request slot
Old cache format exists	Reuse complete legacy entries	Migration does not discard valid progress
Incomplete season could publish	Require coverage report and atomic final replacement	Archive validation rejects failures or missing events

Part Eight Test Publishing with GitHub Firebase and Cloudflare

This part records the publishing alternatives, commands, evidence, blockers, security decisions, and rollback boundaries for the test website.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

GitHub Branch and Pull Request Process

Focused branches and pull requests record the problem, changed behavior, checks performed, and remaining limitations. The repository is the change ledger for code and documentation.

A model change should not be merged as a comment-only cleanup. It requires tests, version notes, and archive consequences.

Record field	Notebook entry
Evidence	Recorded pull requests 2, 4, 5, 6, 7, and 8; CONTRIBUTING.md.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The local snapshot used for this edition does not contain a complete Git commit history.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

GitHub Pages Build Workflow

The test frontend is published from a manually dispatched workflow. The workflow installs pinned dependencies, runs regression tests, checks API health, builds with the /Vex-Rank/ base path, uploads the Pages artifact, and deploys only after the build job succeeds.

The workflow contains no Event VEX secret because the frontend calls the public test Worker.

Record field	Notebook entry
Evidence	website/.github/workflows/github-pages-test.yml.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	A successful Pages deployment still depends on the external API and archive origins.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Firestore Prototype Architecture

The Firestore trial wrapped application routes in a second-generation HTTPS function and proposed Firestore documents for compressed response caching. Minimum instances were zero and maximum instances one to constrain cost.

Firestore rules and indexes were deployed, but the API token was intended to enter through a managed secret, not source code or browser configuration.

Record field	Notebook entry
Evidence	website/firebase deployment files and guide.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Cloud Functions require the Blaze plan; the secure function path was not deployed under the free-plan constraint.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Firestore Trial Decision

The Firestore blocker was discovered before uploading the Event VEX token. Secret Manager and Cloud Functions deployment requirements conflicted with the test project's free-plan constraint.

The team kept the Firestore reference design and moved the test API to Cloudflare Workers and D1.

Record field	Notebook entry
Evidence	Firestore deployment status notes and official Firestore quota documentation.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Firestore remains a documented alternative rather than an active backend.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Cloudflare Worker Architecture

The Worker imports the same route handlers used by the application, allowlists public requests, injects the secret on the server, adds CORS and security headers, and stores modest successful responses in D1.

Historical archives bypass the live sampled ranking route and are served as versioned assets.

Record field	Notebook entry
Evidence	worker.ts, archive-source.ts, schema.sql, wrangler.json.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The Worker is a test deployment and free-tier limits remain operational constraints.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Cloudflare D1 Cache Policy

D1 provides shared cache state across Worker invocations. Cache keys include only allowlisted response-affecting parameters and add a model namespace for rating routes.

Responses larger than about 1.8 MB bypass D1 because the platform's documented maximum string or row size is 2 MB.

Record field	Notebook entry
Evidence	Worker implementation and official D1 limits.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	The 15-minute TTL and threshold need load testing with real traffic.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Secrets and Credential Boundary

The Event VEX API token is stored as a Worker secret or managed server secret. It is never embedded in the static frontend, archive, workflow, or repository.

Only server routes can read the secret. Public clients receive derived public responses.

Record field	Notebook entry
Evidence	Worker Env binding; Firebase defineSecret design; bundle inspection requirement.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Secret rotation and access review should be added to an operational checklist.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

CORS Route and Query Validation

The Worker accepts GET and OPTIONS from the GitHub Pages origin. Route patterns and numeric query parameters are validated before a cache key or upstream request is created.

Unknown routes, modes, and query formats return an explicit error instead of reaching route handlers.

Record field	Notebook entry
Evidence	canonicalUrl and cors helpers.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	CORS does not authenticate non-browser clients and must not be treated as an authorization system.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

Deployment Verification and Rollback

Publishing should stop if tests, API health, archive validation, or the frontend build fail. A model change also requires cache-version review and archive rebuild decisions.

Rollback means redeploying the prior static artifact and Worker version while retaining the last complete versioned archive.

Record field	Notebook entry
Evidence	Workflow dependencies, archive atomic replacement, deployment guides.
Decision status	Implemented, proposed, or retained as described in this entry
Known limitation	Automated rollback and monitoring are not yet implemented.
Contributor consequence	Preserve the stated boundary or update the implementation, tests, and documentation together.

Verification questions

- Where is the credential stored
- What check prevents partial publication
- What exact artifact can be rolled back

GitHub Pages Workflow Evidence

Annotated publishing sequence

```

9  jobs:
10   build:
11     runs-on: ubuntu-latest
12     steps:
13       - uses: actions/checkout@v4
14       - uses: pnpm/action-setup@v4
15         with:
16           version: 11.19.0
17       - uses: actions/setup-node@v4
18         with:
19           node-version: 24
20           cache: pnpm
21           cache-dependency-path: website/pnpm-lock.yaml
22       - run: pnpm install --frozen-lockfile
23         working-directory: website
24       - name: Check data regressions
25         run: node --test tests/data-regressions.test.mjs tests/team-directory.test.mjs
26         working-directory: website
27       - name: Verify test API is deployed
28         run: curl --fail --retry 2 https://vexrank-api-test.vexrank-eason.workers.dev/api/health
29       - name: Build test website
30         run: pnpm exec vite build --config vite.pages.config.ts
31         working-directory: website
32       - uses: actions/upload-pages-artifact@v3
33         with:
34           path: website/dist-pages
35   deploy:
36     needs: build
37     runs-on: ubuntu-latest
38     permissions:
39       pages: write
40       id-token: write
41     environment:
42       name: github-pages
43       url: ${ steps.deployment.outputs.page_url }
44     steps:
45       - name: Deploy test website
46         id: deployment
47         uses: actions/deploy-pages@v4

```

The important dependency is that deployment needs the successful build job. The build job runs tests and a remote API health request before creating and uploading the static artifact.

Firestore Trial Commands and Outcome

Reference deployment sequence

```
# Deploy rules and indexes first so the cache schema is controlled.
firebase deploy --only firestore:rules,firestore:indexes

# The API function needs the Event VEX credential as a managed secret.
firebase functions:secrets:set ROBOT_EVENTS_API_TOKEN

# Deploy only after the secret exists and the project meets the plan requirement.
firebase deploy --only functions
```

Observed result: Firestore rules and indexes deployed, but the function path stopped at the plan requirement for Cloud Functions and Secret Manager. The token was not uploaded and no insecure fallback was introduced.

Cloudflare Deployment Checklist

1. Create or select the Worker and D1 database described by wrangler.json.
2. Apply cloudflare/schema.sql to create the API cache table.
3. Bind D1 and the archive asset service to the Worker.
4. Store ROBOT_EVENTS_API_TOKEN as an encrypted Worker secret.
5. Deploy the Worker and check /api/health before connecting the frontend.
6. Verify representative event, team, ranking, skills, archive manifest, and historical archive requests.
7. Run the GitHub Pages workflow with the verified Worker URL.
8. Inspect browser loading, error, season, and narrow-screen states.
9. Record response sizes, cache headers, model version, and deployment timestamp.
10. Retain the prior Worker version and last complete archive for rollback.

External Platform References Applied to the Design

Platform rule	Design application
GitHub Pages custom workflows upload and deploy an artifact	The repository uses separate build and deploy jobs with a Pages artifact
Firebase secret parameters bind Secret Manager values to selected functions	The trial defined a managed token rather than a checked-in environment value
Cloud Functions for Firebase require the Blaze plan	The Firebase function trial stopped instead of expanding cost scope
Cloudflare Worker secrets expose encrypted values through env	The Worker receives ROBOT_EVENTS_API_TOKEN only at runtime
D1 maximum string BLOB or row size is 2,000,000 bytes	The Worker caches only bodies below 1,800,000 bytes

Part Nine Chronological Engineering Record

This part converts the design and repository evidence into dated trial entries with problem, action, observation, decision, and follow-up fields.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Notebook Entry 1 Architecture and VCR 1.0 concept

Field	Entry
Date	August 2026
Design stage	Architecture and VCR 1.0 concept
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Defined data sources, rating branches, website boundaries, validation laboratory, and open mathematical questions.
Observation	Defined data sources, rating branches, website boundaries, validation laboratory, and open mathematical questions.
Decision	Use the Lucidchart as an architecture authority and record unresolved formulas before implementation.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 2 VCR 2.0 detailed specification

Field	Entry
Date	September 3 2026
Design stage	VCR 2.0 detailed specification
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Expanded the model into public score, uncertainty, dynamic partner shares, four-component blend, event strength, and backtesting requirements.
Observation	Expanded the model into public score, uncertainty, dynamic partner shares, four-component blend, event strength, and backtesting requirements.
Decision	Treat the specification as a launch design and verify that executable routes use the same model.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 3 Integrated website prototype

Field	Entry
Date	September 9 2026
Design stage	Integrated website prototype
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Combined React views, API routes, shared helpers, project documentation, and generated algorithm PDFs.
Observation	Combined React views, API routes, shared helpers, project documentation, and generated algorithm PDFs.
Decision	Use a shared interface and begin regression testing around real event shapes.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 4 Worlds and data reliability trials

Field	Entry
Date	September 11 to 12 2026
Design stage	Worlds and data reliability trials
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Found incomplete divisions, misplaced bracket replays, agenda flattening, false event classifications, and empty-looking failures.
Observation	Found incomplete divisions, misplaced bracket replays, agenda flattening, false event classifications, and empty-looking failures.
Decision	Implement complete pagination, slot-preserving brackets, structured agendas, explicit classification, and visible errors.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 5 Firebase deployment trial

Field	Entry
Date	September 14 2026
Design stage	Firestore configuration succeeded but secure function deployment required Blaze.
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Firestore configuration succeeded but secure function deployment required Blaze.
Observation	Firestore configuration succeeded but secure function deployment required Blaze.
Decision	Do not upload the token; retain the reference design and compare another serverless backend.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 6 GitHub Pages and Cloudflare test stack

Field	Entry
Date	September 14 to 15 2026
Design stage	GitHub Pages and Cloudflare test stack
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Separated the static frontend from the Worker API, added D1 cache identity, secret binding, health checks, and Pages workflow.
Observation	Separated the static frontend from the Worker API, added D1 cache identity, secret binding, health checks, and Pages workflow.
Decision	Select Cloudflare for the test backend under the recorded cost constraint.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 7 VCR 3 candidate settlement

Field	Entry
Date	September 15 2026
Design stage	VCR 3 candidate settlement
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Centralized event processing, bounded components, achievement correction, and an audit ledger; integrated the candidate across ranking and history paths.
Observation	Centralized event processing, bounded components, achievement correction, and an audit ledger; integrated the candidate across ranking and history paths.
Decision	Keep candidate status until chronological calibration and multi-division champion handling are complete.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 8 Complete-season archive

Field	Entry
Date	September 15 to 16 2026
Design stage	Complete-season archive
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Replayed Push Back across 1,681 loaded event payloads, 1,548 rated events, and 110,150 eligible matches.
Observation	Replayed Push Back across 1,681 loaded event payloads, 1,548 rated events, and 110,150 eligible matches.
Decision	Publish only versioned archives with complete coverage and zero recorded failures.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 9 Rate-limited rebuild redesign

Field	Entry
Date	September 16 2026
Design stage	Rate-limited rebuild redesign
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	High Stakes returned HTTP 429 during multi-season rebuilding.
Observation	High Stakes returned HTTP 429 during multi-season rebuilding.
Decision	Persist successful pages, reserve request slots, resume cached progress, and stop before replacing an incomplete archive.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 10 Contributor documentation

Field	Entry
Date	September 16 2026
Design stage	Contributor documentation
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Added source ownership, comments around unusual behavior, test commands, limits, and non-AI contribution guidance.
Observation	Added source ownership, comments around unusual behavior, test commands, limits, and non-AI contribution guidance.
Decision	Treat documentation and regression fixtures as part of the implementation.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Notebook Entry 11 Expanded engineering notebook

Field	Entry
Date	September 16 2026
Design stage	Expanded engineering notebook
Problem or question	Determine whether the current design satisfied the project's correctness, usability, security, reliability, cost, and maintainability criteria.
Action	Added detailed origin, contents, test evidence, archive analysis, deployment references, code listings, appendices, glossary, and source record.
Observation	Added detailed origin, contents, test evidence, archive analysis, deployment references, code listings, appendices, glossary, and source record.
Decision	Use this edition as the contributor and review baseline; update it with new dated trials rather than rewriting history.
Evidence retained	Source files, tests, inspection notes, deployment guides, model documents, archive metadata, or generated artifacts named in the relevant notebook part.
Next trial	Reproduce the result, add a fixture or measured dataset, and record any changed requirement before implementation.

Part Ten Final Evaluation and Next Engineering Work

This part evaluates the project against its criteria and turns unresolved limits into testable next trials.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Evaluation Against the Design Criteria

Criterion	Current evidence	Remaining work
Correctness	Complete pagination fixtures, fail-closed collections, bracket and agenda regressions	More organizer formats and end-to-end historical comparisons
Reproducibility	Versioned model, archive manifest, coverage envelope, chronological replay	Immutable input snapshots and full release automation
Usability	Search, responsive states, visible loading and errors, inspection record	Participant study and formal accessibility audit
Security	Server-side secret, route allowlist, no browser token	Rotation procedure and dependency review
Reliability	Retries, timeouts, process cache, D1 cache, resumable archive download	Monitoring and controlled load tests
Cost	Pages and Worker test stack selected after Firebase plan blocker	Measure free-tier headroom and define escalation budget
Maintainability	Shared VCR core, comments, contributor guide, 19 tests	Clean lint and type baseline; split large UI module
Predictive validity	Documented metrics and complete Push Back replay	Calibrated baseline comparison and held-out season

Recommended Trial 1 Build the Frozen Schedule and Bracket Simulator

Persist an event-specific random seed, qualification schedule, selection assumptions, bracket shape, and 10,000 simulated outcomes. Compare simulated qualification percentile, elimination utility, and title probability with the current field proxy.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 2 Backtest VCR Two and VCR Three Against Baselines

Replay seasons chronologically and compare log loss, Brier score, calibration, rank stability, and later-event performance against Elo, win rate, and OPR-only baselines.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 3 Resolve Multi Division Champions

Model division champions and an overall championship bracket as separate scopes so a division final cannot be mistaken for the overall title or counted twice.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 4 Complete Older Season Archives

Resume one season at a time, inspect coverage reports, rebuild with the current model, and publish only after expected equals loaded and failures are empty.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 5 Clean the Type and Lint Baseline

Replace unsafe any types with validated unknown inputs, resolve hook dependencies, fix accessibility roles and keyboard behavior, and remove unused or ambiguous constructs without changing outputs.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 6 Refactor the Main React Page

Move view-specific state, adapters, and presentation into tested modules. Compare the Pages bundle and interaction behavior before and after the refactor.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 7 Run Accessibility and Usability Trials

Measure keyboard completion, screen-reader labels, table semantics, chart alternatives, contrast, mobile reflow, and participant task success.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 8 Load Test Worker and D1

Record cold and warm latency percentiles, response sizes, cache hit rate, CPU time, subrequests, 429 and 502 behavior, and archive delivery limits.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 9 Add Monitoring and Release Records

Publish model, Worker, archive, and frontend identifiers with health status. Define alert thresholds and rollback steps.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Recommended Trial 10 Establish Stable VCR Release Governance

Require a calibration report, held-out result, version note, complete archive rebuild, clean tests, and reviewer approval before replacing the candidate label.

Planning field	Required record
Hypothesis	State the measurable improvement expected before implementation.
Inputs	Freeze source data, version, configuration, and random seed where applicable.
Baseline	Use the current candidate and at least one simpler comparison.
Metrics	Choose correctness, prediction, latency, usability, accessibility, or cost measures that match the trial.
Acceptance	Define the threshold and failure response before observing the result.
Artifacts	Retain code, fixtures, logs, datasets, charts, decision, and changed documentation.

Appendix A Annotated Code Listings

These listings reproduce focused implementation regions with line numbers and explain the behavior a contributor must preserve.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Code Listing A1 Browser Origin Selection

The helper distinguishes API and archive requests from ordinary Pages assets. Comments explain why a static test build needs two origins and how the /Vex-Rank/ base path is preserved.

website/lib/client-fetch.ts

```

3
4  /** Separate API and asset origins for GitHub Pages; keep same-origin hosting working. */
5  export function siteFetch(path:string, init?:RequestInit) {
6    // Build-time constants: the API can live on a Worker while assets live beneath
7    // a GitHub Pages repository path. Undefined constants support same-origin dev.
8    const apiBase=typeof __VEX_API_BASE__==='undefined'?':__VEX_API_BASE__';
9    const assetBase=typeof __VEX_ASSET_BASE__==='undefined'?':__VEX_ASSET_BASE__';
10   // Versioned archives use the API origin when configured because the Worker
11   // serves its archive assets there; ordinary static files use the Pages base.
12   const archive=/^\s\/rankings-20\d{2}-\d{2}-vcr3\.json$/.test(path);
13   const
url=path.startsWith('/api/')||archive&&apiBase?`${apiBase}${path}`:path.startsWith('/')?`${assetBase}${path.slice(1)}`:path;
14   return fetch(url,init);
15 }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A2 Bounded Concurrent Mapping

The shared cursor is claimed before await, so workers cannot duplicate an index and results remain in input order.

website/lib/vex-api.ts

```

3  const pages = new Map<string, { expires:number; data:any }>();
4  /** Read complete official collections. A failed page must never masquerade as no results. */
5  export async function mapLimit<T, R>(items: T[], limit: number, mapper: (item: T) => Promise<R>): Promise<R[]> {
6    const results = new Array<R>(items.length);
7    // Claim each index before awaiting: workers share the cursor but preserve input order.
8    let cursor = 0;
9    await Promise.all(Array.from({ length: Math.min(limit, items.length) }, async () => {
10     while (cursor < items.length) {
11       const index = cursor++;
12       results[index] = await mapper(items[index]);
13     }
14   }));
15   return results;
16 }
17
18 // Retry timeouts, network errors, 429 and 5xx; other HTTP failures are terminal.
19 // Cache successes for five minutes and cap entries to bound long-lived worker memory.
20 export async function vexJson(url: string, headers: Record<string, string>): Promise<any> {
21   const cached=pages.get(url);
22   if(cached&&cached.expires>Date.now())return cached.data;
23   let lastStatus = 0;
24   for (let attempt = 0; attempt < 3; attempt++) {
25     try {
26       const response = await fetch(url, { headers, signal: AbortSignal.timeout(15000) });
27       if (response.ok) {

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A3 Upstream Retry and Success Cache

Only successful public pages enter the five-minute process cache. Timeouts, 429, and server errors retry; terminal client errors do not.

website/lib/vex-api.ts

```

18 // Retry timeouts, network errors, 429 and 5xx; other HTTP failures are terminal.
19 // Cache successes for five minutes and cap entries to bound long-lived worker memory.
20 export async function vexJson(url: string, headers: Record<string, string>): Promise<any> {
21   const cached=pages.get(url);
22   if(cached&&cached.expires>Date.now())return cached.data;
23   let lastStatus = 0;
24   for (let attempt = 0; attempt < 3; attempt++) {
25     try {
26       const response = await fetch(url, { headers, signal: AbortSignal.timeout(15000) });
27       if (response.ok) {
28         const data=await response.json();
29         if(pages.size>=600)pages.delete(pages.keys().next().value!);
30         pages.set(url,{expires:Date.now()+300000,data});
31         return data;
32       }
33       lastStatus = response.status;
34       if (response.status !== 429 && response.status < 500) throw new Error(`Event.VEX request failed
35       (${response.status}).`);
36       await response.body?.cancel();
37       if(response.status===429&&attempt<2){
38         const seconds=Number(response.headers.get('retry-after'));
39         await new
40         Promise(resolve=>setTimeout(resolve,Math.min(10000,Math.max(2000,Number.isFinite(seconds)?seconds*1000:2000))));
41       }
42       } catch (error) {
43         if (attempt === 2 || (error instanceof Error && error.message.startsWith('Event.VEX request failed'))) throw
44         error;
45       }
46     }
47   }
48 }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A4 Complete Paginated Collection

The helper keeps original filters, reads every page, validates every payload, and returns only after all pages succeed.

website/lib/vex-api.ts

```

48 // Fetch every page with the original filters intact. Reject the whole collection
49 // if any page fails, so callers cannot publish a partial ranking as complete.
50 export async function vexCollection(url: string, headers: Record<string, string>): Promise<any[]> {
51   const parsed = new URL(url);
52   parsed.searchParams.set('per_page', '250');
53   parsed.searchParams.set('page', '1');
54   const first = await vexJson(parsed.href, headers);
55   if (!Array.isArray(first.data)) throw new Error('Event.VEX returned an invalid collection.');
```

```

56   const lastPage = Math.max(1, Number(first.meta?.last_page ?? 1));
57   const rest = await mapLimit(Array.from({ length: lastPage - 1 }, (_, index) => index + 2), 2, async page => {
58     const next = new URL(parsed);
59     next.searchParams.set('page', String(page));
60     const payload = await vexJson(next.href, headers);
61     if (!Array.isArray(payload.data)) throw new Error('Event.VEX returned an invalid collection.');
```

```

62     return payload.data;
63   });
64   return [...first.data, ...rest.flat()];
65 }
```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A5 Bracket Slot Preservation

Round and instance define the slot. Later replay games replace the display match without losing earlier games.

website/lib/bracket.ts

```

3 // Return the latest game for the card and every game for details. Keep null slots:
4 // removing them would shift opponents and draw incorrect bracket connections.
5 export function bracketRound(matches:any[], round:number, slots:number) {
6   return Array.from({length:slots}, (_,index)=>{
7     const games=matches.filter(match=>Number(match.round)===round&&Number(match.instance)===index+1)
8     .sort((a,b)=>Number(a.matchnum)-Number(b.matchnum));
9     return games.length?{...games[games.length-1],games}:null;
10   });
11 }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A6 Historical Match Eligibility

The filter distinguishes valid historical scores from ambiguous zero placeholders and removes duplicate match IDs only after validation.

website/lib/archive-matches.mjs

```

4 // IDs across divisions. Mark an ID seen only after validation so an invalid copy
5 // cannot hide a later valid copy. An explicitly scored 0-0 tie remains valid.
6 export function historicalMatches(divisions) {
7   const seen=new Set();
8   return divisions.flatMap(d=>d.matches??[]).filter(m=>{
9     if(seen.has(m.id)||Number(m.round)<2||Number(m.round)>6||m.alliances?.length!==2)return false;
10    if(!m.alliances.every(a=>Number.isFinite(a.score)&&a.score>=0&&a.teams?.length))return false;
11    if(!m.scored&&m.alliances.every(a=>a.score===0))return false;
12    seen.add(m.id);return true;
13  });
14 }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A7 VCR Match Evidence

The function validates bounded inputs, calculates expectation surprise and margin, and returns a component suitable for settlement.

website/lib/vcr3.mjs

```

25  * Reliability is returned separately and applied when evidence is aggregated.
26  */
27  export function matchEvidence({actual,expected,margin,scoreScale,share=.5,reliability=1}){
28    if(![0,.5,1].includes(actual))throw new RangeError('Invalid
actual');finite(expected,'expected',0,1);finite(margin,'margin');finite(scoreScale,'scoreScale',Number.MIN_VALUE);finite(share,'share',0,1);finite(reliability,'reliability',0,1);
29    const mov=actual===.5?1:1+.25*Math.tanh(Math.abs(margin)/scoreScale);
30    return{residual:(actual-expected)*mov*2*share,reliability};
31  }
32  // Average reliable evidence instead of rewarding match count indefinitely.
33  // Six units of reliable evidence reach full strength; smaller samples taper down.
34  const matchComponent=rows=>{let total=0,mass=0;for(const row of rows){finite(row.residual,'residual',-2.5,2.5);finite(row.reliability,'match reliability',0,1);total+=row.residual*row.reliability;mass+=row.reliability}return 40*(mass?total/mass:0)*Math.min(1,mass/6)};
35  /**
36   * Pure calculation: returns an audit ledger without modifying input or team state.
37   * The caller verifies event completion; completed=true is an assertion, not a check
38   * against upstream data. Qualification/elimination inputs and title probability
39   * are in [0,1]; null contribution/auto residuals mean unavailable evidence.
40   */
41  export function settleEvent(input){
42    const{rating,tier,matches,qualificationActual,qualificationExpected,eliminationActual,eliminationExpected,champion=false,titleProbability,completed,reliability=1,contributionResidual=null,autoResidual=null}=input;
43    finite(rating,'rating');finite(reliability,'reliability',0,1);if(!Object.hasOwn(EVENT_WEIGHTS,tier))throw new RangeError('Unknown tier');if(completed!==true)throw new Error('Only a verified completed event may settle');
44    for(const[k,v]of Object.entries({qualificationActual,qualificationExpected,eliminationActual,eliminationExpected,titleProbability}))finite(v,k,0,1);if(champion&&eliminationActual!==1)throw new Error('Champion must have eliminationActual=1');
45    const statistic=(v,scale,name)=>v===null?0:scale*Math.tanh(finite(v,name,-10,10)/2);
46    const components={match:matchComponent(matches),qualification:21*(qualificationActual-qualificationExpected),elimination:39*(eliminationActual-eliminationExpected),contribution:statistic(contributionResidual,8,'contribution residual'),auto:statistic(autoResidual,4,'auto residual')};
47    // Upside follows tier weight; downside uses its bounded reciprocal. Apply event
48    // reliability once here, independently of reliability on individual match rows.
49    const
base=Object.values(components).reduce((a,b)=>a+b,0),eventWeight=EVENT_WEIGHTS[tier],negativeWeight=clamp(1/eventWeight,.8,1.25),weighted=base*(base>=0?eventWeight:negativeWeight)*reliability;

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A8 VCR Event Settlement

The core combines match, qualification, elimination, contribution, and autonomous evidence before tier weight, reliability, achievement correction, and cap.

website/lib/vcr3.mjs

```

39  * are in [0,1]; null contribution/auto residuals mean unavailable evidence.
40  */
41  export function settleEvent(input){
42
43    const{rating,tier,matches,qualificationActual,qualificationExpected,eliminationActual,eliminationExpected,champion=false,titleProbability,completed,reliability=1,contributionResidual=null,autoResidual=null}=input;
44    finite(rating,'rating');finite(reliability,'reliability',0,1);if(!Object.hasOwn(EVENT_WEIGHTS,tier))throw new
RangeError('Unknown tier');if(completed!==true)throw new Error('Only a verified completed event may settle');
45    for(const[k,v]of
Object.entries({qualificationActual,qualificationExpected,eliminationActual,eliminationExpected,titleProbability}))finite(v,k
,0,1);if(champion&&eliminationActual!==1)throw new Error('Champion must have eliminationActual=1');
46    const statistic=(v,scale,name)=>v===null?0:scale*Math.tanh(finite(v,name,-10,10)/2);
47    const components={match:matchComponent(matches),qualification:21*(qualificationActual-
qualificationExpected),elimination:39*(eliminationActual-
eliminationExpected),contribution:statistic(contributionResidual,8,'contribution
residual'),auto:statistic(autoResidual,4,'auto residual')};
48    // Upside follows tier weight; downside uses its bounded reciprocal. Apply event
49    // reliability once here, independently of reliability on individual match rows.
50    const
base=Object.values(components).reduce((a,b)=>a+b,0),eventWeight=EVENT_WEIGHTS[tier],negativeWeight=clamp(1/eventWeight,.8,1.2
51    5),weighted=base*(base>=0?eventWeight:negativeWeight)*reliability;
52    // Favorites who win cannot lose points; winners with <=25% title probability
53    // receive a positive floor. Record the correction before the final +/-100 cap.
54    const championFloor=champion?(titleProbability<=.25?8*eventWeight*(1-
titleProbability)*reliability:0):null,achievementCorrection=champion?Math.max(0,championFloor-
weighted):0,uncappedDelta=weighted+achievementCorrection,delta=clamp(uncappedDelta,-100,100);
55    return{version:VCR_VERSION,ratingBefore:rating,ratingAfter:rating+delta,delta,components,base,eventWeight,negativeWeight,reli
ability,weighted,championFloor,achievementCorrection,capAdjustment:delta-uncappedDelta};
56  }
57  function
activeTeams(alliance,includeSitting=false){return(alliance?.teams??[]).filter(x=>includeSitting||!x.sitting).map(x=>x.team).f
ilter(Boolean)}
58  function resultFor(score,other){return score===other?.5:score>other?1:0}
59  function stage(match){const name=String(match?.name??'');if(/^final\b/i.test(name))return
1;if(/^sf\b|semifinal\b/i.test(name))return .65;if(/^qf\b|quarterfinal\b/i.test(name))return .45;if(/^r16\b|round of
16\b/i.test(name))return .25;return 0}
60  // Blend frozen strength with evidence from earlier matches in this event.
61  // Bound the partner split so a stronger partner never takes all two-team credit.
62  function dynamicShares(teams,ratings,local){
63    if(teams.length===1)return[1];if(teams.length===2)return teams.map(()=>1/teams.length);
64    const strength=teams.map(t=>(ratings.get(t.id)?1500)+80*(local.get(t.id)?0));const raw=1/(1+Math.exp(-(strength[0]-
strength[1])/220));const gap=Math.abs(strength[0]-strength[1]);const max=gap<100?.55:gap<200?.65:gap<300?.725:.8;const
a=clamp(raw,1-max,max);return[a,1-a];
65  }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A9 Completed Event Processing

The adapter freezes participant ratings, processes matches chronologically, infers limited champion evidence, mutates state once, and writes versioned history.

website/lib/vcr3.mjs

```

69  * Sitting teammates enter the field, but only active teammates get match evidence.
70  */
71  export function processCompletedEvent({event,matches,states,historyByTeam}){
72    const
valid=matches.filter(m=>m?.alliances?.length===2&&m.alliances.every(a=>Number.isFinite(a.score)&&activeTeams(a).length));if(!
valid.length)return;
73    const participantIds=new Set(valid.flatMap(m=>m.alliances.flatMap(a=>activeTeams(a,true).map(t=>t.id)));for(const m
of valid)for(const a of m.alliances)for(const t of
activeTeams(a,true))if(!states.has(t.id))states.set(t.id,{id:t.id,number:t.name,rating:1500,matches:0,wins:0,losses:0,ties:0,
pointsFor:0,pointsAgainst:0,events:new Set(),middleGradeEvents:new Set(),highGradeEvents:new Set()});
74    // Freeze expectations for the entire event. Updating these inside the match loop
75    // would make later opponents depend on already-settled outcomes in this event.
76    const frozen=new
Map([...participantIds].map(id=>[id,states.get(id)?.rating??1500]),field=[...frozen.values()].sort((a,b)=>a-
b),mean=field.reduce((a,b)=>a+b,0)/field.length;
77    // Softmax field-strength proxy for title probability, not a bracket simulation.
78    const expWeights=new Map([...frozen].map(([id,r])=>[id,Math.exp((r-
mean)/200)])),expTotal=[...expWeights.values()].reduce((a,b)=>a+b,0);
79    // qA/qE hold actual/expected qualification results; elim is furthest named stage.
80    // local accumulates residuals only for partner shares, never for expectations.
81    const rows=new Map([...participantIds].map(id=>[id,{matches:[],qA:[],qE:[],elim:0,finalW:0,finalL:0}]}),local=new
Map();
82    // Use event score spread (minimum 12) to limit margin sensitivity across games.
83    const
scores=valid.flatMap(m=>m.alliances.map(a=>a.score)),scoreMean=scores.reduce((a,b)=>a+b,0)/scores.length,scoreScale=Math.max(
12,Math.sqrt(scores.reduce((s,x)=>s+(x-scoreMean)**2,0)/scores.length));
84    for(const match of
valid.sort((a,b)=>String(a.scheduled??a.started??').localeCompare(String(b.scheduled??b.started??')))){
85      const[a,b]=match.alliances,ta=activeTeams(a),tb=activeTeams(b);const
ra=ta.reduce((s,t)=>s+(frozen.get(t.id)?1500,0)/ta.length,rb=tb.reduce((s,t)=>s+(frozen.get(t.id)?1500,0)/tb.length,ea=ex
pectedResult(ra,rb),aa=resultFor(a.score,b.score),qual=qual/i.test(String(match.name??'))),final=/^final\b/i.test(String(mat
ch.name??')));
86      for(const[alliance,opponent,teams,expected,actual]of[[a,b,ta,ea,aa],[b,a,tb,1-ea,1-aa]]){const
shares=dynamicShares(teams,frozen,local);teams.forEach((team,i)=>{const
row=rows.get(team.id),e=matchEvidence({actual,expected,margin:alliance.score-
opponent.score,scoreScale,share:shares[i]});row.matches.push(e);local.set(team.id,(local.get(team.id)??0)+e.residual);if(qual
){row.qA.push(actual);row.qE.push(expected)}row.elim=Math.max(row.elim,stage(match));if(final){actual===1?row.finalW++:actual
===0&&row.finalL++}})
87    }
88    // Finals wins/losses infer champions from match names, not awards. Multi-division
89    // events aggregate those records per team; this is not an overall-bracket resolver.
90    // Missing qualification evidence contributes zero (both means default to 0.5).
91    const tier=eventTier(event);
92    for(const id of participantIds){const
team=states.get(id),row=rows.get(id),champion=row.finalW>row.finalL&&row.finalW>0;if(champion)row.elim=1;else
if(row.finalL>row.finalW&&row.finalL>0)row.elim=.85;const
own=frozen.get(id),below=field.filter(x=>x<own).length,equal=field.filter(x=>x===own).length,percentile=field.length>1?(below
+.5*(equal-1))/(field.length-
1):.5,expectedElim=.1+.55*percentile,titleProbability=(expWeights.get(id)?1)/expTotal,qA=row.qA.length?row.qA.reduce((a,b)=>
a+b,0)/row.qA.length:.5,qE=row.qE.length?row.qE.reduce((a,b)=>a+b,0)/row.qE.length:.5,reliability=Math.min(1,valid.length/6);
93    // Contribution/autonomous estimates are unavailable in this adapter. Keep them

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A10 Archive Validation

The validator rejects wrong season, wrong model, empty ranking, incomplete coverage, or recorded failures.

website/lib/ranking-seasons.mjs

```

11 // This checks envelope metadata, not every team row or exact candidate version:
12 // publication also relies on the builder's validation and archive regression tests.
13 export function validateArchive(data, season) {
14   if(data?.seasonId !== RANKING_SEASONS[season]?.id || !String(data?.modelVersion ?? '').startsWith('VCR-
3.0') || !Array.isArray(data?.rankings) || !data.rankings.length || data.coverage?.failures?.length !== 0 || data.coverage?.loadedEvent
s !== data.coverage?.expectedEvents) throw new Error('The requested VCR 3.0 archive is not complete. ');
15   return data;
16 }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A11 Worker Canonical URL

The function allowlists routes and query fields and adds the model namespace for rating responses.

website/cloudflare/worker.ts

```

12 // Unknown query fields are dropped. Add new data-affecting parameters here as well
13 // as in their route, or distinct requests can incorrectly share cached responses.
14 function canonicalUrl(request: Request) {
15   const incoming = new URL(request.url);
16   if (!/^\/api\/(archive-source(?:\/\d{1,10})?)|events(?:\/\d{1,10})?)|teams\/[0-9]{1,8}[A-Za-z0-9-
17 ]{0,12}|rankings|skills)$/.test(incoming.pathname)) return null;
18   const url = new URL(incoming.pathname, incoming.origin);
19   for (const key of ['season', 'teamId', 'division', 'page']) {
20     const value = incoming.searchParams.get(key);
21     if (value) {
22       if (!/^\/\d{1,10}$/.test(value)) return null;
23       url.searchParams.set(key, value);
24     }
25   }
26   const mode=incoming.searchParams.get('mode');
27   if(mode){if(!['metadata','teams','matches'].includes(mode))return null;url.searchParams.set('mode',mode)}
28   // Model-version namespace prevents a new rating model from reusing old responses.
29   if (url.pathname === '/api/rankings' || url.pathname.startsWith('/api/teams/')) url.searchParams.set('model',
30 VCR_VERSION);
31   return url;
32 }
33
34 async function dispatch(url: URL) {
35   if (url.pathname.startsWith('/api/archive-source')) return archiveSource(url);
36   const request = new Request(url);
37   if (url.pathname === '/api/events') return events(request);
38   if (url.pathname === '/api/rankings') return rankings(request);

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A12 Worker Cache and Error Boundary

The handler checks D1, dispatches the route, caches only safe successful bodies, maps rate limits separately, and returns no-store errors.

website/cloudflare/worker.ts

```

49
50 export default {
51   async fetch(request: Request, env: Env, ctx: ExecutionContext): Promise<Response> {
52     if (request.method === 'OPTIONS') return cors(new Response(null, { status: 204 }));
53     if (request.method !== 'GET') return cors(Response.json({ error: 'Method not allowed' }, { status: 405, headers: {
Allow: 'GET, OPTIONS' } }));
54     const archiveSeasons:Record<string,string>={'197':'2025-26','190':'2024-25','181':'2023-24','173':'2022-23'};
55     const incoming=new URL(request.url),archiveSeason=incoming.searchParams.get('season');
56     if(incoming.pathname=== '/api/archive-manifest')return cors(await env.ARCHIVES.fetch(new Request(new
URL('/manifest.json',incoming.origin))));
57     if(incoming.pathname=== '/api/team-directory'){
58       const asset=await env.ARCHIVES.fetch(new Request(new URL('/team-directory.json',incoming.origin)));
59       if(!asset.ok)return cors(Response.json({error:'Team directory is temporarily unavailable.'},{status:503}));
60       const response=cors(asset);response.headers.set('Cache-Control','public,max-age=3600');return response;
61     }
62     // Historical rankings come only from published static archives. A missing asset
63     // is a 503, not a fallback to the live route's partial event sample.
64     const archivePath=/^\/rankings-20\d{2}-\d{2}-
vcr3\.json$/.test(incoming.pathname)?incoming.pathname=== '/api/rankings'&&archiveSeason&&archiveSeasons[arc
hiveSeason]?`/rankings-${archiveSeasons[archiveSeason]}-vcr3.json`:null;
65     if(archivePath){
66       const asset=await env.ARCHIVES.fetch(new Request(new URL(archivePath,incoming.origin)));
67       if(!asset.ok)return cors(Response.json({error:'This historical archive is not published
yet.'},{status:503,headers:{'Cache-Control':'no-store'}}));
68       return cors(asset);
69     }
70     if (new URL(request.url).pathname === '/api/health') {
71       await env.DB.prepare('SELECT 1').first();
72       return cors(Response.json({ status: 'ok', environment: 'test', database: 'D1', modelVersion: VCR_VERSION }));
73     }

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A13 Historical Rebuild Entry

The builder resumes cached official pages and protects the final archive behind coverage checks.

website/scripts/rebuild-season-archives.mjs

```

20 // Shared reservation clock spaces both download workers' requests by 1.8 seconds.
21 let nextRequest=0;
22 async function get(path) {
23   const cachedFile=resolve(pageCache,createHash('sha256').update(path).digest('hex')+'.json');
24   try{return JSON.parse(await readFile(cachedFile,'utf8'))}catch{}
25   for(let attempt=0;attempt<5;attempt++) {
26     try {
27       const wait=Math.max(0,nextRequest-Date.now());nextRequest=Math.max(nextRequest,Date.now()+1800;if(wait)await
sleep(wait);
28       const r=await fetch(api+path,{signal:AbortSignal.timeout(90000)});
29       if(r.ok){const data=await r.json();await writeFile(cachedFile,JSON.stringify(data));return data}
30       if(r.status===429)throw Object.assign(new Error('Official API rate limit reached; cached progress preserved.
Retry later.'),{rateLimited:true});
31       if(r.status<500&&r.status!==429)throw new Error(`HTTP ${r.status}: ${path}`);
32     } catch(error) { if(error.rateLimited||attempt===4)throw error; }
33     await sleep(Math.min(30000,2000*2**attempt));
34   }
35   throw new Error(`Could not retrieve ${path}`);
36 }
37 async function paged(path){const first=await get(path);if(!Array.isArray(first.data))throw new Error('Invalid source
page');const data=[...first.data];for(let page=2;page<=Number(first.meta?.last_page??1);page++){const next=await
get(`${path}&page=${page}`);if(!Array.isArray(next.data))throw new Error('Invalid source
page');data.push(...next.data)}return data}
38 // Reuse the older direct-API cache only when every division page is present.
39 // Otherwise fetch through the public adapter; historicalMatches filters afterward.
40 async function divisionMatches(event,d){
41   const url='https://events.vex.com/api/v2/events/${event.id}/divisions/${d.id}/matches?per_page=250';
42   const readLegacy=async path=>JSON.parse(await readFile(resolve(`.ranking-
cache/${event.season.id}/${createHash('sha1').update(path).digest('hex')}.json`),'utf8'));
43   try{const first=await readLegacy(url),data=[...first.data];for(let
page=2;page<=Number(first.meta?.last_page??1);page++)data.push(...(await readLegacy(`${url}&page=${page}`)).data);return
data}catch{}
44   return paged(`/api/archive-source/${event.id}?mode=matches&division=${d.id}`);

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A14 Archive Packaging

Packaging validates every configured archive before copying assets and writing the manifest.

website/scripts/prepare-archive-assets.mjs

```

1 import { mkdir, readFile, writeFile } from 'node:fs/promises';
2 import { createHash } from 'node:crypto';
3 import { RANKING_SEASONS, validateArchive } from '../lib/ranking-seasons.mjs';
4 await mkdir('archive-assets', { recursive: true });
5 const manifest = [];
6 for (const [label, config] of Object.entries(RANKING_SEASONS)) {
7   if (!config.archive) continue;
8   let source; try { source = await readFile(`public${config.archive}`, 'utf8') } catch (error) { if (process.argv.includes('--available') && error.code === 'ENOENT') continue; throw error }
9   const data = validateArchive(JSON.parse(source), label);
10  // Keep concise recent-event history in the public leaderboard; the rebuild cache retains source matches.
11  for (const team of
data.rankings) { team.form = team.form.map(({ eventId, event, eventDate, change, rating, tier, champion }) => ({ eventId, event, eventDate, cha
nge, rating, tier, champion })); delete team.displayedStrength }
12  const content = JSON.stringify(data);
13  if (Buffer.byteLength(content) > 25 * 1024 * 1024) throw new Error(`Archive exceeds asset limit: ${config.archive}`);
14  await writeFile(`archive-assets${config.archive}`, content);
15
manifest.push({ seasonId: config.id, season: label, path: config.archive, modelVersion: data.modelVersion, teams: data.rankings.length,
events: data.eventsProcessed, matches: data.matchesProcessed, coverage: data.coverage, asOf: data.asOf, sha256: createHash('sha256').u
pdate(content).digest('hex'), bytes: Buffer.byteLength(content) });
16 }
17 await writeFile('archive-assets/manifest.json', JSON.stringify({ archives: manifest }, null, 2));
18 await mkdir('work', { recursive: true });
19 await writeFile('work/archive-manifest.json', JSON.stringify({ archives: manifest }, null, 2));
20 console.log(manifest.map(m => `${m.season}: ${m.teams} teams / ${m.events} events / ${m.matches} matches`).join('\n'));

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Code Listing A15 GitHub Pages Build and Deploy

The workflow runs tests and health checks before build, uploads a Pages artifact, and deploys only after build success.

website/.github/workflows/github-pages-test.yml

```

7   group: vexrank-test-pages
8   cancel-in-progress: false
9   jobs:
10    build:
11      runs-on: ubuntu-latest
12      steps:
13        - uses: actions/checkout@v4
14        - uses: pnpm/action-setup@v4
15          with:
16            version: 11.19.0
17        - uses: actions/setup-node@v4
18          with:
19            node-version: 24
20            cache: pnpm
21            cache-dependency-path: website/pnpm-lock.yaml
22        - run: pnpm install --frozen-lockfile
23          working-directory: website
24        - name: Check data regressions
25          run: node --test tests/data-regressions.test.mjs tests/team-directory.test.mjs
26          working-directory: website
27        - name: Verify test API is deployed
28          run: curl --fail --retry 2 https://vexrank-api-test.vexrank-eason.workers.dev/api/health
29        - name: Build test website
30          run: pnpm exec vite build --config vite.pages.config.ts
31          working-directory: website

```

Review question	Contributor check
Input contract	Identify validated and unvalidated inputs before changing behavior.
Failure behavior	Confirm failure remains visible and cannot become a valid-looking partial result.
Side effects	Identify caches, state mutation, file replacement, secrets, or deployment actions.
Regression	Run or add the smallest fixture that proves the intended invariant.

Appendix B Complete Test Matrix

This appendix consolidates automated, build, browser, live integration, historical, and planned predictive checks.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Automated Test Matrix Rows 1 Through 7

ID	Test	Area	Result	Time ms
1	Historical scores remain usable when the upstream scored flag is false	historical match eligibility	Passed	6.8675
2	Each season has its own archive and incomplete or wrong season results are rejected	archive identity and coverage	Passed	1.1143
3	Semifinal replays retain their bracket slot and missing byes leave a gap	bracket identity	Passed	1.6892
4	A Worlds qualifier is not the World Championship	event classification	Passed	2.7254
5	Mentioning a signature event does not make a scrimmage a signature	event classification	Passed	0.3512
6	High school names do not imply school based tournament format	event format classification	Passed	0.3001
7	All pages are retained and elimination round filters survive pagination	complete collection loading	Passed	79.8972

Automated Test Matrix Rows 8 Through 14

ID	Test	Area	Result	Time ms
8	Failed later pages reject rather than publish a partial collection	fail closed collection behavior	Passed	2.5680
9	Table agenda keeps activity start end and location together	agenda table parsing	Passed	2.0999
10	Single day agendas without day headings still retain timed entries	agenda text parsing	Passed	92.3847
11	Numeric family search includes every suffix including unregistered teams	team discovery	Passed	4.0437
12	Names and organizations support partial case insensitive search	team discovery	Passed	0.4119
13	All three filters combine and countries normalize	directory filters	Passed	1.1565
14	Locations include inactive only countries and dependent regions	directory filter values	Passed	2.2824

Automated Test Matrix Rows 15 Through 19

ID	Test	Area	Result	Time ms
15	Single active robot can receive all alliance credit	partner share validation	Passed	2.8706
16	Complementary shares tolerate ordinary floating point rounding	numeric validation	Passed	1.3181
17	Unexpected champions gain at every tier	achievement policy	Passed	3.1027
18	Legitimate non champion loss remains negative	achievement policy boundary	Passed	0.4941
19	Completed event processor applies champion guarantee and produces versioned audit	integrated event settlement	Passed	28.0056

Manual and Integration Test Matrix

Test	Method	Observed result	Next repetition
Worlds division coverage	Inspect event routes and interface	Event 64025 showed 11 divisions and 865 teams	After route or pagination changes
Worker health	GET /api/health	Status ok with D1 and model identifier	Before every Pages deployment
Worlds event 64026	Live event request	585 teams and all eight divisions	After upstream or route change
Cache repeat	Repeat team request	About 80 ms in one observation	Controlled cold and warm load test
Static build	Vite Pages build	2,362 modules transformed	Every frontend change
Lint	pnpm lint	Failed on known backlog	Every cleanup batch

Appendix C Deployment Runbooks

These concise runbooks turn the deployment narrative into repeatable checks while preserving the distinction between test and production readiness.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Runbook C1 Pre Merge Review

1. State the problem and expected behavior
2. Add or identify regression fixtures
3. Run all 19 fixture tests
4. Run lint and report the existing baseline separately
5. Build the affected frontend or application target
6. Inspect loading error and narrow-screen states for UI changes
7. Document model or cache-key impact
8. Exclude credentials dependencies caches and unrelated generated data

Completion record

Field	Value to record
Operator	Name or team
Date and time	Timestamp and timezone
Source revision	Commit or pull request
Artifact version	Frontend Worker archive and model identifiers
Checks	Command or URL and result
Decision	Proceed stop or roll back

Runbook C2 GitHub Pages Test Publish

1. Verify the Worker health endpoint
2. Verify representative API and archive requests
3. Dispatch the Pages workflow
4. Confirm test and build jobs pass
5. Open the deployed base path under /Vex-Rank/
6. Inspect navigation and network origins
7. Record artifact and workflow identifiers
8. Retain the prior artifact for rollback

Completion record

Field	Value to record
Operator	Name or team
Date and time	Timestamp and timezone
Source revision	Commit or pull request
Artifact version	Frontend Worker archive and model identifiers
Checks	Command or URL and result
Decision	Proceed stop or roll back

Runbook C3 Cloudflare Worker Publish

1. Review route and query allowlist
2. Apply D1 schema changes
3. Verify secret binding names
4. Deploy Worker
5. Check health and D1
6. Check CORS from the Pages origin
7. Check cache miss and hit headers
8. Check a response near the cache-size boundary
9. Record Worker and model version

Completion record

Field	Value to record
Operator	Name or team
Date and time	Timestamp and timezone
Source revision	Commit or pull request
Artifact version	Frontend Worker archive and model identifiers
Checks	Command or URL and result
Decision	Proceed stop or roll back

Runbook C4 Historical Archive Publish

1. Select one supported season
2. Resume or refresh cached official pages intentionally
3. Wait for complete coverage
4. Verify expected equals loaded and failures are empty
5. Run archive tests
6. Validate season and model identifiers
7. Prepare versioned assets and manifest
8. Inspect sample teams and counts
9. Publish without replacing the last complete archive until final validation

Completion record

Field	Value to record
Operator	Name or team
Date and time	Timestamp and timezone
Source revision	Commit or pull request
Artifact version	Frontend Worker archive and model identifiers
Checks	Command or URL and result
Decision	Proceed stop or roll back

Runbook C5 Incident and Rollback

1. Record time route status and model version
2. Separate upstream failure from code failure
3. Disable or roll back the changed Worker or frontend artifact
4. Keep the last validated archive available
5. Avoid caching error responses
6. Reproduce with a minimal fixture
7. Add a regression before redeploying
8. Update the notebook with the observation and decision

Completion record

Field	Value to record
Operator	Name or team
Date and time	Timestamp and timezone
Source revision	Commit or pull request
Artifact version	Frontend Worker archive and model identifiers
Checks	Command or URL and result
Decision	Proceed stop or roll back

Appendix D Data Dictionary and Glossary

This appendix defines the terms and fields used across the interface, archive, algorithm, and deployment records.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Glossary Terms 1 Through 8

Term	Definition
Archive	A versioned complete-season JSON artifact that passes season, model, ranking, and coverage validation.
Canonical record	A normalized representation used after validating an upstream source record.
Confidence or RD	A stored uncertainty quantity; lower values in the current archive generally represent more accumulated evidence.
D1	Cloudflare's SQL database used by the test Worker for shared response caching.
Elimination utility	A numeric value assigned to the furthest bracket stage reached and used consistently for actual and expected results.
Event Strength Score	A proposed 0-100 summary of frozen event context in the VCR 1 and VCR 2 designs.
Event Weight	A tier-based multiplier applied to tournament-level event evidence, not ordinary match evidence.
Fail closed	Reject incomplete or invalid data instead of returning it as if it were complete.

Glossary Terms 9 Through 16

Term	Definition
Frozen input	A value captured before an event result and not recomputed using later outcomes.
Historical replay	Processing completed events in chronological order to reconstruct rating state.
Model version	The identifier that names the exact rating behavior used to produce a result.
Partner share	The fraction of alliance evidence assigned to one active team, bounded by the model.
Qualification actual	A normalized value derived from a team's official division qualification rank.
Reliability	A factor from zero to one representing usable evidence and data quality.
Settlement	The completed-event calculation that combines components and returns a rating delta and audit ledger.
VCR	VEX Competitive Rating, the project's versioned estimate of competitive strength and achievement.

Archive Field Dictionary

Field	Meaning	Caution
rankings	Published team rows sorted for the archive	Order can change with model version
eventsProcessed	Completed events that produced eligible rating evidence	Can be lower than loaded calendar events
matchesProcessed	Eligible historical matches used in replay	Filtered placeholders and invalid records are excluded
coverage.expectedEvents	Events in the fetched season calendar	Depends on upstream calendar completeness
coverage.loadedEvents	Calendar event payloads successfully fetched	Must equal expected for publication
coverage.failures	Fetch or processing failures	Must be empty for publication
modelVersion	Executable model identifier	Must match the archive validator
rating	Internal candidate rating after replay and display decay	Not the VCR 2 conservative 0-1000 public transform
confidence	Stored uncertainty or RD-like field	Interface naming should explain direction
form	Recent event history embedded for display	Not a complete separate awards dataset

Appendix E References and Source Record

This appendix identifies the project artifacts and external platform documentation used to prepare the notebook.

The pages in this part preserve the design question, evidence, decision, verification method, and remaining uncertainty so a contributor can reconstruct why the implementation took its present form.

Primary Project References

1. VEX Rank System Architecture and VCR v1.0, six-page Lucidchart PDF export, August 2026, supplied project file.
2. VEX Competitive Rating Full Algorithm Specification, version 2.0, September 2026, `output/pdf/vex_competitive_rating_full_algorithm.pdf`.
3. VCR 3.0 candidate event settlement and achievement, `docs/VCR-3.0-revision.md`.
4. VCR 3.0 candidate reference, `versions/vcr-3.0-candidate.md`.
5. VEX Rank repository overview and VCR 2.0 design rules, `README.md`.
6. Contributor source map, constraints, checks, archive process, and deployment notes, `CONTRIBUTING.md`.
7. Website inspection record, `website/tests/INSPECTION.md`.
8. Firebase deployment trial record, `website/firebase/DEPLOYMENT.md`.
9. Cloudflare deployment record, `website/cloudflare/DEPLOYMENT.md`.
10. GitHub Pages test workflow, `website/.github/workflows/github-pages-test.yml`.
11. Executable VCR 3 candidate, `website/lib/vcr3.mjs`.
12. Official API pagination and retry helper, `website/lib/vex-api.ts`.
13. Cloudflare Worker route and cache boundary, `website/cloudflare/worker.ts`.
14. Regression suites, `website/tests/archive.test.mjs`, `data-regressions.test.mjs`, `team-directory.test.mjs`, and `vcr3.test.mjs`.
15. Published Push Back VCR 3 candidate archive, `website/archive-assets/rankings-2025-26-vcr3.json`.
16. Historical archive manifest, `versions/vcr3-historical-archives.json` and `website/archive-assets/manifest.json`.

External Platform References

1. GitHub Docs. Configuring a publishing source for your GitHub Pages site.
<https://docs.github.com/en/pages/getting-started-with-github-pages/configuring-a-publishing-source-for-your-github-pages-site>. Accessed September 16 2026.
2. GitHub Docs. Using custom workflows with GitHub Pages. <https://docs.github.com/en/enterprise-cloud@latest/pages/getting-started-with-github-pages/using-custom-workflows-with-github-pages>. Accessed September 16 2026.
3. GitHub Docs. Deployment environments. <https://docs.github.com/en/actions/concepts/workflows-and-actions/deployment-environments>. Accessed September 16 2026.
4. Firebase. Configure your environment for Cloud Functions for Firebase.
<https://firebase.google.com/docs/functions/config-env>. Accessed September 16 2026.
5. Firebase. Quotas and limits for Cloud Functions for Firebase.
<https://firebase.google.com/docs/functions/quotas>. Accessed September 16 2026.
6. Firebase. Get started with Cloud Firestore Security Rules.
<https://firebase.google.com/docs/firestore/security/get-started>. Accessed September 16 2026.
7. Cloudflare Developers. Workers secrets. <https://developers.cloudflare.com/workers/configuration/secrets/>. Accessed September 16 2026.
8. Cloudflare Developers. D1 limits. <https://developers.cloudflare.com/d1/platform/limits/>. Accessed September 16 2026.

Evidence Provenance and Reproduction Notes

Automated test evidence was reproduced locally on September 16 2026 with the bundled Node runtime. The complete run contained 19 passing tests in approximately 2.1 seconds. The static Pages build was also reproduced and generated the sizes recorded in Part Six. The lint command was reproduced and failed on the documented baseline categories.

Archive statistics were computed directly from the published Push Back VCR 3 candidate JSON. The analysis included all 12,573 ranking rows. Percentiles use linear interpolation on sorted values, means use all numeric rows, and standard deviations are population values. Country, region, grade, tier, and champion-form counts are descriptive summaries of the archive fields.

Live integration counts and deployment outcomes are dated project observations recorded in inspection and deployment documents. They should be repeated before an operational decision because public data, infrastructure, and free-tier limits can change.

The Lucidchart figures are reproduced from the supplied PDF. The diagrams are used as evidence of design intent. Statements about implemented behavior are based on repository code and tests rather than assuming that every diagram box was completed.

Document Maintenance Procedure

1. Add a dated notebook entry for every material trial, failure, algorithm change, archive build, or deployment decision.
2. State whether each claim is observed, designed, or recommended.
3. Update the detailed table of contents and heading hierarchy after structural changes.
4. Refresh measured test, build, lint, archive, and integration data instead of copying stale counts.
5. Add or revise code listings only when they explain a design boundary or contributor risk.
6. Keep formulas tied to their model version and do not merge VCR 1, VCR 2, and VCR 3 rules into one unnamed method.
7. Update platform references when a deployment constraint or limit changes.
8. Render the complete DOCX and PDF and inspect every page for clipping, broken tables, blank pages, missing glyphs, and stale fields.
9. Publish the new files only after the final page count, reference list, and evidence dates are verified.

Part Ten Current Engineering State and Design Decisions

This section brings the earlier design record forward to the GitHub main branch at commit 6a03cd0 on September 23 2026. The GitHub repository began on September 9; the earlier local website history began on September 1. Every historical count below is tied to its dated source rather than treated as a permanent property of the service.

Evidence	Date and boundary	What it establishes
Local website Git history	Sep 1-11; 49 commits	Early portal, event hub, rankings, archive, bracket and reliability design
GitHub main history	Sep 9-23; 101 commits at 6a03cd0	Published code and merge chronology
GitHub pull requests	29 numbered records; 28 in the repository listing	Review and merge trail; PR 1 closed unmerged and PR 29 open
Current tests	Sep 23; 37/37 fixtures pass	Executable behavior of this checkout
Current static build	Sep 23; 2,366 modules	Frontend build completes with a bundle size warning
Current lint	Sep 23; exit 1	Existing source/type warnings and errors remain

September 1 to September 23 Timeline

Date	Observed milestone	Evidence
Sep 1	Portal, live Event VEX connections, skills and team pages	Local website commits f402ff8 through 952d28d
Sep 2	Event hub, agendas, historical rankings and team trajectories	Local website commits f0ff33f through bad1229
Sep 3	Event classes and multi-division elimination bracket	Local website commits db39f20 through 36d8061
Sep 9	GitHub repository created and website, PDFs, source docs imported	GitHub initial and documentation commits
Sep 11-15	Worlds reliability, test backend, VCR 3 candidate and archives	Local ddc5680; GitHub PRs 2, 4, 5, 6 and 7
Sep 16	Contributor comments, search, redesign, routing and CI	GitHub PRs 8 through 10 and direct commits
Sep 17	Scoring reconciliation, reliability, themes and interaction trials	GitHub PRs 11 through 24
Sep 22-23	Custom-domain path, API origin, Discord, homepage layout and revert	GitHub commits 9e55113 onward; PRs 25-29

The September 1-3 commits are from the separate local website repository. GitHub itself was created on September 9. This distinction prevents an early local design experiment from being mistaken for a public release date.

Project Need Revisited Against the Current Site

The original need was a competition view that let participants move from official events to teams and seasons, compare performance with context, and inspect a rating model rather than accept an unexplained order. The current site has a searchable team directory, event hub, skills and rankings views, team histories, and a versioned VCR candidate. Those features answer different parts of the need: finding a team, understanding what happened at an event, and interpreting the rating estimate.

The user-facing distinction between a sampled current-season ranking and a completed-season archive remains central. The live route replays the latest 36 completed events; the versioned historical file represents a separate archive build with coverage metadata. A contributor should never make a partial live response look like a complete season result. The Worker returns an unavailable response for a missing archive instead of quietly falling back to the live sample.

A proposed gallery, logo, and team link would require a new submission and moderation system. The September 17 roadmap notes that the official shaped team payload contains no image or link field. That future product choice is distinct from the current official-data viewer.

Current System Boundary and Request Flow

The repository contains one shared React interface with a Vinext application and a separate Vite static entry. GitHub Pages can serve the static build; the Cloudflare Worker serves allowed API routes and versioned archive assets. The Worker reuses route handlers and stores short-lived successful API responses in D1. RobotEvents credentials remain on the service side. These boundaries explain why a browser build can be published without shipping the upstream token.

Boundary	Current code	Failure behavior
Browser	app/page.tsx and lib/client-fetch.ts	Shows load/retry state and reports likely blocked cross-origin requests
Routes	app/api and cloudflare/worker.ts	Only allowlisted paths/query values dispatch
Upstream	lib/vex-api.ts	Failed later pages reject rather than silently returning partial results
Archive	cloudflare/archive-source.ts and versioned assets	Missing publication returns 503
Cache	Cloudflare D1 api_cache	15-minute storage for responses under the size guard

Browser API and Archive Routing

The browser helper routes /api paths to the configured API origin. A VCR 3 archive path also uses that origin because the Worker serves archive assets. Other static paths use the Pages asset base. This is a small but important design boundary: a copied page URL can load the UI while data fails if the API origin is blocked or misconfigured.

Current source excerpt: *website/lib/client-fetch.ts*, lines 4-14

```

4  /** Separate API and asset origins for GitHub Pages; keep same-origin hosting working. */
5  export function siteFetch(path:string, init?:RequestInit) {
6    // Build-time constants: the API can live on a Worker while assets live beneath
7    // a GitHub Pages repository path. Undefined constants support same-origin dev.
8    const apiBase=typeof __VEX_API_BASE__==='undefined'?':__VEX_API_BASE__';
9    const assetBase=typeof __VEX_ASSET_BASE__==='undefined'?':__VEX_ASSET_BASE__';
10   // Versioned archives use the API origin when configured because the Worker
11   // serves its archive assets there; ordinary static files use the Pages base.
12   const archive=/^\s\/rankings-20\d{2}-\d{2}-vcr3\.json$/.test(path);
13   const
14   url=path.startsWith('/api/')||archive&&apiBase?`${apiBase}${path}`:path.startsWith('/')?`${assetBase}${path.slice(1)}`:path;
15   return fetch(url,init);

```

The regular expression is a deliberate archive boundary. A future archive naming change requires a corresponding routing update and a test; otherwise the browser may request the file from Pages while the Worker owns it.

Versioned Rating and Presentation Calculation

The executable event model is the VCR 3 candidate in `website/lib/vcr3.mjs`. Its scoring presentation helpers are centralized in `website/lib/vcr-scoring.mjs`. On September 17 the project found that live rankings, graph endpoints, and archives were using inconsistent event sets, decay, and confidence floors. The current helper defines a 75-day recency half-life, a 35-point confidence floor, and the decayed rating from raw event changes. This is a display/aggregation rule around settlement, not a replacement for the event processor.

Current source excerpt: `website/lib/vcr-scoring.mjs`, lines 10-33

```

10  /** Half-life in days for down-weighting older results. */
11  export const RECENCY_HALF_LIFE_DAYS = 75;
12
13  /** Weight in (0,1] for a result, decaying as it ages relative to `now`. */
14  export function recencyWeight(date, now) {
15    const ageDays = Math.max(0, (now.getTime() - new Date(date).getTime()) / 86400000);
16    return 2 ** (-ageDays / RECENCY_HALF_LIFE_DAYS);
17  }
18
19  /** Rating uncertainty: shrinks as a team accumulates matches, floored at 35. */
20  export function confidenceFor(matches) {
21    return Math.max(35, Math.round(120 / Math.sqrt(Math.max(1, matches / 4))));
22  }
23
24  /**
25   * The published rating: 1500 plus each result's change, decayed by age.
26   *
27   * This is the single source of truth for "what number does a team have". Both
28   * the ranking list and the per-event graph must go through it, or the graph's
29   * final point will not equal the rating shown in the table - which is exactly
30   * the defect this was written to fix.
31   */
32  export function decayedRating(history, now) {
33    return 1500 + history.reduce(

```

When modifying rating display, test the ranking row, graph endpoint, and archive builder together. The selected event set and cutoff date must also match; sharing a formula alone does not make two views comparable.

VCR 3 Candidate and Algorithm Version Boundaries

The VCR v1.0 Lucidchart pages remain evidence of an early design. The VCR 2.0 PDFs document a later model with a 55/18/10/17 component blend. The merged VCR 3 candidate uses event settlement with frozen expectations, bounded components, an event-level weight, an explicit winner correction, and an audit ledger. The specification was dated September 15 and the candidate implementation merged through PR 5. Its own file still carries a pre-merge status sentence; Git history is authoritative for merge timing.

The model status is still candidate because a passing scenario suite is not historical predictive calibration. The README's opening VCR 2.0 description is older than the contributor guide's executable-model note. A reviewer should read the version on an archive record or API result, then use the matching implementation and specification. The September 23 checkout identifies VCR-3.0-candidate.2 in its archive manifest.

Version	Documented role	Important caution
VCR 1.0	Lucidchart design flow	Not a statement of current weights
VCR 2.0	Detailed earlier specification/PDF	README introduction has not been fully reconciled
VCR 3 candidate.2	Shared implementation and one published Push Back archive manifest	Calibration and complete multi-season archive set remain open

Historical Archive State on September 23

The checked-in versions manifest lists one Push Back VCR-3.0-candidate.2 archive: season ID 197, 12,573 teams, 1,548 rated events, and 110,150 matches. Its coverage envelope records 1,681 expected and 1,681 loaded event payloads with no failures, an as-of date of June 1 2026, and a SHA-256 digest. These are values in the manifest, not a fresh recount of the remote Worker asset today. The previous notebook's distribution analysis was performed against the September 16 local archive snapshot.

The builder caches downloaded pages and event payloads so an interrupted run can resume. A successful coverage check refers to the fetched season calendar; it does not independently prove that the upstream calendar included every real competition. Old cache data also has no automatic expiry, so a corrected upstream result requires a deliberate refresh and rebuild. The four configured historical season IDs are 197, 190, 181, and 173, but only season 197 appears in the checked-in version manifest.

A source-backed archive publication requires the data file, matching manifest/version, coverage report, digest, Worker asset packaging, and a live fetch after deployment. Tests should reject wrong-season and incomplete coverage before any user-facing publication.

Shareable Navigation and Cold Start

The September 16 routing work moved view state into the address-bar hash. GitHub Pages can serve /Vex-Rank/ but does not provide the same dynamic route fallback as the application server. Hash routes preserve a cold-loaded team or event link without requiring server path routing. The September 17 fix explicitly tested a browser that had never opened the site, which catches a failure that an in-session navigation test can miss.

Current source excerpt: website/lib/routes.mjs, lines 9-35

```

 9  */
10
11 /** Build the address-bar hash for the current view and selection. */
12 export function routeToHash(view, team, event) {
13   if (view === 'team') return team?.number ? `#/teams/${encodeURIComponent(team.number)}` : '#/teams';
14   if (view === 'event') return event?.id ? `#/events/${encodeURIComponent(String(event.id))}` : '#/events';
15   if (view === 'home') return '#/';
16   return `#/${view}`;
17 }
18
19 /**
20  * Parse a hash into a view plus any identifier it carries.
21  * Returns null for anything unrecognised so the caller can ignore it rather
22  * than navigating somewhere arbitrary.
23  */
24 export function hashToRoute(hash) {
25   const parts = String(hash ?? '').replace(/^#\//, '').split('/').filter(Boolean);
26   if (!parts.length) return { view: 'home' };
27   let head, param;
28   try {
29     head = decodeURIComponent(parts[0]);
30     param = parts[1] ? decodeURIComponent(parts[1]) : undefined;
31   } catch {
32     return null; // malformed percent-encoding
33   }
34   if (head === 'events') return param ? { view: 'event', eventId: param } : { view: 'events' };
35   if (head === 'teams') return param ? { view: 'team', teamNumber: param } : { view: 'teams' };

```

The parser rejects malformed encoded hashes and distinguishes list views from detail IDs. Contributors should preserve those cases when adding a new view; a route that works only after clicking inside the app is incomplete.

Interaction Trials and Accessible Motion

The September 17 sequence is an unusually visible design trial: a handwriting face became a stroke-by-stroke SVG trace; the timing was slowed toward the final character, then accelerated by about 20 percent; stray ink from parked strokes was corrected, followed by a specific H stroke-order adjustment. The history shows iteration through observed visual defects rather than a single final effect. The component checks prefers-reduced-motion and displays the finished number without animation for that preference.

Current source excerpt: website/components/signed-number.tsx, lines 27-42

```

27   if (!svg) return;
28   const paths = [...svg.querySelectorAll('path')];
29   const characterCount = Array.from(text).length;
30   if (!paths.length) return;
31
32   // Reduced motion: leave it fully written rather than animating or hiding.
33   if (window.matchMedia?.('(prefers-reduced-motion: reduce)').matches) {
34     paths.forEach(path => { path.style.strokeDasharray = 'none'; path.style.strokeDashoffset = '0'; });
35     return;
36   }
37
38   const PEN_SPEED = 0.92; // ms per user unit, at the start of the signature
39   const PEN_LIFT = 56;    // pause between strokes, as the hand repositions
40   const FINAL_DRAG = 1.35; // extra slowness by the last character (2.35x total)
41   const animations: Animation[] = [];
42   let at = 180;           // let the profile header settle first

```

Reduced-motion behavior and before-paint setup are functional requirements. A future visual change should be inspected with animation enabled and reduced motion, because fixture tests do not establish visual timing or absence of a one-frame flash.

Worker Cache and Failure Semantics

The Worker canonicalizes allowed routes and query fields before building a shared D1 cache key. It adds the VCR model version to ranking and team requests, preventing a changed model from reusing an old cached score. Successful responses under 1.8 million encoded bytes are cached for 15 minutes. Larger responses still return to the visitor. A failed upstream call returns an error response and is not written as successful cached data.

Current source excerpt: [website/cloudflare/worker.ts](#), lines 13-32

```

13 // as in their route, or distinct requests can incorrectly share cached responses.
14 function canonicalUrl(request: Request) {
15   const incoming = new URL(request.url);
16   if (!/^\/api\/(archive-source(?:\/\d{1,10})?)?|events(?:\/\d{1,10})?|teams\/[0-9]{1,8}[A-Za-z0-9-
17 ]{0,12}|rankings|skills)$/.test(incoming.pathname)) return null;
18   const url = new URL(incoming.pathname, incoming.origin);
19   for (const key of ['season', 'teamId', 'division', 'page']) {
20     const value = incoming.searchParams.get(key);
21     if (value) {
22       if (!/^d{1,10}$/.test(value)) return null;
23       url.searchParams.set(key, value);
24     }
25   }
26   const mode=incoming.searchParams.get('mode');
27   if(mode){if(!['metadata','teams','matches'].includes(mode))return null;url.searchParams.set('mode',mode)}
28   // Model-version namespace prevents a new rating model from reusing old responses.
29   if (url.pathname === '/api/rankings' || url.pathname.startsWith('/api/teams/')) url.searchParams.set('model',
30     VCR_VERSION);
31   return url;
32 }
33
34 async function dispatch(url: URL) {

```

Adding a query parameter to a route requires adding it to canonicalUrl if it changes response content. Otherwise distinct requests can collide in the cache even though the route handler itself is correct.

The September 22 custom-domain commits changed the site base path and CORS origin. These are release-sensitive settings: the browser base path, Worker allowed origin, and actual deployed hostname must be checked together. CORS is a browser access rule and does not authenticate a public GET API.

The September 23 homepage work included a design reversal. PRs 26 and 27 merged a blog panel beside vertical Signature Events and a featured-event date. PR 28 then merged the restoration of the earlier horizontal event section and three feature blocks, while keeping the Discord link and date. PR 29 preserves the blog layout as an open, unmerged draft for a later decision. The restored horizontal layout is the behavior in main at this snapshot.

Current Automated Verification

On September 23 2026, the latest GitHub checkout at 6a03cd0 ran all website *.test.mjs fixtures with the bundled Node 24.19 runtime: 37 tests passed, zero failed, in 1.042 seconds. This suite covers archive eligibility/validation, event parsing, route encoding, directory search, rating decay, VCR settlement and audit, and historical score handling. These tests verify deterministic local behavior; they do not prove current upstream completeness or predictive calibration.

The static Pages build transformed 2,366 modules and succeeded in 1.46 seconds. Its output was 0.63 kB HTML, 225.86 kB CSS, and 666.72 kB JavaScript before gzip. The build warned that one chunk exceeded 500 kB. The most recent lint run in this revision exited with code 1, including a no-new-array error, unresolved Firebase function types in this website-level run, a team-directory type mismatch, and numerous explicit-any and other warnings. The CI workflow treats lint as advisory with continue-on-error, while build and tests are required steps.

Check	Result	Interpretation
node --test tests/*.test.mjs	37 passed; 0 failed	Current fixture suite passes
Static Vite Pages build	Success; 2,366 modules	Frontend bundles, with >500 kB warning
pnpm lint	Exit 1	Advisory CI debt; not a clean lint report
GitHub CI run 35845951399	Success for 6a03cd0	Remote required steps passed
GitHub test deployment run 35846097394	Success for 6a03cd0	Workflow reported successful deployment

GitHub Pages Firebase and Cloudflare Publishing Record

The September 14 deployment design first prepared a Firebase second-generation function and Firestore. Firestore rules/index deployment succeeded, but the API function could not be deployed under the Spark plan because its secret handling required Blaze. The project then connected a GitHub Pages static test build to a Cloudflare Worker and D1, merged in PR 4 on September 15. Firebase remains a recorded trial, not the current API host.

The deployment document still says GitHub Pages is not published and describes an earlier private-repository blocker. That status text is dated history: the GitHub repository was public in the September 23 API response, and the VEXRank test deployment workflow run 35846097394 completed successfully against 6a03cd0. The workflow includes tests, Worker health verification, a static build, artifact upload, and a Pages deploy. A successful workflow is evidence of the publishing process; independent end-user reachability and custom-domain routing should still be checked for an operational claim.

The source change on September 22 put the root path on vex-rank.com and allowed that origin in the Worker. The source-level release procedure is: merge reviewed code, let CI verify build/tests, confirm Worker and D1 health, dispatch the Pages workflow, verify its run and page URL, then test API/data views from the published origin. Archive rebuild and Worker deploy remain separate operations from publishing the static site.

Open Questions and Next Engineering Trials

Question	Evidence now	Trial or decision needed
Is VCR 3 predictive?	Scenario tests pass; no historical calibration result in checked-in record	Chronological holdout backtest with calibration and subgroup reports
Are every past season archives ready?	Manifest lists only season 197	Rebuild, validate and publish each configured season separately
Does the custom domain remove blocked API calls?	Root path/CORS code merged; Pages workflow succeeded	Test a fresh browser with strict blocker settings on deployed host
Is CI clean?	Build/tests pass; lint exits 1 and is advisory	Triage lint debt, then make it a required gate
Can the frontend load faster?	JS 665.87 kB before gzip and build warning	Measure route loading and split expensive views if worthwhile
Can teams submit photos/links?	Official shaped API lacks those fields	Design identity, moderation, storage and consent before accepting uploads

Part Eleven Local Design and Build Chronology

The following 49 commits come from the separate website Git repository in the shared workspace, beginning September 1. They document the early design and build sequence before the GitHub repository was created on September 9. A local commit date is an engineering record, not a GitHub publication date. The GitHub history in Part Twelve is separately identified.

Local work on 2026 09 01

The first day established the portal, official Event VEX connection, event filters, skills, live team rankings, team archives and event locking. The sequence moved from a usable front end to official data and then to team-specific views. These are local source changes; they do not prove a public launch on September 1.

Local time	Commit	Recorded change
01:50	f402ff84	Build VEXRank competition portal
12:19	d3063888	Add stat leaders and advanced event filters
12:43	17a2c9a0	Expand official VEX event filters
13:05	733ab658	Link country and event region filters
13:20	1e7749ec	Connect live Event.VEX data
13:53	e217f0c1	Add complete event profiles and calendar
14:18	6897935a	Replace demo leaderboard with live teams
17:33	c239990d	Add full Push Back season rankings
18:35	c8dd620d	Add official skills leaderboards
19:11	14386a93	Add complete team archive profiles
19:26	2643a8f4	Add season-based team histories
19:37	8f6ad8bd	Fix team event and award details
20:11	cd514164	Lock event rankings before competition
21:11	952d28d8	Fix event filters and Override label

Source: local website Git log. Commit hashes can be inspected in that repository; the descriptions above summarize subjects and chronological relationships rather than claiming undocumented user tests.

Local work on 2026 09 02

The second day concentrated on event detail: daily agendas, organizer data, results, skills standings, historical season filters, navigation persistence and rating trajectories. Several small commits corrected parsing assumptions after earlier implementations, an example of the test-and-revise loop in the engineering process.

Local time	Commit	Recorded change
02:37	f0ff33fc	Add complete event information hub
02:48	db61c0cf	Sync organizer event details from Event VEX
02:55	74f49249	Format event agenda into daily schedules
03:40	e06a03d1	Compact event agenda entries
12:04	b3b985d3	Split event results and add skills standings
14:30	e9bff2e1	Add past seasons to event calendar
14:38	3dc65c6f	Normalize varied event agenda formats
15:17	429cd29f	Persist navigation across page refreshes
15:26	d786b5b2	Fix event rankings matches and agenda parsing
15:35	949e8f37	Reconstruct table agendas and show match alliances
15:43	2b673941	Fix agenda rows scores and event links
16:11	a0c994ec	Label cancelled events correctly
16:58	85c65048	Expand world rankings to five seasons
17:01	3af9d91d	Use season filters for historical rankings
17:14	7690b201	Use real rankings with grade and location filters
17:37	5fc348ae	Fix ranking season cache and historical importer
17:52	a8cc347b	Keep rankings visible when grade data is unavailable
20:09	b8f87ee2	Limit world rankings to archived seasons
20:22	36eb8bfa	Preserve ranking context through team profiles
20:34	bad12290	Add interactive team rating trajectories

Source: local website Git log. Commit hashes can be inspected in that repository; the descriptions above summarize subjects and chronological relationships rather than claiming undocumented user tests.

Local work on 2026 09 03

The third day refined classification, official team IDs, loading speed, and elimination brackets. The sequence from first bracket view through connector layout, alliance order, division selector and final classification shows visual and data trials before a stable presentation.

Local time	Commit	Recorded change
00:06	db39f202	Preserve season-specific team grades
00:39	af66c0c9	Correct verified historical team grades
01:48	75664c01	Use official event levels for signature classification
01:53	6d36534a	Fix event team profile navigation
02:00	d507b770	Update event status colors
02:11	511dc1fa	Speed up event and team loading
02:28	731b5240	Fix event team profile loading
20:03	e5fa56ef	Use official team IDs for event profiles
20:45	a43b8277	Add elimination bracket view
21:04	a60cacd3	Polish elimination bracket connectors
21:14	5ab25ad7	Fix bracket match layout and alliance order
22:16	39ea8f1f	Add division result selector and themed scrollbars
23:24	ba76aba3	Center overall finals and label divisions
23:39	36d80614	Fix elimination data and event classification

Source: local website Git log. Commit hashes can be inspected in that repository; the descriptions above summarize subjects and chronological relationships rather than claiming undocumented user tests.

Local work on 2026 09 11

The September 11 inspection repaired Worlds results, bracket replay placement and loading reliability after the early event-hub implementation. The dated inspection note in `website/tests/INSPECTION.md` records both verified fixes and remaining data limits.

Local time	Commit	Recorded change
20:56	ddc56804	Fix complete event results, bracket replay slots and loading reliability

Source: local website Git log. Commit hashes can be inspected in that repository; the descriptions above summarize subjects and chronological relationships rather than claiming undocumented user tests.

Part Twelve GitHub Commit by Commit Engineering Record

Each non-merge GitHub commit through 6a03cd0 has a dated record on the following pages. Merge commits are indexed in Part Thirteen with their pull requests and in the complete commit register. Changed-file counts and line totals come from Git diffs. An engineering implication is identified as an inference when the commit itself does not contain a design note. Current tests validate the final checkout, not every intermediate commit retroactively.

GitHub change 01 2026-09-09 3a633af

Initial commit

Field	Recorded value
Commit time	September 09 2026 at 13:43 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +2 / -0 lines
Full identifier	3a633af7c6e7d58efc5e5e61f060290faecf3cd2

Recorded change: Initial commit. The largest file-level change was README.md (+2/-0).

Changed path	+	-
README.md	2	0

Representative added source from README.md:

- + # Vex-Rank
- + Better event.vex for the competitors

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/3a633af7c6e7d58efc5e5e61f060290faecf3cd2>

GitHub change 02 2026-09-09 63877ed

Add VEX ranking website, algorithm PDFs, and project documentation

Field	Recorded value
Commit time	September 09 2026 at 13:50 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	94 changed paths; +17,381 / -0 lines
Full identifier	63877ed5fb611a110389d68937291f57c35918ee

Recorded change: Add VEX ranking website, algorithm PDFs, and project documentation. The largest file-level change was website/pnpm-lock.yaml (+6501/-0).

Changed path	+	-
website/pnpm-lock.yaml	6501	0
website/components/ui/sidebar.tsx	723	0
build_full_algorithm_pdf.py	525	0
website/app/page.tsx	506	0
output/pdf/vex_competitive_rating_full_algorithm.pdf	468	0
website/components/ui/chart.tsx	373	0
build_vcr_pdf.py	324	0

The diff also touches 87 additional paths; use the full linked commit for the complete file list.

Representative added source from docs/SETUP.md:

```
+ # VEX Ranking
+ VEX ranking website, ranking scripts, and algorithm documentation.
+ ## Project contents
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/63877ed5fb611a110389d68937291f57c35918ee>

GitHub change 03 2026-09-09 55481c6

Add algorithm documentation links

Field	Recorded value
Commit time	September 09 2026 at 13:58 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +9 / -1 lines
Full identifier	55481c6ed7b4471edd2f635d3cb68717f6d6e0df

Recorded change: Add algorithm documentation links. The largest file-level change was README.md (+9/-1).

Changed path	+	-
README.md	9	1

Representative added source from README.md:

```
+ Better event.vex for competitors.
+ ## Documentation
+ - [Project setup and structure](docs/SETUP.md)
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/55481c6ed7b4471edd2f635d3cb68717f6d6e0df>

GitHub change 04 2026-09-09 62b834c

Link algorithm flowchart in setup documentation

Field	Recorded value
Commit time	September 09 2026 at 13:58 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +1 / -0 lines
Full identifier	62b834c2de4bd9a61fb8b2f34188d8ef96781dc6

Recorded change: Link algorithm flowchart in setup documentation. The largest file-level change was docs/SETUP.md (+1/-0).

Changed path	+	-
docs/SETUP.md	1	0

Representative added source from docs/SETUP.md:

```
+ - [Algorithm flowchart (Lucidchart)](https://lucid.app/lucidchart/155358d5-0829-4c12-bbc1-64ef25e11112/edit?viewport_loc=-510%2C10%2C2540...
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/62b834c2de4bd9a61fb8b2f34188d8ef96781dc6>

GitHub change 05 2026-09-09 9b32fda

Expand README with VCR algorithm overview

Field	Recorded value
Commit time	September 09 2026 at 14:02 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +104 / -4 lines
Full identifier	9b32fda68a1a009a0e0b2a4a88e3bce6163327af

Recorded change: Expand README with VCR algorithm overview. The largest file-level change was README.md (+104/-4).

Changed path	+	-
README.md	104	4

Representative added source from README.md:

- + Vex-Rank is a VEX V5 ranking and statistics project built around **VCR 2.0 (VEX Competitive Rating)**. VCR estimates current competitive ...
- + ## How VCR works
- + Each team has an internal strength rating (μ) centered at 1500 and a rating deviation (σ) that represents uncertainty. The public s...

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/9b32fda68a1a009a0e0b2a4a88e3bce6163327af>

GitHub change 06 2026-09-11 a765593

Fix Worlds results and improve website data/loading reliability

Field	Recorded value
Commit time	September 11 2026 at 20:57 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	14 changed paths; +357 / -209 lines
Full identifier	a76559316eb5cf7a4a2909bb37399ce0b4ff352c

Recorded change: Fix Worlds results and improve website data/loading reliability. The largest file-level change was website/app/page.tsx (+89/-88).

Changed path	+	-
website/app/page.tsx	89	88
website/tests/data-regressions.test.mjs	76	0
website/app/api/events/[id]/route.ts	11	59
website/lib/vex-api.ts	58	0
website/app/api/events/route.ts	5	33
website/app/api/rankings/route.ts	13	14
website/lib/agenda.ts	27	0

The diff also touches 7 additional paths; use the full linked commit for the complete file list.

Representative added source from website/app/api/events/[id]/route.ts:

```
+ import { vexCollection, vexJson, mapLimit } from '@lib/vex-api';
+ const getAllTeams = (id:string, headers:Record<string,string>) =>
  vexCollection(`${API_ROOT}/events/${id}/teams`,headers);
+ const getAll = (url:string, headers:Record<string,string>) => vexCollection(`${API_ROOT}${url}`,headers);
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/a76559316eb5cf7a4a2909bb37399ce0b4ff352c>

GitHub change 07 2026-09-14 261ebc2

Prepare GitHub Pages and Firebase test deployment; document Spark billing blocker

Field	Recorded value
Commit time	September 14 2026 at 14:54 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	19 changed paths; +2,558 / -6 lines
Full identifier	261ebc24793c3a1b4fe52cf0469123cd62830b80

Recorded change: Prepare GitHub Pages and Firebase test deployment; document Spark billing blocker. The largest file-level change was website/firebase/functions/pnpm-lock.yaml (+2303/-0).

Changed path	+	-
website/firebase/functions/pnpm-lock.yaml	2303	0
website/firebase/functions/src/index.ts	79	0
.github/workflows/github-pages-test.yml	44	0
website/firebase/DEPLOYMENT.md	37	0
website/firebase/functions/src/cloudflare-adapter.ts	20	0
website/vite.pages.config.ts	18	0
website/app/page.tsx	7	6

The diff also touches 12 additional paths; use the full linked commit for the complete file list.

Representative added source from .github/workflows/github-pages-test.yml:

```
+ name: VEXRank test deployment
+ on:
+ workflow_dispatch:
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/261ebc24793c3a1b4fe52cf0469123cd62830b80>

GitHub change 08 2026-09-14 f1804c3

Connect GitHub Pages test build to live Cloudflare Workers and D1 backend

Field	Recorded value
Commit time	September 14 2026 at 21:22 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	8 changed paths; +146 / -3 lines
Full identifier	f1804c376dbb26d985f23c8673c87c0f0940642c

Recorded change: Connect GitHub Pages test build to live Cloudflare Workers and D1 backend. The largest file-level change was website/cloudflare/worker.ts (+68/-0).

Changed path	+	-
website/cloudflare/worker.ts	68	0
website/cloudflare/DEPLOYMENT.md	51	0
website/cloudflare/wrangler.json	11	0
website/cloudflare/schema.sql	7	0
.github/workflows/github-pages-test.yml	4	1
website/vite.pages.config.ts	2	1
website/firebase/DEPLOYMENT.md	2	0

The diff also touches 1 additional paths; use the full linked commit for the complete file list.

Representative added source from .github/workflows/github-pages-test.yml:

```
+ - name: Check data regressions
+ run: node --test tests/data-regressions.test.mjs
+ working-directory: website
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/f1804c376dbb26d985f23c8673c87c0f0940642c>

GitHub change 09 2026-09-15 88eb420

Keep homepage Signature events independent of calendar season filters

Field	Recorded value
Commit time	September 15 2026 at 13:27 UTC-04:00
Author	Eason Li
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +18 / -2 lines
Full identifier	88eb4200ad7061fde58fa023020e4305e0f3b437

Recorded change: Keep homepage Signature events independent of calendar season filters. The largest file-level change was website/app/page.tsx (+18/-2).

Changed path	+	-
website/app/page.tsx	18	2

Representative added source from website/app/page.tsx:

```
+ const eventCalendarCache=new Map<string,any>();
+ const [homeEvents, setHomeEvents] = useState<any[]>([]);
+ cachedJson(`/api/events?season=${seasonId}&classification=v49`,eventCalendarCache,seasonId).then((payload:any) =>
{
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/88eb4200ad7061fde58fa023020e4305e0f3b437>

GitHub change 10 2026-09-15 8b114ef

docs: link VCR 3.0 implementation

Field	Recorded value
Commit time	September 15 2026 at 13:51 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +2 / -0 lines
Full identifier	8b114efc08d375361e1a5988e9e95088536eec94

Recorded change: docs: link VCR 3.0 implementation. The largest file-level change was README.md (+2/-0).

Changed path	+	-
README.md	2	0

Representative added source from README.md:

```
+ - [VCR 3.0 candidate: corrected event settlement](docs/VCR-3.0-revision.md) – implemented in the live ranking API,
team history, and hist...
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/8b114efc08d375361e1a5988e9e95088536eec94>

GitHub change 11 2026-09-15 979d37d

feat: apply VCR 3.0 to global rankings

Field	Recorded value
Commit time	September 15 2026 at 13:51 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +8 / -15 lines
Full identifier	979d37dc70466576fb49415c0a5f1507b8121da0

Recorded change: feat: apply VCR 3.0 to global rankings. The largest file-level change was website/app/api/rankings/route.ts (+8/-15).

Changed path	+	-
website/app/api/rankings/route.ts	8	15

Representative added source from website/app/api/rankings/route.ts:

```
+ import { processCompletedEvent, VCR_VERSION } from '@lib/vcr3.mjs';
+ return data.map(match=>({...match,eventId:event.id,gradeHint:gradeFromContext(`${event.name}
${division.name}`)}));
+ const historyByTeam=new Map<number,any[]>(),matchesByEvent=new Map<number,any[]>();for(const match of
matches){if(!matchesByEvent.has(mat...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/979d37dc70466576fb49415c0a5f1507b8121da0>

GitHub change 12 2026-09-15 e8e1600

feat: apply VCR 3.0 to team history

Field	Recorded value
Commit time	September 15 2026 at 13:51 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +6 / -12 lines
Full identifier	e8e160013a24a9d541db4418bacc1735e09b9338

Recorded change: feat: apply VCR 3.0 to team history. The largest file-level change was website/app/api/teams/[number]/route.ts (+6/-12).

Changed path	+	-
website/app/api/teams/[number]/route.ts	6	12

Representative added source from website/app/api/teams/[number]/route.ts:

```
+ import { processCompletedEvent, VCR_VERSION } from '@lib/vcr3.mjs';
+ const ratingStates=new Map<number,any>(),ratingHistoryByTeam=new Map<number,any[]>();
+
+ if(eventMatches.length)processCompletedEvent({event,matches:eventMatches,states:ratingStates,historyByTeam:ratingHistoryByTeam});
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/e8e160013a24a9d541db4418bacc1735e09b9338>

GitHub change 13 2026-09-15 90ac608

feat: apply VCR 3.0 to historical rankings

Field	Recorded value
Commit time	September 15 2026 at 13:52 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +8 / -8 lines
Full identifier	90ac608f0cdf8837cafb1aa601489cc85a38f27c

Recorded change: feat: apply VCR 3.0 to historical rankings. The largest file-level change was website/scripts/build-historical-ranking.mjs (+8/-8).

Changed path	+	-
website/scripts/build-historical-ranking.mjs	8	8

Representative added source from website/scripts/build-historical-ranking.mjs:

```
+ import { processCompletedEvent, VCR_VERSION } from '../lib/vcr3.mjs';
+ const matchGroups=await mapLimit(jobs,1,async({event,division})=>{const
url=`${root}/events/${event.id}/divisions/${division.id}/matches?...
+ const states=new Map(),historyByTeam=new Map(),matchesByEvent=new Map();for(const match of
matches){if(!matchesByEvent.has(match.eventId))...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/90ac608f0cdf8837cafb1aa601489cc85a38f27c>

GitHub change 14 2026-09-15 a7feaf4

feat: add VCR 3.0 reference event model

Field	Recorded value
Commit time	September 15 2026 at 13:52 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +61 / -0 lines
Full identifier	a7feaf427933e39253f4006410df91c628e95133

Recorded change: feat: add VCR 3.0 reference event model. The largest file-level change was algorithm/vcr-event.mjs (+61/-0).

Changed path	+	-
algorithm/vcr-event.mjs	61	0

Representative added source from algorithm/vcr-event.mjs:

```
+ // VCR 3.0 candidate: pure event settlement, with caller-supplied frozen evidence.
+ export const EVENT_WEIGHTS = Object.freeze({ C: .70, B: .82, A: .95, 'Bronze S': 1.15, 'Silver S': 1.35, 'Gold S':
1.60, Worlds: 1.85 });
+ const clamp = (x, lo, hi) => Math.max(lo, Math.min(hi, x));
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/a7feaf427933e39253f4006410df91c628e95133>

GitHub change 15 2026-09-15 fad6a89

test: add VCR 3.0 invariants

Field	Recorded value
Commit time	September 15 2026 at 13:52 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +55 / -0 lines
Full identifier	fad6a898da02706db70a103fa1277708b4f69053

Recorded change: test: add VCR 3.0 invariants. The largest file-level change was algorithm/vcr-event.test.mjs (+55/-0).

Changed path	+	-
algorithm/vcr-event.test.mjs	55	0

Representative added source from algorithm/vcr-event.test.mjs:

```
+ import test from 'node:test';
+ import assert from 'node:assert/strict';
+ import { EVENT_WEIGHTS, settleEvent, matchEvidence, expectedResult } from './vcr-event.mjs';
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/fad6a898da02706db70a103fa1277708b4f69053>

GitHub change 16 2026-09-15 0454fda

docs: publish VCR 3.0 algorithm specification

Field	Recorded value
Commit time	September 15 2026 at 13:52 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +121 / -0 lines
Full identifier	0454fda93822e0fb55f3961e75b781d76457d908

Recorded change: docs: publish VCR 3.0 algorithm specification. The largest file-level change was docs/VCR-3.0-revision.md (+121/-0).

Changed path	+	-
docs/VCR-3.0-revision.md	121	0

Representative added source from docs/VCR-3.0-revision.md:

```
+ # VCR 3.0 candidate: event settlement and achievement
+ Date: 2026-09-15. Status: implemented in the website calculation paths and scenario-tested; not deployed or
historically calibrated.
+ ## Source and findings
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/0454fda93822e0fb55f3961e75b781d76457d908>

GitHub change 17 2026-09-15 7f43b66

docs: record VCR 3.0 candidate provenance

Field	Recorded value
Commit time	September 15 2026 at 13:52 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +16 / -0 lines
Full identifier	7f43b66c2e59be19417ded3142f6f11eb42b51f1

Recorded change: docs: record VCR 3.0 candidate provenance. The largest file-level change was versions/vcr-3.0-candidate.md (+16/-0).

Changed path	+	-
versions/vcr-3.0-candidate.md	16	0

Representative added source from versions/vcr-3.0-candidate.md:

```
+ # VCR 3.0 candidate
+ Status: implemented in website calculation paths; not deployed or historically calibrated.
+ Predecessor: VCR 2.0 on `main`. This version remains isolated on its own branch and pull request so the previous
  model and history remain...
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/7f43b66c2e59be19417ded3142f6f11eb42b51f1>

GitHub change 18 2026-09-15 035e21b

feat: add shared VCR 3.0 processor

Field	Recorded value
Commit time	September 15 2026 at 13:53 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +53 / -0 lines
Full identifier	035e21ba48254a720b6c9952b8be8be462b0466c

Recorded change: feat: add shared VCR 3.0 processor. The largest file-level change was website/lib/vcr3.mjs (+53/-0).

Changed path	+	-
website/lib/vcr3.mjs	53	0

Representative added source from website/lib/vcr3.mjs:

```
+ export const VCR_VERSION = 'VCR-3.0-candidate.2';
+ export const EVENT_WEIGHTS = Object.freeze({ C: .70, B: .82, A: .95, 'Bronze S': 1.15, 'Silver S': 1.35, 'Gold S': 1.60, Worlds: 1.85 });
+ const clamp=(x,lo,hi)=>Math.max(lo,Math.min(hi,x));
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/035e21ba48254a720b6c9952b8be8be462b0466c>

GitHub change 19 2026-09-15 4a2186b

test: cover live VCR 3.0 processor

Field	Recorded value
Commit time	September 15 2026 at 13:53 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +6 / -0 lines
Full identifier	4a2186bd497a49d4ee1ab64ddb549c0e620bcc9

Recorded change: test: cover live VCR 3.0 processor. The largest file-level change was website/tests/vcr3.test.mjs (+6/-0).

Changed path	+	-
website/tests/vcr3.test.mjs	6	0

Representative added source from website/tests/vcr3.test.mjs:

```
+ import test from 'node:test'; import assert
  from 'node:assert/strict'; import {EVENT_WEIGHTS, settleEvent, matchEvidence, processCompletedEvent, V...
+ const base=(o={})=>({rating:1400,tier:'Gold
  S',completed:true,matches:Array.from({length:8},()=>matchEvidence({actual:0,expected:.8,margi...
+ test('unexpected champions gain at every tier',()=>{for(const tier of
  Object.keys(EVENT_WEIGHTS))assert.ok(settleEvent(base({tier})).delt...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/4a2186bd497a49d4ee1ab64ddb549c0e620bcc9>

GitHub change 20 2026-09-15 cb5a298

fix: verify VCR 3.0 deployment and valid alliance credit

Field	Recorded value
Commit time	September 15 2026 at 14:02 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +4 / -1 lines
Full identifier	cb5a298ed9ea9166be40d331cc97318e0296d688

Recorded change: fix: verify VCR 3.0 deployment and valid alliance credit. The largest file-level change was website/cloudflare/worker.ts (+4/-1).

Changed path	+	-
website/cloudflare/worker.ts	4	1

Representative added source from website/cloudflare/worker.ts:

```
+ import { VCR_VERSION } from '../lib/vcr3.mjs';
+ if (url.pathname === '/api/rankings' || url.pathname.startsWith('/api/teams/')) url.searchParams.set('model',
VCR_VERSION);
+ return cors(Response.json({ status: 'ok', environment: 'test', database: 'D1', modelVersion: VCR_VERSION }));
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/cb5a298ed9ea9166be40d331cc97318e0296d688>

GitHub change 21 2026-09-15 d0bf136

fix: verify VCR 3.0 deployment and valid alliance credit

Field	Recorded value
Commit time	September 15 2026 at 14:02 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +2 / -2 lines
Full identifier	d0bf136758ea959b5002deec275921e73fed953c

Recorded change: fix: verify VCR 3.0 deployment and valid alliance credit. The largest file-level change was website/lib/vcr3.mjs (+2/-2).

Changed path	+	-
website/lib/vcr3.mjs	2	2

Representative added source from website/lib/vcr3.mjs:

```
+ if(![0,.5,1].includes(actual))throw new RangeError('Invalid
actual');finite(expected,'expected',0,1);finite(margin,'margin');finite(score...
+ const matchComponent=rows=>{let total=0,mass=0;for(const row of rows){finite(row.residual,'residual',-
2.5,2.5);finite(row.reliability,'ma...
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/d0bf136758ea959b5002deec275921e73fed953c>

GitHub change 22 2026-09-15 32028f1

fix: verify VCR 3.0 deployment and valid alliance credit

Field	Recorded value
Commit time	September 15 2026 at 14:02 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +2 / -0 lines
Full identifier	32028f15a2dd5108fa9a503278f34bd42aee3858

Recorded change: fix: verify VCR 3.0 deployment and valid alliance credit. The largest file-level change was website/tests/vcr3.test.mjs (+2/-0).

Changed path	+	-
website/tests/vcr3.test.mjs	2	0

Representative added source from website/tests/vcr3.test.mjs:

```
+ test('single active robot can receive all alliance credit',()=>{const
evidence=matchEvidence({actual:1,expected:0,margin:100,scoreScale:1...
+ test('complementary shares tolerate ordinary floating-point
rounding',()=>{assert.doesNotThrow(()=>matchEvidence({actual:0,expected:.9,ma...
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/32028f15a2dd5108fa9a503278f34bd42aee3858>

GitHub change 23 2026-09-15 3a09c7c

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +33 / -0 lines
Full identifier	3a09c7c17edb441c6ce7ab227dbdb6204235817c

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/cloudflare/archive-source.ts (+33/-0).

Changed path	+	-
website/cloudflare/archive-source.ts	33	0

Representative added source from website/cloudflare/archive-source.ts:

```
+ import { vexCollection, vexJson, mapLimit } from '../lib/vex-api';
+ const root='https://events.vex.com/api/v2';
+ const historicalSeasons=new Set([197,190,181,173]);
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/3a09c7c17edb441c6ce7ab227dbdb6204235817c>

GitHub change 24 2026-09-15 49984c1

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +18 / -4 lines
Full identifier	49984c130179399a15477ed7edfcc56ef5b1bc6f

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/cloudflare/worker.ts (+18/-4).

Changed path	+	-
website/cloudflare/worker.ts	18	4

Representative added source from website/cloudflare/worker.ts:

```
+ import { archiveSource } from './archive-source';
+ interface Env { DB: D1Database; ROBOT_EVENTS_API_TOKEN: string; ARCHIVES: Fetcher }
+ if (!/^\/api\/(archive-source(?:\/\d{1,10})?)|events(?:\/\d{1,10})?|teams\/[0-9]{1,8}[A-Za-z0-9-
]{0,12}|rankings|skills)$/.test(incoming.p...
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/49984c130179399a15477ed7edfcc56ef5b1bc6f>

GitHub change 25 2026-09-15 745214f

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +2 / -0 lines
Full identifier	745214f38bf3db3e44be7c8c90d9a1eb98076c19

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/cloudflare/wrangler.json (+2/-0).

Changed path	+	-
website/cloudflare/wrangler.json	2	0

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/745214f38bf3db3e44be7c8c90d9a1eb98076c19>

GitHub change 26 2026-09-15 54d8bb8

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +12 / -0 lines
Full identifier	54d8bb88fa193846a9148eafb960c1e60eb198f8

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/lib/archive-matches.mjs (+12/-0).

Changed path	+	-
website/lib/archive-matches.mjs	12	0

Representative added source from website/lib/archive-matches.mjs:

```
+ // Historical upstream records often say scored:false despite recorded scores.
+ // Use actual score evidence, and exclude ambiguous 0-0 placeholders.
+ export function historicalMatches(divisions) {
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/54d8bb88fa193846a9148eafb960c1e60eb198f8>

GitHub change 27 2026-09-15 deb01cc

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +12 / -0 lines
Full identifier	deb01cc1dcdb1d6d0e9a1607d5a72a6a67dcdd6e

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/lib/ranking-seasons.mjs (+12/-0).

Changed path	+	-
website/lib/ranking-seasons.mjs	12	0

Representative added source from website/lib/ranking-seasons.mjs:

```
+ export const RANKING_SEASONS = Object.freeze({
+ '2026-27 Override': {id:204,archive:null},
+ '2025-26 Push Back': {id:197,archive:'/rankings-2025-26-vcr3.json'},
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/deb01cc1dcdb1d6d0e9a1607d5a72a6a67dcdd6e>

GitHub change 28 2026-09-15 c2182cf

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:03 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +3 / -1 lines
Full identifier	c2182cf032de413de8e0019597c7ba2594197eff

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/lib/client-fetch.ts (+3/-1).

Changed path	+	-
website/lib/client-fetch.ts	3	1

Representative added source from website/lib/client-fetch.ts:

```
+ const archive=/^\rankings-20\d{2}-\d{2}-vcr3\.json$/.test(path);
+ const
url=path.startsWith('/api/')||archive&&apiBase?`${apiBase}${path}`:path.startsWith('/')?`${assetBase}${path.slice(1)}`:path;
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/c2182cf032de413de8e0019597c7ba2594197eff>

GitHub change 29 2026-09-15 8f89ed5

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:04 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +21 / -12 lines
Full identifier	8f89ed5e632f7ff50295c28a1dc7418d347e95f2

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/app/page.tsx (+21/-12).

Changed path	+	-
website/app/page.tsx	21	12

Representative added source from website/app/page.tsx:

```
+ import { RANKING_SEASONS, validateArchive } from '@lib/ranking-seasons.mjs';
+ function useAvailableRankingSeasons(){
+   const [available,setAvailable]=useState(['2026-27 Override','2025-26 Push Back']);
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/8f89ed5e632f7ff50295c28a1dc7418d347e95f2>

GitHub change 30 2026-09-15 264932c

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:04 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +72 / -0 lines
Full identifier	264932ca31a1c080e3b4d08529efff37e17d8690

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/scripts/rebuild-season-archives.mjs (+72/-0).

Changed path	+	-
website/scripts/rebuild-season-archives.mjs	72	0

Representative added source from website/scripts/rebuild-season-archives.mjs:

```
+ import { mkdir, readFile, writeFile, rename } from 'node:fs/promises';
+ import { resolve } from 'node:path';
+ import { processCompletedEvent, VCR_VERSION } from '../lib/vcr3.mjs';
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/264932ca31a1c080e3b4d08529efff37e17d8690>

GitHub change 31 2026-09-15 90c8fda

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:04 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +21 / -0 lines
Full identifier	90c8fda86f19d7aec26f5f7773183b77469c955b

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/scripts/prepare-archive-assets.mjs (+21/-0).

Changed path	+	-
website/scripts/prepare-archive-assets.mjs	21	0

Representative added source from website/scripts/prepare-archive-assets.mjs:

```
+ import { mkdir, readFile, writeFile } from 'node:fs/promises';
+ import { createHash } from 'node:crypto';
+ import { RANKING_SEASONS, validateArchive } from '../lib/ranking-seasons.mjs';
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/90c8fda86f19d7aec26f5f7773183b77469c955b>

GitHub change 32 2026-09-15 a6a709c

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:04 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +17 / -0 lines
Full identifier	a6a709c6d4b1041c5b8ba44161d688cc2cc09dba

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/tests/archive.test.mjs (+17/-0).

Changed path	+	-
website/tests/archive.test.mjs	17	0

Representative added source from website/tests/archive.test.mjs:

```
+ import test from 'node:test';
+ import assert from 'node:assert/strict';
+ import { historicalMatches } from '../lib/archive-matches.mjs';
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/a6a709c6d4b1041c5b8ba44161d688cc2cc09dba>

GitHub change 33 2026-09-15 123f488

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:04 UTC-04:00
Author	Eason Li
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +5 / -0 lines
Full identifier	123f488df3769f971caf1ad6ce39147d17459677

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was website/docs/HISTORICAL-VCR3.md (+5/-0).

Changed path	+	-
website/docs/HISTORICAL-VCR3.md	5	0

Representative added source from website/docs/HISTORICAL-VCR3.md:

```
+ Get-Content:
+ Line |
+ 2 | Get-Content -Raw -LiteralPath 'website/docs/HISTORICAL-VCR3.md'
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/123f488df3769f971caf1ad6ce39147d17459677>

GitHub change 34 2026-09-15 25363ed

feat: replay and publish versioned VCR 3.0 historical archives

Field	Recorded value
Commit time	September 15 2026 at 20:05 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +24 / -0 lines
Full identifier	25363edcbcef4e6448d81a31242b0a440b1d7aef

Recorded change: feat: replay and publish versioned VCR 3.0 historical archives. The largest file-level change was versions/vcr3-historical-archives.json (+24/-0).

Changed path	+	-
versions/vcr3-historical-archives.json	24	0

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/25363edcbcef4e6448d81a31242b0a440b1d7aef>

GitHub change 35 2026-09-15 14a75ca

fix: handle historical archive validation before state updates

Field	Recorded value
Commit time	September 15 2026 at 20:06 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	14a75ca23e41861ae2e9dfaaa9e1ea425d66b586

Recorded change: fix: handle historical archive validation before state updates. The largest file-level change was website/app/page.tsx (+1/-1).

Changed path	+	-
website/app/page.tsx	1	1

Representative added source from website/app/page.tsx:

```
+ useEffect(()=>{if(season==='2026-27 Override'||archives[season])return;let active=true;setArchiveError('');const
base=RANKING_SEASONS[sea...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/14a75ca23e41861ae2e9dfaaa9e1ea425d66b586>

GitHub change 36 2026-09-15 16e8bdd

perf: pace historical requests and resume from cached pages

Field	Recorded value
Commit time	September 15 2026 at 20:10 UTC-04:00
Author	Eason Li
Engineering area	Rating and data processing
Diff scale	1 changed paths; +17 / -5 lines
Full identifier	16e8bdd0246913159091f3e57c31802c61bd3ccf

Recorded change: perf: pace historical requests and resume from cached pages. The largest file-level change was website/scripts/rebuild-season-archives.mjs (+17/-5).

Changed path	+	-
website/scripts/rebuild-season-archives.mjs	17	5

Representative added source from website/scripts/rebuild-season-archives.mjs:

```
+ import { createHash } from 'node:crypto';
+ const pageCache=resolve('.ranking-cache/archive-pages');await mkdir(pageCache,{recursive:true});
+ let nextRequest=0;
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/16e8bdd0246913159091f3e57c31802c61bd3ccf>

GitHub change 37 2026-09-16 e1c5ec8

docs: explain ranking internals and contributor workflow

Field	Recorded value
Commit time	September 16 2026 at 01:42 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	13 changed paths; +203 / -0 lines
Full identifier	e1c5ec8631e3c0d9a16270a0962ee14ce8b2cc8d

Recorded change: docs: explain ranking internals and contributor workflow. The largest file-level change was CONTRIBUTING.md (+93/-0).

Changed path	+	-
CONTRIBUTING.md	93	0
website/lib/vcr3.mjs	46	0
website/scripts/rebuild-season-archives.mjs	14	0
website/lib/agenda.ts	8	0
website/app/api/rankings/route.ts	7	0
website/lib/vex-api.ts	7	0
website/cloudflare/worker.ts	6	0

The diff also touches 6 additional paths; use the full linked commit for the complete file list.

Representative added source from CONTRIBUTING.md:

```
+ # Contributing to Vex-Rank
+ You can develop this project with a text editor, Node.js, pnpm and Git. No AI tools are required. Start with the
+ source map below, reprod...
+ ## Local setup
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/e1c5ec8631e3c0d9a16270a0962ee14ce8b2cc8d>

GitHub change 38 2026-09-16 7b02a13

Update team search examples to 8829C and 55288A

Field	Recorded value
Commit time	September 16 2026 at 02:35 UTC-04:00
Author	Eason Li
Engineering area	Competition data and presentation
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	7b02a137619207984fc993aada5dd21d974cac89

Recorded change: Update team search examples to 8829C and 55288A. The largest file-level change was website/app/page.tsx (+1/-1).

Changed path	+	-
website/app/page.tsx	1	1

Representative added source from website/app/page.tsx:

```
+ <div className="mt-8 grid gap-3 border border-white/10 bg-[#101319] p-4 md:grid-cols-[2fr_1fr_1fr]"><label
  className="relative"><Search c...
```

Design implication: The affected view depends on heterogeneous official event records. Source IDs, divisions, missing positions, and partial upstream pages require explicit handling.

Verification to preserve: Inspect an ordinary event and a Worlds multi-division example; run the event regression fixtures.

Source: <https://github.com/easonli29/Vex-Rank/commit/7b02a137619207984fc993aada5dd21d974cac89>

GitHub change 39 2026-09-16 b8b0337

Search full V5RC team directory with country, region and grade filters

Field	Recorded value
Commit time	September 16 2026 at 11:49 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	9 changed paths; +184 / -19 lines
Full identifier	b8b03376a5a5b24607d215ce7d9b28e2cba7c8a7

Recorded change: Search full V5RC team directory with country, region and grade filters. The largest file-level change was website/components/team-directory.tsx (+51/-0).

Changed path	+	-
website/components/team-directory.tsx	51	0
website/scripts/build-team-directory.mjs	41	0
website/tests/team-directory.test.mjs	27	0
website/app/page.tsx	7	18
website/lib/team-directory.mjs	23	0
website/cloudflare/TEAM-DIRECTORY.md	20	0
website/cloudflare/team-directory-manifest.json	9	0

The diff also touches 2 additional paths; use the full linked commit for the complete file list.

Representative added source from .github/workflows/github-pages-test.yml:

```
+ run: node --test tests/data-regressions.test.mjs tests/team-directory.test.mjs
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/b8b03376a5a5b24607d215ce7d9b28e2cba7c8a7>

GitHub change 40 2026-09-16 c3e01a9

feat: editorial redesign, scroll motion, and capped result lists

Field	Recorded value
Commit time	September 16 2026 at 16:04 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	2 changed paths; +306 / -73 lines
Full identifier	c3e01a933721b79e957e9502b8ae7e0d90bffffb1

Recorded change: feat: editorial redesign, scroll motion, and capped result lists. The largest file-level change was website/app/page.tsx (+129/-65).

Changed path	+	-
website/app/page.tsx	129	65
website/app/globals.css	177	8

Representative added source from website/app/globals.css:

```
+ @import
  url('https://fonts.googleapis.com/css2?family=Hanken+Grotesk:ital,wght@0,300..800;1,300..800&family=Source+Serif+4:opsz,wght@8..6...
+ --font-sans: var(--font-claude-sans);
+ --font-mono: var(--font-claude-mono);
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/c3e01a933721b79e957e9502b8ae7e0d90bffffb1>

GitHub change 41 2026-09-16 0f73d15

fix: resolve merge conflict left unresolved in page.tsx

Field	Recorded value
Commit time	September 16 2026 at 16:17 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	1 changed paths; +0 / -21 lines
Full identifier	0f73d1519fc58695ed8a18e799eb734c3d88800a

Recorded change: fix: resolve merge conflict left unresolved in page.tsx. The largest file-level change was website/app/page.tsx (+0/-21).

Changed path	+	-
website/app/page.tsx	0	21

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/0f73d1519fc58695ed8a18e799eb734c3d88800a>

GitHub change 42 2026-09-16 9b16bb7

chore: ignore local .claude agent config

Field	Recorded value
Commit time	September 16 2026 at 16:23 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	1 changed paths; +1 / -0 lines
Full identifier	9b16bb7a3cc9fceed0fe7bb1a9287b17c48c3739

Recorded change: chore: ignore local .claude agent config. The largest file-level change was .gitignore (+1/-0).

Changed path	+	-
.gitignore	1	0

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/9b16bb7a3cc9fceed0fe7bb1a9287b17c48c3739>

GitHub change 43 2026-09-16 b2a127e

fix: stop vinext treating the static-site entry as a Pages router

Field	Recorded value
Commit time	September 16 2026 at 16:35 UTC-04:00
Author	n3veTz
Engineering area	Publishing and service boundary
Diff scale	8 changed paths; +12 / -12 lines
Full identifier	b2a127eeb7a9607b85a85ef636e931a84136c4b7

Recorded change: fix: stop vinext treating the static-site entry as a Pages router. The largest file-level change was website/pages/main.tsx (+0/-6).

Changed path	+	-
website/pages/main.tsx	0	6
website/static-site/main.tsx	6	0
CONTRIBUTING.md	1	1
website/pages/index.html	0	2
website/pages/test.css	0	2
website/static-site/index.html	2	0
website/static-site/test.css	2	0

The diff also touches 1 additional paths; use the full linked commit for the complete file list.

Representative added source from CONTRIBUTING.md:

```
+ | Pages entry point | `static-site/main.tsx`, `vite.pages.config.ts` | Static frontend using the shared main UI
and separate API origin |
```

Design implication: The change crosses the browser, API, cache, or deployment boundary. Confirm the relevant origin, route, secret handling, cache key, and failure response before promoting it.

Verification to preserve: Check a production build and a live health/request path; a successful local build alone does not verify a deployed service.

Source: <https://github.com/easonli29/Vex-Rank/commit/b2a127eeb7a9607b85a85ef636e931a84136c4b7>

GitHub change 44 2026-09-16 c96decc

chore: make the linter actionable and clear the debt it was hiding

Field	Recorded value
Commit time	September 16 2026 at 16:39 UTC-04:00
Author	n3veTz
Engineering area	Verification and contributor workflow
Diff scale	2 changed paths; +6 / -28 lines
Full identifier	c96decc9d12889e11d8ba533bf1094b219989f26

Recorded change: chore: make the linter actionable and clear the debt it was hiding. The largest file-level change was website/app/page.tsx (+5/-27).

Changed path	+	-
website/app/page.tsx	5	27
website/.oxlintrc.json	1	1

Representative added source from website/app/page.tsx:

```
+ import { ArrowLeft, ArrowRight, ArrowUp, ArrowUpRight, BarChart3, CalendarDays, ChevronRight, Filter, Gauge,
  Globe2, MapPin, Menu, Search...
+ const openTeam = (team: any) => { void prefetchTeam(team.number,team.seasonId,team.id);setSelectedTeam(team);
  setTeamReturnView(view); go...
+ const openEvent = (event:any) => { void prefetchEvent(event.id);setSelectedEvent(event); go('event'); };
```

Design implication: The recorded change makes behavior reproducible or changes the guardrails contributors use before a merge. The exact fixture and CI command matter more than the commit title alone.

Verification to preserve: Run the relevant tests and build, and distinguish advisory lint from a required gate.

Source: <https://github.com/easonli29/Vex-Rank/commit/c96decc9d12889e11d8ba533bf1094b219989f26>

GitHub change 45 2026-09-16 5d7e356

ci: verify build, tests and lint on every pull request

Field	Recorded value
Commit time	September 16 2026 at 16:39 UTC-04:00
Author	n3veTz
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +47 / -0 lines
Full identifier	5d7e35629c1f928e3dc33a588feea37b54d05600

Recorded change: ci: verify build, tests and lint on every pull request. The largest file-level change was .github/workflows/ci.yml (+47/-0).

Changed path	+	-
.github/workflows/ci.yml	47	0

Representative added source from .github/workflows/ci.yml:

```
+ name: CI
+ on:
+ pull_request:
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/5d7e35629c1f928e3dc33a588feea37b54d05600>

GitHub change 46 2026-09-16 12d852c

a11y: lift muted text above the WCAG AA contrast floor

Field	Recorded value
Commit time	September 16 2026 at 16:40 UTC-04:00
Author	n3veTz
Engineering area	Competition data and presentation
Diff scale	2 changed paths; +49 / -49 lines
Full identifier	12d852ca775dd4c97e6e99440293950b1118fb46

Recorded change: a11y: lift muted text above the WCAG AA contrast floor. The largest file-level change was website/app/page.tsx (+45/-45).

Changed path	+	-
website/app/page.tsx	45	45
website/components/team-directory.tsx	4	4

Representative added source from website/app/page.tsx:

```
+ <Search className="h-4 w-4 text-white/50" /><input value={globalSearch} onChange={e =>
  setGlobalSearch(e.target.value)} className="h-9 w-...
+ <footer className="mt-16 border-t border-white/10 bg-[#0d1015]"><div className="mx-auto flex max-w-[1440px] flex-
  col gap-5 px-5 py-8 text...
+ <div className="mb-6 flex items-end justify-between gap-4"><div><p className="mb-1.5 text-[11px] italic text-
  white/45">2026-27 V5RC seaso...
```

Design implication: The affected view depends on heterogeneous official event records. Source IDs, divisions, missing positions, and partial upstream pages require explicit handling.

Verification to preserve: Inspect an ordinary event and a Worlds multi-division example; run the event regression fixtures.

Source: <https://github.com/easonli29/Vex-Rank/commit/12d852ca775dd4c97e6e99440293950b1118fb46>

GitHub change 47 2026-09-16 597af57

feat: shareable URLs, working Back button, and deep links

Field	Recorded value
Commit time	September 16 2026 at 16:45 UTC-04:00
Author	n3veTz
Engineering area	Navigation and deep linking
Diff scale	1 changed paths; +57 / -2 lines
Full identifier	597af57918b86d41fd79e9304641bc80971026ba

Recorded change: feat: shareable URLs, working Back button, and deep links. The largest file-level change was website/app/page.tsx (+57/-2).

Changed path	+	-
website/app/page.tsx	57	2

Representative added source from website/app/page.tsx:

```
+ import { useEffect, useMemo, useRef, useState } from 'react';
+ /** Hash routing, not path routing: GitHub Pages has no SPA fallback, so a cold
+ load of /Vex-Rank/teams/31260X would 404. A hash survives any static host. */
```

Design implication: A page state must survive refresh, browser Back, and a cold visit on the static host. The address bar and rendered view should agree after this change.

Verification to preserve: Test a copied link in a new browser context, refresh, Back/Forward, malformed hashes, and encoded team numbers.

Source: <https://github.com/easonli29/Vex-Rank/commit/597af57918b86d41fd79e9304641bc80971026ba>

GitHub change 48 2026-09-16 70213f9

refactor: extract hash routing into a tested module

Field	Recorded value
Commit time	September 16 2026 at 16:48 UTC-04:00
Author	n3veTz
Engineering area	Verification and contributor workflow
Diff scale	3 changed paths; +78 / -28 lines
Full identifier	70213f9fe71a2bfaaf20c124d6b8d953be98296a

Recorded change: refactor: extract hash routing into a tested module. The largest file-level change was website/lib/routes.mjs (+38/-0).

Changed path	+	-
website/lib/routes.mjs	38	0
website/tests/routes.test.mjs	35	0
website/app/page.tsx	5	28

Representative added source from website/app/page.tsx:

```
+ import { routeToHash, hashToRoute } from '@lib/routes.mjs';
+ import { useEffect, useMemo, useState } from 'react';
+ // No guard needed for popstate: the browser has already updated the hash by
```

Design implication: The recorded change makes behavior reproducible or changes the guardrails contributors use before a merge. The exact fixture and CI command matter more than the commit title alone.

Verification to preserve: Run the relevant tests and build, and distinguish advisory lint from a required gate.

Source: <https://github.com/easonli29/Vex-Rank/commit/70213f9fe71a2bfaaf20c124d6b8d953be98296a>

GitHub change 49 2026-09-16 eab79b1

refactor: extract country and event-region vocabularies

Field	Recorded value
Commit time	September 16 2026 at 16:49 UTC-04:00
Author	n3veTz
Engineering area	Verification and contributor workflow
Diff scale	3 changed paths; +54 / -17 lines
Full identifier	eab79b144eb8a5e817c1cc5325a4e0b706574b77

Recorded change: refactor: extract country and event-region vocabularies. The largest file-level change was website/tests/event-regions.test.mjs (+30/-0).

Changed path	+	-
website/tests/event-regions.test.mjs	30	0
website/lib/event-regions.mjs	23	0
website/app/page.tsx	1	17

Representative added source from website/app/page.tsx:

```
+ import { vexCountries, vexEventRegions, eventRegionsForCountry } from '@lib/event-regions.mjs';
```

Design implication: The recorded change makes behavior reproducible or changes the guardrails contributors use before a merge. The exact fixture and CI command matter more than the commit title alone.

Verification to preserve: Run the relevant tests and build, and distinguish advisory lint from a required gate.

Source: <https://github.com/easonli29/Vex-Rank/commit/eab79b144eb8a5e817c1cc5325a4e0b706574b77>

GitHub change 50 2026-09-17 41a1a3e

fix: retry transient API failures and stop claiming 'no events' when a load fails

Field	Recorded value
Commit time	September 17 2026 at 11:38 UTC-04:00
Author	n3veTz
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +32 / -4 lines
Full identifier	41a1a3e79891328f8f54b85f0165b0f53f883765

Recorded change: fix: retry transient API failures and stop claiming 'no events' when a load fails. The largest file-level change was website/app/page.tsx (+32/-4).

Changed path	+	-
website/app/page.tsx	32	4

Representative added source from website/app/page.tsx:

```
+ /**
+ * Retries transient upstream failures. The test worker returns intermittent
+ * 502s, which previously surfaced to the user as a bare error or - worse, on
```

Design implication: The change crosses the browser, API, cache, or deployment boundary. Confirm the relevant origin, route, secret handling, cache key, and failure response before promoting it.

Verification to preserve: Check a production build and a live health/request path; a successful local build alone does not verify a deployed service.

Source: <https://github.com/easonli29/Vex-Rank/commit/41a1a3e79891328f8f54b85f0165b0f53f883765>

GitHub change 51 2026-09-17 4639d7e

docs: add roadmap with findings against the open items

Field	Recorded value
Commit time	September 17 2026 at 12:47 UTC-04:00
Author	n3veTz
Engineering area	Documentation and knowledge transfer
Diff scale	1 changed paths; +44 / -0 lines
Full identifier	4639d7e0dfcec0b36ea0f2c4258dac27f455939e

Recorded change: docs: add roadmap with findings against the open items. The largest file-level change was ROADMAP.md (+44/-0).

Changed path	+	-
ROADMAP.md	44	0

Representative added source from ROADMAP.md:

```
+ # VEXRank roadmap
+ Legend:
+ - Bullet point is the points
```

Design implication: The diff changes the explanation or navigational record used by contributors. Review the named document against the executable version at this date, especially where an older algorithm description remains accessible.

Verification to preserve: Open the changed links and compare formulas, status labels, and setup instructions with the matching source revision.

Source: <https://github.com/easonli29/Vex-Rank/commit/4639d7e0dfcec0b36ea0f2c4258dac27f455939e>

GitHub change 52 2026-09-17 0c23a74

fix: make the ranking list self-consistent and align the graph with it

Field	Recorded value
Commit time	September 17 2026 at 12:52 UTC-04:00
Author	n3veTz
Engineering area	Rating and data processing
Diff scale	4 changed paths; +114 / -10 lines
Full identifier	0c23a7459c8a83961348a31eb92dd3fa26a8e51e

Recorded change: fix: make the ranking list self-consistent and align the graph with it. The largest file-level change was website/tests/vcr-scoring.test.mjs (+47/-0).

Changed path	+	-
website/tests/vcr-scoring.test.mjs	47	0
website/lib/vcr-scoring.mjs	37	0
website/app/api/teams/[number]/route.ts	26	7
website/app/api/rankings/route.ts	4	3

Representative added source from website/app/api/rankings/route.ts:

```
+ import { confidenceFor, decayedRating as computeRating } from '@lib/vcr-scoring.mjs';
+ const rated=[...states.values()].filter(team=>team.matches>=4).map(team=>{const
confidence=confidenceFor(team.matches);const history=hist...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/0c23a7459c8a83961348a31eb92dd3fa26a8e51e>

GitHub change 53 2026-09-17 6682453

fix: one scoring definition everywhere, drop dead retry, explain the sampling limit

Field	Recorded value
Commit time	September 17 2026 at 13:17 UTC-04:00
Author	n3veTz
Engineering area	Rating and data processing
Diff scale	5 changed paths; +15 / -7 lines
Full identifier	66824539ead4b1074573a5ec41295cdaca45b297

Recorded change: fix: one scoring definition everywhere, drop dead retry, explain the sampling limit. The largest file-level change was website/scripts/build-historical-ranking.mjs (+6/-3).

Changed path	+	-
website/scripts/build-historical-ranking.mjs	6	3
ROADMAP.md	5	0
website/scripts/rebuild-season-archives.mjs	3	2
website/app/page.tsx	1	1
website/app/api/rankings/route.ts	0	1

Representative added source from ROADMAP.md:

```
+ - Fourth cause found and it is intentional, not a bug (`/api/rankings` ends its event filter with `.slice(-36)`:  
the live ranking replays...  
+ ### Scoring constants disagreed between live and archives  
+ - Live rankings and historical archives used different maths (`app/api/rankings/route.ts` floored confidence at 35  
while `scripts/build-h...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/66824539ead4b1074573a5ec41295cdaca45b297>

GitHub change 54 2026-09-17 1dba5cb

feat: pointer-tracked card highlights and better scroll behaviour

Field	Recorded value
Commit time	September 17 2026 at 14:04 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	3 changed paths; +105 / -3 lines
Full identifier	1dba5cb80d340fb52a61e9138070fffd3c5ca47a

Recorded change: feat: pointer-tracked card highlights and better scroll behaviour. The largest file-level change was website/app/globals.css (+79/-0).

Changed path	+	-
website/app/globals.css	79	0
website/app/page.tsx	25	2
website/components/team-directory.tsx	1	1

Representative added source from website/app/globals.css:

```
+ /* Scrolling ----- */
+ /* The reduced-motion block above already forces scroll-behavior back to auto,
+ so this is safe to set unconditionally. */
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/1dba5cb80d340fb52a61e9138070fffd3c5ca47a>

GitHub change 55 2026-09-17 84cfce5

fix: season band no longer pops in as an empty slab on a team profile

Field	Recorded value
Commit time	September 17 2026 at 14:10 UTC-04:00
Author	n3veTz
Engineering area	Competition data and presentation
Diff scale	2 changed paths; +18 / -2 lines
Full identifier	84cfce5b1f5d33a84ef513913db2beb906f5ed1f

Recorded change: fix: season band no longer pops in as an empty slab on a team profile. The largest file-level change was website/app/globals.css (+11/-0).

Changed path	+	-
website/app/globals.css	11	0
website/app/page.tsx	7	2

Representative added source from website/app/globals.css:

```
+ /* Hold the whole band, not just its contents. The band is ~745px tall, so
+ its top edge sits on screen long before it is scrolled to - and painting
+ the olive slab and heading while the figures inside were still at
```

Design implication: The affected view depends on heterogeneous official event records. Source IDs, divisions, missing positions, and partial upstream pages require explicit handling.

Verification to preserve: Inspect an ordinary event and a Worlds multi-division example; run the event regression fixtures.

Source: <https://github.com/easonli29/Vex-Rank/commit/84cfce5b1f5d33a84ef513913db2beb906f5ed1f>

GitHub change 56 2026-09-17 6d68e89

feat: four colour themes with a picker

Field	Recorded value
Commit time	September 17 2026 at 14:19 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	4 changed paths; +164 / -54 lines
Full identifier	6d68e89e95bd62997d819844fcb7deca75937d01

Recorded change: feat: four colour themes with a picker. The largest file-level change was website/app/page.tsx (+92/-46).

Changed path	+	-
website/app/page.tsx	92	46
website/app/globals.css	67	3
website/components/team-directory.tsx	4	4
website/static-site/index.html	1	1

Representative added source from website/app/globals.css:

```
+ background: var(--c-accent);
+ /* Palette -----
+ Every surface and the accent live here so a theme is a token swap and
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/6d68e89e95bd62997d819844fcb7deca75937d01>

GitHub change 57 2026-09-17 311c587

feat: team numbers are signed on in a handwriting face

Field	Recorded value
Commit time	September 17 2026 at 14:47 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	2 changed paths; +38 / -9 lines
Full identifier	311c58767ae411679ee471c7fab7aad622f6861c

Recorded change: feat: team numbers are signed on in a handwriting face. The largest file-level change was website/app/globals.css (+37/-8).

Changed path	+	-
website/app/globals.css	37	8
website/app/page.tsx	1	1

Representative added source from website/app/globals.css:

```
+ @import
  url('https://fonts.googleapis.com/css2?family=Hanken+Grotesk:ital,wght@0,300..800;1,300..800&family=Source+Serif+4:opsz,wght@8..6...
+ /* Team numbers on a profile are signed rather than set: Caveat is a
+ handwriting face whose digits stay legible, which most scripts do not -
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/311c58767ae411679ee471c7fab7aad622f6861c>

GitHub change 58 2026-09-17 bca7c00

feat: trace team numbers stroke by stroke instead of revealing them

Field	Recorded value
Commit time	September 17 2026 at 14:55 UTC-04:00
Author	n3veTz
Engineering area	Competition data and presentation
Diff scale	4 changed paths; +160 / -11 lines
Full identifier	bca7c002c565e260f0803384833f165c1d62af85

Recorded change: feat: trace team numbers stroke by stroke instead of revealing them. The largest file-level change was website/components/signed-number.tsx (+86/-0).

Changed path	+	-
website/components/signed-number.tsx	86	0
website/lib/stroke-glyphs.mjs	66	0
website/app/globals.css	6	10
website/app/page.tsx	2	1

Representative added source from website/app/globals.css:

```
+ /* The number is drawn, not set: SignedNumber renders one path per pen stroke
+ and traces them in order. Height rather than font-size drives the scale. */
+ display: block;
```

Design implication: The affected view depends on heterogeneous official event records. Source IDs, divisions, missing positions, and partial upstream pages require explicit handling.

Verification to preserve: Inspect an ordinary event and a Worlds multi-division example; run the event regression fixtures.

Source: <https://github.com/easonli29/Vex-Rank/commit/bca7c002c565e260f0803384833f165c1d62af85>

GitHub change 59 2026-09-17 2acff59

feat: slow the signature and ease off toward the final character

Field	Recorded value
Commit time	September 17 2026 at 15:01 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +19 / -5 lines
Full identifier	2acff59f8555ae6d8dc293541a1f8d7ef67f42fb

Recorded change: feat: slow the signature and ease off toward the final character. The largest file-level change was website/components/signed-number.tsx (+19/-5).

Changed path	+	-
website/components/signed-number.tsx	19	5

Representative added source from website/components/signed-number.tsx:

```
+ const characterCount = Array.from(text).length;
+ const PEN_SPEED = 1.15;    // ms per user unit, at the start of the signature
+ const PEN_LIFT = 70;       // pause between strokes, as the hand repositions
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/2acff59f8555ae6d8dc293541a1f8d7ef67f42fb>

GitHub change 60 2026-09-17 ddfec42

tune: quicken the signature by about 20 percent

Field	Recorded value
Commit time	September 17 2026 at 15:07 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +2 / -2 lines
Full identifier	ddfec42cfb29c8fee5db299aba389caccca2a1cc

Recorded change: tune: quicken the signature by about 20 percent. The largest file-level change was website/components/signed-number.tsx (+2/-2).

Changed path	+	-
website/components/signed-number.tsx	2	2

Representative added source from website/components/signed-number.tsx:

```
+ const PEN_SPEED = 0.92; // ms per user unit, at the start of the signature
+ const PEN_LIFT = 56;    // pause between strokes, as the hand repositions
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/ddfec42cfb29c8fee5db299aba389caccca2a1cc>

GitHub change 61 2026-09-17 a0ebdb0

fix: jump to the top on navigation instead of scrolling there smoothly

Field	Recorded value
Commit time	September 17 2026 at 15:14 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +9 / -1 lines
Full identifier	a0ebdb006b33fde1764fb68d734bbaad15cb8b76

Recorded change: fix: jump to the top on navigation instead of scrolling there smoothly. The largest file-level change was website/app/page.tsx (+9/-1).

Changed path	+	-
website/app/page.tsx	9	1

Representative added source from website/app/page.tsx:

```
+ const go = (next: View) => {
+   setView(next); setMobile(false);
+   // Instant, not smooth. A smooth scroll from deep in a long list takes
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/a0ebdb006b33fde1764fb68d734bbaad15cb8b76>

GitHub change 62 2026-09-17 501e688

fix: undrawn strokes no longer leave stray ink before they are written

Field	Recorded value
Commit time	September 17 2026 at 15:20 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	1 changed paths; +15 / -4 lines
Full identifier	501e6888e02e167537a6a05c6f284d38948599af

Recorded change: fix: undrawn strokes no longer leave stray ink before they are written. The largest file-level change was website/components/signed-number.tsx (+15/-4).

Changed path	+	-
website/components/signed-number.tsx	15	4

Representative added source from website/components/signed-number.tsx:

```
+ const STROKE_WIDTH = 7;  
+ // The gap must be longer than the path, not merely equal to it.  
+ // A one-value stroke-dasharray of L expands to [L, L] - a pattern of
```

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/501e6888e02e167537a6a05c6f284d38948599af>

GitHub change 63 2026-09-17 81786ee

fix: tell the reader when a content blocker is stopping the data

Field	Recorded value
Commit time	September 17 2026 at 15:27 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	2 changed paths; +32 / -3 lines
Full identifier	81786ee8ae1c934df34120c69d7cc592c4c24687

Recorded change: fix: tell the reader when a content blocker is stopping the data. The largest file-level change was website/app/page.tsx (+26/-3).

Changed path	+	-
website/app/page.tsx	26	3
ROADMAP.md	6	0

Representative added source from ROADMAP.md:

- + ### Content blockers break the site, and only a deployment change truly fixes it
- + - Every data request is third-party, which is what blockers drop (The site is served from GitHub Pages while the API lives on a separate ...
- + - Mitigated in the client, not solved (Blocked requests are now detected and reported as "a browser content blocker or privacy extension ...

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/81786ee8ae1c934df34120c69d7cc592c4c24687>

GitHub change 64 2026-09-17 960f693

fix: bind the catch parameter in the rankings error handler

Field	Recorded value
Commit time	September 17 2026 at 15:29 UTC-04:00
Author	n3veTz
Engineering area	Rating and data processing
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	960f693852676e9252e82a636fe41496b587da34

Recorded change: fix: bind the catch parameter in the rankings error handler. The largest file-level change was website/app/page.tsx (+1/-1).

Changed path	+	-
website/app/page.tsx	1	1

Representative added source from website/app/page.tsx:

```
+ fetchWithBackoff('/api/rankings?data=v49').then(response=>response.ok?response.json():Promise.reject(new
Error('Rankings request failed'))...
```

Design implication: The affected path contributes to rating calculation, archive integrity, or how scores are presented. Preserve model-version labels and compare outputs across live and historical paths when changing it.

Verification to preserve: Check rating fixtures, archive validation, and the displayed value against the graph and selected event set.

Source: <https://github.com/easonli29/Vex-Rank/commit/960f693852676e9252e82a636fe41496b587da34>

GitHub change 65 2026-09-17 4cb9e0c

feat: remove the global search from the header

Field	Recorded value
Commit time	September 17 2026 at 15:39 UTC-04:00
Author	n3veTz
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +1 / -5 lines
Full identifier	4cb9e0cea2a1e00685302d2f6d5fddcdb54914f5

Recorded change: feat: remove the global search from the header. The largest file-level change was website/app/page.tsx (+1/-5).

Changed path	+	-
website/app/page.tsx	1	5

Representative added source from website/app/page.tsx:

```
+ <div className="ml-auto" />
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/4cb9e0cea2a1e00685302d2f6d5fddcdb54914f5>

GitHub change 66 2026-09-17 7feb6c5

feat: give a team's location its own line under the number

Field	Recorded value
Commit time	September 17 2026 at 15:47 UTC-04:00
Author	n3veTz
Engineering area	Competition data and presentation
Diff scale	1 changed paths; +3 / -2 lines
Full identifier	7feb6c553ed0ece8b1140767c06936f2d43f9862

Recorded change: feat: give a team's location its own line under the number. The largest file-level change was website/app/page.tsx (+3/-2).

Changed path	+	-
website/app/page.tsx	3	2

Representative added source from website/app/page.tsx:

```
+ <div className="rise mt-10 flex flex-wrap items-baseline gap-x-3 gap-y-1 text-[11px] font-bold uppercase tracking-
[.22em] text-white/50" ...
+ {profile.region&&<p className="rise mt-4 flex items-center gap-2.5 text-lg leading-snug text-white/75 sm:text-2xl"
style={{'--i':2}} as Re...
+ <div className="rise mt-6 flex flex-col gap-6 border-t border-white/10 pt-6 sm:flex-row sm:items-end" style={{'--
i':3}} as React.CSSProper...
```

Design implication: The affected view depends on heterogeneous official event records. Source IDs, divisions, missing positions, and partial upstream pages require explicit handling.

Verification to preserve: Inspect an ordinary event and a Worlds multi-division example; run the event regression fixtures.

Source: <https://github.com/easonli29/Vex-Rank/commit/7feb6c553ed0ece8b1140767c06936f2d43f9862>

GitHub change 67 2026-09-17 8a24908

fix: draw H's crossbar before its stems

Field	Recorded value
Commit time	September 17 2026 at 16:10 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	8a249087e7c7ba7c0e381672ae0d9fdd93191475

Recorded change: fix: draw H's crossbar before its stems. The largest file-level change was website/lib/stroke-glyphs.mjs (+1/-1).

Changed path	+	-
website/lib/stroke-glyphs.mjs	1	1

Representative added source from website/lib/stroke-glyphs.mjs:

```
+ H: ['M10,50 L50,50', 'M10,6 L10,95', 'M50,6 L50,95'], // crossbar first, then both stems
```

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/8a249087e7c7ba7c0e381672ae0d9fdd93191475>

GitHub change 68 2026-09-17 2f9719d

fix: H draws left stem, crossbar, then right stem

Field	Recorded value
Commit time	September 17 2026 at 16:12 UTC-04:00
Author	n3veTz
Engineering area	Interface and documentation
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	2f9719d94a387c53a64dd40570bc7c33fa8b47e0

Recorded change: fix: H draws left stem, crossbar, then right stem. The largest file-level change was website/lib/stroke-glyphs.mjs (+1/-1).

Changed path	+	-
website/lib/stroke-glyphs.mjs	1	1

Representative added source from website/lib/stroke-glyphs.mjs:

```
+ H: ['M10,6 L10,95', 'M10,50 L50,50', 'M50,6 L50,95'], // left stem, crossbar, right stem
```

Design implication: The change refines the website's visual or contributor-facing behavior. It should be judged against the intended user task and existing layout hierarchy.

Verification to preserve: Inspect the changed view at desktop and narrow widths and keep the documented behavior aligned with the implementation.

Source: <https://github.com/easonli29/Vex-Rank/commit/2f9719d94a387c53a64dd40570bc7c33fa8b47e0>

GitHub change 69 2026-09-17 ee81570

fix: shared links work in a browser that has never opened the site

Field	Recorded value
Commit time	September 17 2026 at 16:17 UTC-04:00
Author	n3veTz
Engineering area	Navigation and deep linking
Diff scale	1 changed paths; +11 / -7 lines
Full identifier	ee81570d9062e7ce42adf82c12343b4030f7c7fa

Recorded change: fix: shared links work in a browser that has never opened the site. The largest file-level change was website/app/page.tsx (+11/-7).

Changed path	+	-
website/app/page.tsx	11	7

Representative added source from website/app/page.tsx:

```
+
if(saved){if(saved.view)setView(saved.view);if(saved.selectedTeam)setSelectedTeam(saved.selectedTeam);if(saved.selectedEvent)setSelectedE...
+ // Outside the try, and outside the saved branch: an explicit URL must win
+ // over restored state, and must still work when there is no saved state at
```

Design implication: A page state must survive refresh, browser Back, and a cold visit on the static host. The address bar and rendered view should agree after this change.

Verification to preserve: Test a copied link in a new browser context, refresh, Back/Forward, malformed hashes, and encoded team numbers.

Source: <https://github.com/easonli29/Vex-Rank/commit/ee81570d9062e7ce42adf82c12343b4030f7c7fa>

GitHub change 70 2026-09-22 9e55113

Use root path for vex-rank.com

Field	Recorded value
Commit time	September 22 2026 at 23:50 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +2 / -2 lines
Full identifier	9e551130d669c7b4cf949f566805d78fae01c5db

Recorded change: Use root path for vex-rank.com. The largest file-level change was website/vite.pages.config.ts (+2/-2).

Changed path	+	-
website/vite.pages.config.ts	2	2

Representative added source from website/vite.pages.config.ts:

```
+ base: '/',
+ __VEX_ASSET_BASE__:JSON.stringify('/')
```

Design implication: The change crosses the browser, API, cache, or deployment boundary. Confirm the relevant origin, route, secret handling, cache key, and failure response before promoting it.

Verification to preserve: Check a production build and a live health/request path; a successful local build alone does not verify a deployed service.

Source: <https://github.com/easonli29/Vex-Rank/commit/9e551130d669c7b4cf949f566805d78fae01c5db>

GitHub change 71 2026-09-22 4466dc6

Allow vex-rank.com to use ranking API

Field	Recorded value
Commit time	September 22 2026 at 23:50 UTC-04:00
Author	Eason Li
Engineering area	Publishing and service boundary
Diff scale	1 changed paths; +1 / -1 lines
Full identifier	4466dc677d2ddf1f0df4dafcde994a6233410a4e

Recorded change: Allow vex-rank.com to use ranking API. The largest file-level change was website/cloudflare/worker.ts (+1/-1).

Changed path	+	-
website/cloudflare/worker.ts	1	1

Representative added source from website/cloudflare/worker.ts:

```
+ result.headers.set('Access-Control-Allow-Origin', 'https://vex-rank.com');
```

Design implication: The diff affects a deployed origin, service route, cache, workflow, or credential boundary. The recorded source should be checked with the corresponding hosting configuration and actual deployment result.

Verification to preserve: Check a production build, relevant workflow, origin policy, and live health/request path before treating the code as deployed.

Source: <https://github.com/easonli29/Vex-Rank/commit/4466dc677d2ddf1f0df4dafcde994a6233410a4e>

GitHub change 72 2026-09-23 12268ff

Add Discord invite icon beside theme selector

Field	Recorded value
Commit time	September 23 2026 at 04:58 UTC-04:00
Author	Eason Li
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +3 / -0 lines
Full identifier	12268ffb6e438b7ca3fb97188d3171de72cb0880

Recorded change: Add Discord invite icon beside theme selector. The largest file-level change was website/app/page.tsx (+3/-0).

Changed path	+	-
website/app/page.tsx	3	0

Representative added source from website/app/page.tsx:

```
+ <a href="https://discord.gg/Gu4TUEmhhY" target="_blank" rel="noopener noreferrer" aria-label="Join the VEXRank
Discord" title="Join the V...
+ <svg viewBox="0 0 24 24" className="h-5 w-5 fill-current" aria-hidden="true"><path d="M20.317 4.3698a19.7913
19.7913 0 00-4.8851-1.5152.0...
+ </a>
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/12268ffb6e438b7ca3fb97188d3171de72cb0880>

GitHub change 73 2026-09-23 03c75b7

Place blog section beside vertical Signature Events

Field	Recorded value
Commit time	September 23 2026 at 05:06 UTC-04:00
Author	Eason Li
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +10 / -2 lines
Full identifier	03c75b780b99b692e0be72e20a2cdef5a4ba80b1

Recorded change: Place blog section beside vertical Signature Events. The largest file-level change was website/app/page.tsx (+10/-2).

Changed path	+	-
website/app/page.tsx	10	2

Representative added source from website/app/page.tsx:

```
+ <section className="mx-auto grid max-w-[1440px] gap-4 px-5 py-8 lg:grid-cols-[1.65fr_1fr] lg:px-8">
+ <section aria-labelledby="home-blog-title" className="min-w-0">
+ <div className="mb-5"><p className="text-[11px] italic text-white/45">From the blog</p><h2 id="home-blog-title"
  className="font-display m...
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/03c75b780b99b692e0be72e20a2cdef5a4ba80b1>

GitHub change 74 2026-09-23 7bf2bc8

Show featured event date and align blog panel

Field	Recorded value
Commit time	September 23 2026 at 05:25 UTC-04:00
Author	Eason Li
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +3 / -3 lines
Full identifier	7bf2bc8de80ce1e91757f3c8a371951d074f241f

Recorded change: Show featured event date and align blog panel. The largest file-level change was website/app/page.tsx (+3/-3).

Changed path	+	-
website/app/page.tsx	3	3

Representative added source from website/app/page.tsx:

```
+ {primaryEvent?<article className="float-card relative min-h-[360px] overflow-hidden border border-white/10 bg-
[linear-gradient(160deg,var...
+ <section aria-labelledby="home-blog-title" className="flex min-w-0 flex-col">
+ <div className="min-h-[360px] flex-1 border border-white/10 bg-[var(--c-surface)] p-7 sm:p-9"><p className="text-
sm text-white/55">Update...
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/7bf2bc8de80ce1e91757f3c8a371951d074f241f>

GitHub change 75 2026-09-23 dd63f3e

Revert blog homepage layout while keeping featured date

Field	Recorded value
Commit time	September 23 2026 at 05:50 UTC-04:00
Author	Eason Li
Engineering area	Interface design and interaction
Diff scale	1 changed paths; +2 / -10 lines
Full identifier	dd63f3edd68830795b5f44bbb0e3eb30c2f760c8

Recorded change: Revert blog homepage layout while keeping featured date. The largest file-level change was website/app/page.tsx (+2/-10).

Changed path	+	-
website/app/page.tsx	2	10

Representative added source from website/app/page.tsx:

```
+ <section className="mx-auto max-w-[1440px] px-5 py-8 lg:px-8"><div className="mb-5 flex items-center justify-between"><div><p className="...
+ <section className="mx-auto grid max-w-[1440px] gap-4 px-5 py-8 md:grid-cols-3 lg:px-8"><StatFeature icon={<Gauge
/>} title="VCR team rat...
```

Design implication: This is an interface trial or layout refinement. Read the diff together with the adjacent design iterations to see what changed after visual review and whether navigation or accessibility behavior was preserved.

Verification to preserve: Inspect the changed view at desktop and narrow widths, check keyboard access and reduced motion where relevant, and verify the linked action still works.

Source: <https://github.com/easonli29/Vex-Rank/commit/dd63f3edd68830795b5f44bbb0e3eb30c2f760c8>

Part Thirteen Pull Request and Merge Register

GitHub pull requests connect design changes to review and publication dates. The recorded merge time is the point at which that pull request entered main. PR 1 was closed without a merge; its VCR 3 candidate branch is a design and provenance record, while the later PR 5 is the merged implementation. PR 28 restored the previous live homepage layout; PR 29 remains an unmerged design draft. The current list was retrieved on September 23.

Pull request decisions 1 and 2

PR 1 VCR 3.0 candidate: traceable event-settlement revision

Opened	Merged or closed	Branch
2026-09-09 18:27 UTC	2026-09-16 06:39 UTC	codex/vcr-3.0-candidate

Recorded purpose and review context: ## Problem and behavior The existing team-history calculation assumes 1500-rated opponents and has no separate tournament outcome credit. This candidate provides an executable replacement event-settlement core with stronger elimination evidence, bounded statistics, one Event Weight application, and an explicit positive minimum for unexpected champions. ## Versioning Adds four new files only: scoring core, regression tests, detailed VCR 3.0 revision, and versions/vcr-3.0-candidate.md. VCR 2.0 documentation and website calculations remain intact. No live integration or deployment is included. ## Validation `node --test algorithm/vcr-event.test.mjs` : all 9 tests pass, covering every tier, champion floors, non-champion losses, margin symmetry, ...

Publication result: Closed without a merge; do not count it as published behavior.

Source: <https://github.com/easonli29/Vex-Rank/pull/1>

PR 2 Fix Worlds results, bracket replays and website loading reliability

Opened	Merged or closed	Branch
2026-09-12 00:58 UTC	2026-09-15 07:39 UTC	codex/worlds-results-site-inspection

Recorded purpose and review context: ## Summary - Load every official event/match page and every Worlds division, including championship finals. - Keep failed/rate-limited requests out of successful caches; reduce eager background fetching and add retry/error handling. - Correct Worlds classification, bracket replay placement, agenda time/table parsing and season-specific team navigation. - Improve keyboard focus, responsive spacing, cancelled-event filtering and stale-data handling. ## Verification - 8 regression tests pass. - TypeScript check and production build pass. - Event 64025: all 11 divisions returned elimination records; 865 teams loaded. - Browser checked High Stakes Worlds filtering, division match scores, final series and replay grouping. ## Scope and limitations...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/2>

Pull request decisions 4 and 5

PR 4 Connect GitHub Pages test website to Cloudflare Workers and D1

Opened	Merged or closed	Branch
2026-09-14 18:55 UTC	2026-09-15 07:39 UTC	codex/firebase-test-deployment

Recorded purpose and review context: The test website now calls a separate live Cloudflare Worker for event, team, ranking and skills data. D1 shares cached responses and persists event-rank locks. The Event.VEX API key is stored as a Worker secret and is absent from the browser bundle. API: <https://vexrank-api-test.vexrank-eason.workers.dev> The manual Pages workflow checks data regressions and API health, builds the website for `/Vex-Rank/`, and publishes through GitHub Pages. Existing routes, ranking model and archive scope are preserved. Previous inspection fixes from #2 are included. The Firebase function configuration is documented as superseded. Validation: Worker bundle and frontend build pass; data regressions 8/8 pass. Live checks returned 707 current events, 581 rank...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/4>

PR 5 Implement VCR 3.0 candidate in ranking pipelines

Opened	Merged or closed	Branch
2026-09-15 17:54 UTC	2026-09-15 17:57 UTC	codex/vcr-3.0-implemented

Recorded purpose and review context: ## Summary Implements the newest VCR 3.0 candidate as a traceable version without replacing `main`. - Adds the shared event-settlement processor used by the website. - Connects it to global rankings, team history, and the historical ranking builder. - Applies event-tier weight once at event settlement. - Gives elimination performance the larger tournament share. - Uses dynamic alliance credit based primarily on event-local evidence, with frozen global rating as a smaller prior. - Guarantees that a sufficiently unexpected tournament champion cannot finish the event with a negative rating change. - Adds 75-day recency decay to leaderboard aggregation. - Publishes the full constants, formulas, assumptions, audit fields, and version record. - A...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/5>

Pull request decisions 6 and 7

PR 6 Apply VCR 3.0 to completed-season archives

Opened	Merged or closed	Branch
2026-09-16 00:05 UTC	2026-09-16 00:06 UTC	codex/vcr3-historical-seasons

Recorded purpose and review context: Older season views previously loaded a static leaderboard calculated with the earlier model. Replays completed seasons through the shared VCR 3.0 processor, freezes decay at each season end, and serves validated versioned archive assets through the existing API. Overall and statistical season selectors use the published archive manifest. Push Back is fully rebuilt: 12,573 teams, 110,150 eligible matches, 1,548 rated events; all 1,681 completed event records retrieved. High Stakes, Over Under and Spin Up are supported by the resumable builder but are not published: the upstream service returned HTTP 429 during High Stakes retrieval. Missing seasons never fall back to a small event sample or an older-model leaderboard. Includes paginated hist...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/6>

PR 7 Reduce API traffic during historical archive rebuilds

Opened	Merged or closed	Branch
2026-09-16 00:11 UTC	2026-09-16 00:11 UTC	codex/vcr3-archive-download-resume

Recorded purpose and review context: Historical downloads hit the upstream rate limit. Pace source requests 1.8 seconds apart, persist successful individual pages, reuse complete legacy match-page caches, and take event metadata from the season calendar. This preserves completed work across interrupted downloads and reduces repeated upstream calls. The live algorithm and published archive values are unchanged. The remaining historical seasons are running through this resumable path; Push Back has already been published.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/7>

Pull request decisions 8 and 9

PR 8 Document core code and contributor workflow

Opened	Merged or closed	Branch
2026-09-16 05:42 UTC	2026-09-16 06:29 UTC	codex/contributor-code-comments

Recorded purpose and review context: Contributors currently have to infer ranking assumptions, archive safety checks and deployment boundaries from dense implementation code. This adds explanatory comments in 12 core files and a root CONTRIBUTING.md with local setup, a source map, data flow, checks and maintenance instructions. Comments explain internal rating units, frozen expectations, mutable event state, reliability and winner corrections, pagination failures, cache keys, archive completeness, bracket replays and organizer schedule parsing. The guide explicitly documents current limitations, including sampled live rankings, legacy archive output and differences between calendar classification and rating tiers. Validation: - All 15 existing data, VCR and archive regression ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/8>

PR 9 feat: editorial redesign, scroll motion, and capped result lists

Opened	Merged or closed	Branch
2026-09-16 20:08 UTC	2026-09-16 20:11 UTC	feat/editorial-ui-redesign

Recorded purpose and review context: Typography and surfaces: - Claude-like type stack, with Styrene/Tiempos named first for anyone who licenses them and Hanken Grotesk / Source Serif 4 as open fallbacks. body previously referenced --font-geist, which is undefined outside the next/font build, so the standalone pages entry had been rendering Arial. - Soften corners: --radius 0.125rem -> 0.75rem, varying by element role. Preflight zeroes border-radius on button/input/select at specificity (0,0,1), so those rules carry real specificity instead of :where(). Editorial pass: - Replace 16 repeated eyebrow+icon section headers, 19 bordered stat tiles and the dashed empty state with ruled, typographic equivalents. - Drop 88 font-black runs for a real weight scale, remove the decorative...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/9>

Pull request decisions 10 and 11

PR 10 Deep links, contrast fixes, working dev server, and CI

Opened	Merged or closed	Branch
2026-09-16 20:51 UTC	2026-09-16 20:52 UTC	feat/routing-and-dx

Recorded purpose and review context: Quality-of-life work across developer experience, accessibility and navigation. Seven commits, each independently reviewable and verified. ## The two that fix things that were actually broken **`pnpm dev` did not work at all. ** `website/pages/` collided with vinext's Pages-Router convention, so it server-rendered `pages/main.tsx` — which calls `createRoot(document...)` at module scope. Every app-router route returned 500 with `document is not defined`. The command CONTRIBUTING tells new contributors to run failed on step one. Renamed to `website/static-site/`; the config *filename* is unchanged so all deployment docs and the Pages workflow keep working as written. **No URL involvement anywhere in navigation. ** No `pushState`, no `searchParams`...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/10>

PR 11 Fix intermittent API failures surfacing as 'no events'

Opened	Merged or closed	Branch
2026-09-17 15:39 UTC	2026-09-17 15:42 UTC	feat/routing-and-dx

Recorded purpose and review context: Fixes a user-reported "events aren't loading" on the deployed site. ## Root cause The API is healthy, but returns **intermittent 502s**. Five consecutive calls to `/api/events?season=197&classification=v49`: `` 502 (2.4s) → 200 (4.7s) → 200 → 200 → 200 `` The first success taking 4.7s points at an upstream cold start. A single unlucky request was enough to break the page. ## Two problems made it worse than it needed to be **1. One transient 502 went straight to the user. ** `cachedJson` now goes through `fetchWithBackoff` — up to 3 attempts, exponential backoff with jitter so parallel requests don't retry in lockstep. Only 5xx and network errors are retried: a 4xx won't fix itself, and a timeout has already cost 60s, so retrying compounds ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/11>

Pull request decisions 12 and 13

PR 12 Colour themes, ranking correctness, and interaction polish

Opened	Merged or closed	Branch
2026-09-17 18:20 UTC	2026-09-17 18:21 UTC	feat/routing-and-dx

Recorded purpose and review context: Ships the "Choose Able Color Theme" roadmap item, plus the accumulated UI, routing and algorithm work from this branch. ## Colour themes The palette wasn't actually swappable — 34 uses of the accent hex, 32 of the panel hex and more were hardcoded through `page.tsx` and `team-directory.tsx`. Those are now **8 tokens** (page, chrome, header, surface, surface-2, accent, band, hero), so a theme is a token swap with no component changes: 75 class replacements plus the two SVG `stroke` props on the rating chart. **Midnight** (default), **Ember**, **Abyss**, **Moss** — all dark on purpose. The interface leans on white-alpha utilities (`text-white/55` and friends) throughout, so a light theme would need those rewritten first and would ship broken ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/12>

PR 13 Sign team numbers on with a handwriting face

Opened	Merged or closed	Branch
2026-09-17 18:47 UTC	2026-09-17 18:48 UTC	feat/routing-and-dx

Recorded purpose and review context: A team profile now writes its number out left to right, like a signature, rather than having it appear. ## Font **Caveat**, for the profile lockup only. Team numbers are alphanumeric (`31260X`), and most signature scripts carry decorative figure sets that make digits ambiguous — Caveat keeps them legible while still reading as handwriting. Tables and cards stay on the sans. Script at row size would cost scanability for no gain, so this is the one place it appears. The lockup needed reshaping for a script rather than a grotesque: no uppercase and no negative tracking, both of which wreck the joins, plus a larger clamp because a script carries far less weight per em than the sans it replaces. ## The write-on A left-to-right **clip reveal**, n...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/13>

Pull request decisions 14 and 15

PR 14 Trace team numbers stroke by stroke

Opened	Merged or closed	Branch
2026-09-17 18:56 UTC	2026-09-17 18:57 UTC	feat/routing-and-dx

Recorded purpose and review context: Replaces the clip reveal with real pen strokes. ## Why the previous approach wasn't writing The clip reveal only uncovered a finished typeface left to right — it **spread** the ink rather than drawing it. Writing follows the pen's ***centreline***, in stroke order, and a normal font can't supply that: fonts store the ***outline*** of a glyph, so tracing one draws the boundary of every stem, both sides at once. ## What's here ``lib/stroke-glyphs.mjs`` authors 0–9 and A–Z as centreline paths, in the form plotter and engraving fonts use. Each glyph is an ordered list of strokes, each drawn in the direction a right-handed writer would move — ``X`` is two strokes, ``4`` is two, ``E`` is four, ``8`` is one continuous figure-eight. ``components/signed-number.tsx`` ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/14>

PR 15 Slow the signature and ease off toward the final character

Opened	Merged or closed	Branch
2026-09-17 19:01 UTC	2026-09-17 19:02 UTC	feat/routing-and-dx

Recorded purpose and review context: Base pen speed ``0.62 → 1.15`` ms per user unit, and the pen now decelerates across the signature instead of holding one rate throughout. ## The curve The drag multiplier is keyed to the ***character*** index, not the stroke index — a hand eases off through the whole of the final character, not merely its last stroke. ``drag = 1 + 1.35 * through^1.7`` The exponent matters: a linear ramp makes the whole thing feel sluggish from the first stroke. Raising it keeps the early characters near full speed and concentrates the slowdown where it reads as deliberate. | character | drag | |---|---| | 3 | 1.00× | | 1 | 1.09× | | 2 | 1.28× | | 6 | 1.57× | | 0 | 1.92× | | X | 2.35× | The inter-stroke lift scales with the same factor, so the hand also hesi...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/15>

Pull request decisions 16 and 17

PR 16 Quicken the signature by about 20 percent

Opened	Merged or closed	Branch
2026-09-17 19:07 UTC	2026-09-17 19:08 UTC	feat/routing-and-dx

Recorded purpose and review context: Pen speed $1.15 \rightarrow 0.92$ ms per user unit, inter-stroke lift $70 \rightarrow 56$ ms. Both scale together: dropping only the stroke speed would leave the pauses proportionally longer and make the writing read as hesitant rather than quicker. The deceleration curve is untouched, so the shape of the motion is unchanged and only its overall tempo moves — measured ratio between first and last stroke is still exactly $2.35 \times$ (0.92 against 2.16 ms/unit). `31260X` now writes in $2.33s$, down from 2.86s. 37/37 tests pass, build succeeds.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/16>

PR 17 Jump to the top on navigation instead of gliding there

Opened	Merged or closed	Branch
2026-09-17 19:14 UTC	2026-09-17 19:15 UTC	feat/routing-and-dx

Recorded purpose and review context: Opening a team profile from part-way down the rankings left you looking at the *middle* of the profile while the page was still travelling — so the team number was written above the fold, and the signature was half over by the time it came into view. ## Cause `go()` was already scrolling to the top, but *smoothly*. Measured from `y=2533`: `0ms y=1272 180ms y=439 360ms y=136 540ms y=30 720ms y=0`. The signature starts at 180ms, so the first three strokes happened off-screen. Not a missing scroll — a scroll that was too slow to keep up with the view it was revealing. ## Fix Navigation should jump; only in-page scrolling should glide. `behavior` is now `'instant'`, which also overrides the `html { scroll-behavior: smooth }` added earlier...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/17>

Pull request decisions 18 and 19

PR 18 Fix stray ink on undrawn signature strokes

Opened	Merged or closed	Branch
2026-09-17 19:21 UTC	2026-09-17 19:22 UTC	feat/routing-and-dx

Recorded purpose and review context: A team number ending in `Y` showed a mark at the Y before it was written. ## Cause `stroke-dasharray` with a single value `L` expands to `[L, L]` — a dash of L followed by a gap of L, **repeating with period 2L**. Setting the offset to exactly `L` therefore lands a zero-length dash at path position 0, and `stroke-linecap: round` paints that as a dot the diameter of the stroke. It affected every glyph, not only Y. Y is simply where it shows: its second stroke starts at the junction, mid-air, with no other ink nearby — elsewhere the mark sits on an extremity or gets covered once neighbouring strokes are drawn. ## The fix, and the wrong turn on the way My first attempt pushed the offset one stroke-width past the start. That removed the dot and...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/18>

PR 19 Report content-blocker failures instead of a generic error

Opened	Merged or closed	Branch
2026-09-17 19:29 UTC	2026-09-17 19:33 UTC	feat/routing-and-dx

Recorded purpose and review context: ## The problem Every data request is third-party: the site is served from GitHub Pages while the API lives on a separate `\*.workers.dev` origin. Blockers drop those — `workers.dev` is on several lists because it's free and widely abused, and strict modes block cross-origin XHR outright. The request then fails at the **network layer with a TypeError and no status** — indistinguishable from being offline, but very distinguishable from a server error. It previously surfaced as "Events could not be loaded" or "Rankings are temporarily unavailable", which sends the reader hunting for a fault that isn't there. ## What this does Network-level failures are now identified and reported as a blocker or extension, with an offline variant when `navigato...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/19>

Pull request decisions 20 and 21

PR 20 Remove the global search from the header

Opened	Merged or closed	Branch
2026-09-17 19:39 UTC	2026-09-17 19:40 UTC	feat/routing-and-dx

Recorded purpose and review context: The header search duplicated the per-view search on Rankings and Teams — which is where a search belongs, since those have the filters to narrow the result. Those are untouched. Removed with its state and handler rather than left dangling: ``globalSearch``, ``setGlobalSearch`` and ``submitGlobal`` are all gone, so nothing unreferenced is left behind. The ``Search`` icon import stays — still used by the two in-page fields. One layout detail: the form carried ``ml-auto``, which is what pushed the theme picker and mobile toggle to the right edge. A spacer takes over that job, so the header is otherwise unchanged. Verified: header is now logo → nav → theme picker → menu with no input; the in-page search still renders. 37/37 tests pass, lint unchanged at ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/20>

PR 21 Give a team's location its own line under the number

Opened	Merged or closed	Branch
2026-09-17 19:47 UTC	2026-09-17 19:47 UTC	feat/routing-and-dx

Recorded purpose and review context: City, region and country sat in the tracked micro-label row **above** the number — at 11px, alongside "V5RC team profile" and the active flag — where it read as metadata rather than as something about the team. It now has its own line directly under the lockup at ***18px, rising to 24px*** from the ``sm`` breakpoint, with a pin in the accent colour. The eyebrow row keeps only what it's good at: the profile label and the active state. The hero stagger rennumbers accordingly, so the location fades in between the number and the name rather than alongside the block below it. Verified on ``16610A``: eyebrow reads "V5RC team profile / Active", and "Toronto, Ontario, Canada" renders as the next sibling after the lockup. 37/37 tests pass, lint unchanged at ...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/21>

Pull request decisions 22 and 23

PR 22 Draw H's crossbar before its stems

Opened	Merged or closed	Branch
2026-09-17 20:10 UTC	2026-09-17 20:10 UTC	feat/routing-and-dx

Recorded purpose and review context: H was authored left stem, right stem, crossbar. It now draws the horizontal first, then both verticals. Verified in draw order on `1940H` by sorting the H's three paths by their animation delay: `` 1. M10,50 L50,50 <- crossbar 2. M10,6 L10,95 3. M50,6 L50,95 `` 37/37 tests pass.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/22>

PR 23 H draws left stem, crossbar, then right stem

Opened	Merged or closed	Branch
2026-09-17 20:12 UTC	2026-09-17 20:13 UTC	feat/routing-and-dx

Recorded purpose and review context: The previous commit took "horizontal first" literally and put the crossbar ahead of both stems. The intended order is the conventional one: left vertical, horizontal, right vertical. Verified in draw order on `1940H` by animation delay: `` M10,6 L10,95 1080ms left stem M10,50 L50,50 1404ms crossbar M50,6 L50,95 1656ms right stem `` 37/37 tests pass.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/23>

Pull request decisions 24 and 25

PR 24 Shared links work in a browser that has never opened the site

Opened	Merged or closed	Branch
2026-09-17 20:18 UTC	2026-09-17 20:18 UTC	feat/routing-and-dx

Recorded purpose and review context: A team link pasted into another browser landed on the home view. ## Cause The hash-routing block sat ****inside**** the `if(saved){...}` branch of the `sessionStorage`` restore, so it only ran when that browser already had saved navigation state. A first-time visitor — which is exactly what the recipient of a shared link is — has empty `sessionStorage``, so `saved`` was `null``, the branch was skipped, and the URL was never read at all. ## Why my earlier testing missed it I verified "cold loads" by hard-reloading a browser that had been clicking around beforehand, so it always had state to restore. ****Reloading is not the same test as arriving fresh****, and only the second one matches how a link is actually used. The fix here is as much about the tes...

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/24>

PR 25 Add Discord invite to the header

Opened	Merged or closed	Branch
2026-09-23 09:01 UTC	2026-09-23 09:01 UTC	codex/discord-header-link

Recorded purpose and review context: Adds a Discord icon immediately left of the theme selector. The icon opens the requested invite in a new tab. No other site behavior or styling is changed.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/25>

Pull request decisions 26 and 27

PR 26 Make room for blog posts beside upcoming Signature Events

Opened	Merged or closed	Branch
2026-09-23 09:07 UTC	2026-09-23 09:09 UTC	codex/home-blog-and-signatures

Recorded purpose and review context: Moves the existing upcoming Signature Event cards into a vertical right column below the hero and ranking. Adds a left blog area with an honest placeholder for future posts. Removes the three promotional feature blocks. Event cards and navigation retain their current behavior; existing type, colours, and sizing are reused. Kept on a separate branch for easy rollback.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/26>

PR 27 Show featured event date and align blog panel

Opened	Merged or closed	Branch
2026-09-23 09:26 UTC	2026-09-23 09:27 UTC	codex/home-date-blog-alignment

Recorded purpose and review context: Replaces the decorative 01 on the featured event with that event's month and day. Makes the blog placeholder fill its column so its bottom edge lines up with the vertical Signature Event list. Keeps existing type, colours, event navigation, and layout structure.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/27>

Pull request decisions 28 and 29

PR 28 Take blog layout off the live homepage

Opened	Merged or closed	Branch
2026-09-23 09:50 UTC	2026-09-23 09:57 UTC	codex/revert-home-blog-layout

Recorded purpose and review context: Restores the earlier horizontal Signature Events section and the three feature blocks. Keeps the Discord link and the featured event's date watermark. The blog layout will be preserved on a separate unmerged branch for later.

Publication result: Merged into main.

Source: <https://github.com/easonli29/Vex-Rank/pull/28>

PR 29 Hold blog homepage layout for later

Opened	Merged or closed	Branch
2026-09-23 09:56 UTC	Open at snapshot	codex/home-blog-on-hold

Recorded purpose and review context: Unmerged draft containing the blog panel and vertical Signature Event list. The live-site revert is PR #28; this draft preserves the blog design for a future decision. The featured event date and Discord link are preserved in both versions.

Publication result: Open for review; do not count it as published behavior.

Source: <https://github.com/easonli29/Vex-Rank/pull/29>

Part Fourteen Complete Audit and Reference Appendix

This appendix provides a complete GitHub commit index, current source/test pointers, and a repeatable way to extend the notebook. All 101 commit identities, including 26 merges, are listed so the detailed records can be reconciled with GitHub. Times in the register use the author timestamp and its recorded offset; merge timing is also given separately in the pull request register.

Complete GitHub Commit Register

Date	SHA	Commit subject
2026-09-09	3a633af7	Initial commit
2026-09-09	63877ed5	Add VEX ranking website, algorithm PDFs, and project documentation
2026-09-09	55481c6e	Add algorithm documentation links
2026-09-09	62b834c2	Link algorithm flowchart in setup documentation
2026-09-09	9b32fda6	Expand README with VCR algorithm overview
2026-09-11	a7655931	Fix Worlds results and improve website data/loading reliability
2026-09-14	261ebc24	Prepare GitHub Pages and Firebase test deployment; document Spark billing blocker
2026-09-14	f1804c37	Connect GitHub Pages test build to live Cloudflare Workers and D1 backend
2026-09-15	c3a0847c	Merge pull request #4 from easonli29/codex/firebase-test-deployment
2026-09-15	88eb4200	Keep homepage Signature events independent of calendar season filters
2026-09-15	8b114efc	docs: link VCR 3.0 implementation
2026-09-15	979d37dc	feat: apply VCR 3.0 to global rankings
2026-09-15	e8e16001	feat: apply VCR 3.0 to team history
2026-09-15	90ac608f	feat: apply VCR 3.0 to historical rankings
2026-09-15	a7feaf42	feat: add VCR 3.0 reference event model
2026-09-15	fad6a898	test: add VCR 3.0 invariants
2026-09-15	0454fda9	docs: publish VCR 3.0 algorithm specification
2026-09-15	7f43b66c	docs: record VCR 3.0 candidate provenance
2026-09-15	035e21ba	feat: add shared VCR 3.0 processor
2026-09-15	4a2186bd	test: cover live VCR 3.0 processor
2026-09-15	cdf6f047	Merge pull request #5 from easonli29/codex/vcr-3.0-implemented
2026-09-15	cb5a298e	fix: verify VCR 3.0 deployment and valid alliance credit
2026-09-15	d0bf1367	fix: verify VCR 3.0 deployment and valid alliance credit
2026-09-15	32028f15	fix: verify VCR 3.0 deployment and valid alliance credit
2026-09-15	3a09c7c1	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	49984c13	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	745214f3	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	54d8bb88	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	deb01cc1	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	c2182cf0	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	8f89ed5e	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	264932ca	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	90c8fda8	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	a6a709c6	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	123f488d	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	25363edc	feat: replay and publish versioned VCR 3.0 historical archives
2026-09-15	14a75ca2	fix: handle historical archive validation before state updates
2026-09-15	be181fce	Merge pull request #6 from easonli29/codex/vcr3-historical-seasons
2026-09-15	16e8bdd0	perf: pace historical requests and resume from cached pages
2026-09-15	66c9aa07	Merge pull request #7 from easonli29/codex/vcr3-archive-download-resume
2026-09-16	e1c5ec86	docs: explain ranking internals and contributor workflow
2026-09-16	ef75e139	Merge pull request #8 from easonli29/codex/contributor-code-comments
2026-09-16	7b02a137	Update team search examples to 8829C and 55288A
2026-09-16	b8b03376	Search full V5RC team directory with country, region and grade filters
2026-09-16	c3e01a93	feat: editorial redesign, scroll motion, and capped result lists
2026-09-16	2a6ad332	Merge branch 'main' into feat/editorial-ui-redesign
2026-09-16	38394049	Merge pull request #9 from StevenQQian/feat/editorial-ui-redesign
2026-09-16	0f73d151	fix: resolve merge conflict left unresolved in page.tsx
2026-09-16	9b16bb7a	chore: ignore local .claude agent config
2026-09-16	b2a127ee	fix: stop vinext treating the static-site entry as a Pages router
2026-09-16	c96decc9	chore: make the linter actionable and clear the debt it was hiding
2026-09-16	5d7e3562	ci: verify build, tests and lint on every pull request
2026-09-16	12d852ca	a11y: lift muted text above the WCAG AA contrast floor
2026-09-16	597af579	feat: shareable URLs, working Back button, and deep links
2026-09-16	70213f9f	refactor: extract hash routing into a tested module
2026-09-16	eab79b14	refactor: extract country and event-region vocabularies
2026-09-16	38765dd3	Merge pull request #10 from StevenQQian/feat/routing-and-dx
2026-09-17	41a1a3e7	fix: retry transient API failures and stop claiming 'no events' when a load fails
2026-09-17	40e14673	Merge pull request #11 from StevenQQian/feat/routing-and-dx
2026-09-17	4639d7e0	docs: add roadmap with findings against the open items
2026-09-17	0c23a745	fix: make the ranking list self-consistent and align the graph with it
2026-09-17	66824539	fix: one scoring definition everywhere, drop dead retry, explain the sampling limit
2026-09-17	1dba5cb8	feat: pointer-tracked card highlights and better scroll behaviour
2026-09-17	84cfce5b	fix: season band no longer pops in as an empty slab on a team profile
2026-09-17	6d68e89e	feat: four colour themes with a picker
2026-09-17	5b17e215	Merge pull request #12 from StevenQQian/feat/routing-and-dx
2026-09-17	311c5876	feat: team numbers are signed on in a handwriting face

VEX Rank Engineering Notebook Dated History Edition

Date	SHA	Commit subject
2026-09-17	5bda4105	Merge pull request #13 from StevenQQian/feat/routing-and-dx
2026-09-17	bca7c002	feat: trace team numbers stroke by stroke instead of revealing them
2026-09-17	eb18c605	Merge pull request #14 from StevenQQian/feat/routing-and-dx
2026-09-17	2acff59f	feat: slow the signature and ease off toward the final character
2026-09-17	c73ad6df	Merge pull request #15 from StevenQQian/feat/routing-and-dx
2026-09-17	ddfec42c	tune: quicken the signature by about 20 percent
2026-09-17	3e780f4d	Merge pull request #16 from StevenQQian/feat/routing-and-dx
2026-09-17	a0ebdb00	fix: jump to the top on navigation instead of scrolling there smoothly
2026-09-17	4525e61a	Merge pull request #17 from StevenQQian/feat/routing-and-dx
2026-09-17	501e6888	fix: undrawn strokes no longer leave stray ink before they are written
2026-09-17	82fb90d5	Merge pull request #18 from StevenQQian/feat/routing-and-dx
2026-09-17	81786ee8	fix: tell the reader when a content blocker is stopping the data
2026-09-17	960f6938	fix: bind the catch parameter in the rankings error handler
2026-09-17	52f6fb3a	Merge pull request #19 from StevenQQian/feat/routing-and-dx
2026-09-17	4cb9e0ce	feat: remove the global search from the header
2026-09-17	134e712d	Merge pull request #20 from StevenQQian/feat/routing-and-dx
2026-09-17	7feb6c55	feat: give a team's location its own line under the number
2026-09-17	2329e6ab	Merge pull request #21 from StevenQQian/feat/routing-and-dx
2026-09-17	8a249087	fix: draw H's crossbar before its stems
2026-09-17	d5c2139b	Merge pull request #22 from StevenQQian/feat/routing-and-dx
2026-09-17	2f9719d9	fix: H draws left stem, crossbar, then right stem
2026-09-17	6f6767a7	Merge pull request #23 from StevenQQian/feat/routing-and-dx
2026-09-17	ee81570d	fix: shared links work in a browser that has never opened the site
2026-09-17	1ba6b81d	Merge pull request #24 from StevenQQian/feat/routing-and-dx
2026-09-22	9e551130	Use root path for vex-rank.com
2026-09-22	4466dc67	Allow vex-rank.com to use ranking API
2026-09-23	12268ffb	Add Discord invite icon beside theme selector
2026-09-23	73bfd070	Merge pull request #25 from easonli29/codex/discord-header-link
2026-09-23	03c75b78	Place blog section beside vertical Signature Events
2026-09-23	e1349b18	Merge pull request #26 from easonli29/codex/home-blog-and-signatures
2026-09-23	7bf2bc8d	Show featured event date and align blog panel
2026-09-23	335a3cf8	Merge pull request #27 from easonli29/codex/home-date-blog-alignment
2026-09-23	dd63f3ed	Revert blog homepage layout while keeping featured date
2026-09-23	6a03cd05	Revert blog layout from live site; keep draft PR #29 unmerged

Current Source and Verification References

GitHub repository: <https://github.com/easonli29/Vex-Rank> (main at 6a03cd0 on September 23 2026).

GitHub pull request register: <https://github.com/easonli29/Vex-Rank/pulls?q=is%3Apr> (PRs 1-29 through September 23; number 3 is an issue, not a pull request).

Current CI run 35845951399: <https://github.com/easonli29/Vex-Rank/actions/runs/35845951399>.

Current test deployment run 35846097394: <https://github.com/easonli29/Vex-Rank/actions/runs/35846097394>.

Repository source: [website/lib/vcr3.mjs](#), [website/lib/vcr-scoring.mjs](#), [website/lib/routes.mjs](#), [website/lib/client-fetch.ts](#), [website/cloudflare/worker.ts](#), [website/components/signed-number.tsx](#).

Deployment configuration: [.github/workflows/ci.yml](#), [.github/workflows/github-pages-test.yml](#), [website/cloudflare/DEPLOYMENT.md](#), [website/firebase/DEPLOYMENT.md](#).

Archive provenance: [versions/vcr3-historical-archives.json](#) and [website/docs/HISTORICAL-VCR3.md](#).

Algorithm versions: [docs/VCR-3.0-revision.md](#), [versions/vcr-3.0-candidate.md](#), [output/pdf/vex_competitive_rating_full_algorithm.pdf](#), and the supplied VCR v1.0 Lucidchart PDF.

Local website history: workspace [website/.git](#), commits f402ff8 through ddc5680, September 1-11 2026.

How to continue this dated record

Record a new entry only after identifying its commit, pull request, observation time, model version, and verification evidence. Separate an implemented behavior from a design proposal and a locally observed result from a deployed one. Re-run deterministic tests and build checks at the new head, refresh archive counts only from a validated dataset, update the table of contents, and render every page before publishing another edition.

Part Fifteen - September 27 2026 Release Record

Hide test banner and unfinished Stat leaders

Decision: Remove the VEXRank test deployment strip. Hide Stat leaders from desktop and mobile navigation and the homepage while its data is not ready. Keep the implementation in source for future work. The two remaining homepage feature cards form a two-column row.

Compatibility: Old `#/stats` links, saved Stat leaders sessions, and old team return destinations resolve to Rankings. A regression test covers the old link.

Implementation: PR #31 changes `website/static-site/main.tsx`, `website/app/page.tsx`, `website/lib/routes.mjs`, and `website/tests/routes.test.mjs`. The existing stylesheet remains because it also defines font variables.

Verification: GitHub CI run #45 succeeded. PR #31 was squash-merged into main at `389693de8f88ee9ea85f8b6f92d8da03014ceb63`. GitHub Pages deployment run #27 completed successfully from that commit. Public-site inspection on September 27 found no test banner, Stat leaders navigation, or homepage card. Events, Rankings, Teams, the Discord link, and both remaining feature cards were visible. This does not revalidate underlying event or team data.

Pull request: <https://github.com/easonli29/Vex-Rank/pull/31>

CI: <https://github.com/easonli29/Vex-Rank/actions/runs/36359005901>

Deployment: <https://github.com/easonli29/Vex-Rank/actions/runs/36359153840>

Public site: <https://vex-rank.com/>